



FIGURE 51-36
Ligament injury after trauma
Coronal CT reconstruction demonstrates abnormal asymmetry between the dens and the lateral masses of C1 indicating transverse ligament rupture.

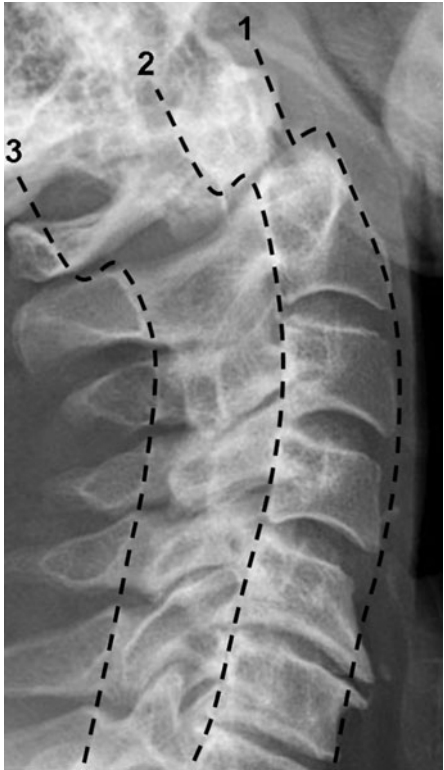


FIGURE 51-37
Odontoid fracture
Sagittal CT demonstrates disruption of the main reference cervical lines. 1: Anterior vertebral body line; 2: Posterior vertebral body line; 3: Spinolaminar line.



FIGURE 51-38
Pathologic fracture
Sagittal T1-weighted MRI (A) demonstrates wedge-shaped T6 vertebral body (arrow). Sagittal postcontrast T1-weighted MRI (B) depicts tumor extension into the epidural space and the involvement of the posterior arch (*), which are highly suggestive of metastatic or primary bone tumor.

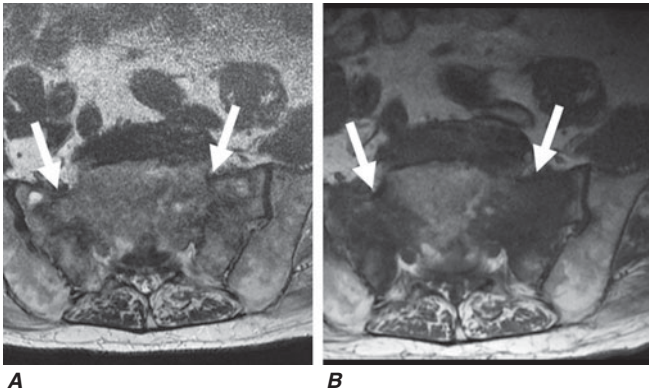
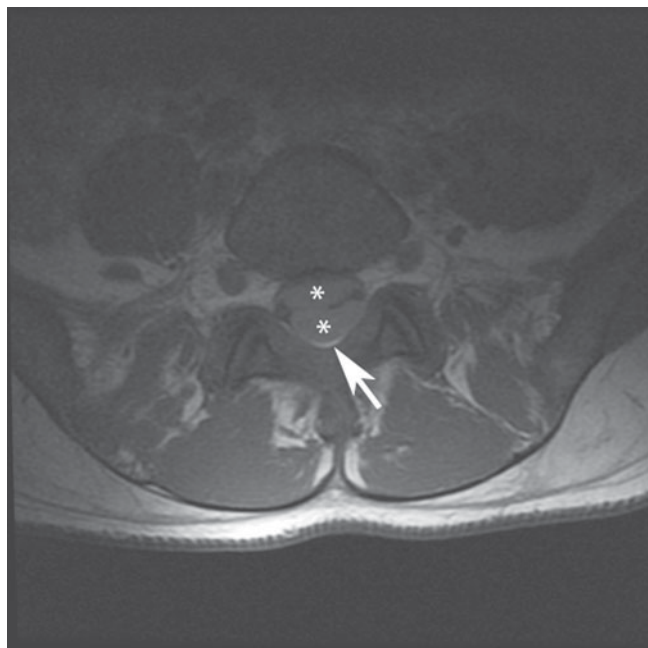


FIGURE 51-39
Sacral insufficiency fracture
Axial T2-weighted MRI (A) and T1-weighted MRI (B) demonstrate symmetric high T2 and low T1 signal involving the sacral alae longitudinally (arrows).

**A****B****FIGURE 51-40****Subdural hematoma**

Sagittal T2-weighted MRI (**A**) and axial noncontrast T1-weighted MRI (**B**) demonstrate subdural collection in the lumbosacral region (**). Note that the epidural fat is compressed but not involved (*arrow*).

**A****B****FIGURE 51-41****Teardrop fracture**

Sagittal CT (**A**) demonstrates fracture line separating the antero-inferior corner of C6 (*arrow*). Sagittal T2-weighted MRI (**B**) displays cord injury (*arrow*).

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SECTION IV

CHRONIC FATIGUE SYNDROME

CHAPTER 52

CHRONIC FATIGUE SYNDROME



Gijs Bleijenberg ■ Jos W.M. van der Meer

DEFINITION

Chronic fatigue syndrome (CFS) is a disorder characterized by persistent and unexplained fatigue resulting in severe impairment in daily functioning. Besides intense fatigue, most patients with CFS report concomitant symptoms such as pain, cognitive dysfunction, and unrefreshing sleep. Additional symptoms can include headache, sore throat, tender lymph nodes, muscle aches, joint aches, feverishness, difficulty sleeping, psychiatric problems, allergies, and abdominal cramps. Criteria for the diagnosis of CFS have been developed by the U.S. Centers for Disease Control and Prevention ([Table 52-1](#)).

TABLE 52-1

DIAGNOSTIC CRITERIA FOR CHRONIC FATIGUE SYNDROME

Characterized by Persistent or Relapsing Unexplained Chronic Fatigue

Fatigue lasts for at least 6 months
Fatigue is of new or definite onset
Fatigue is not the result of an organic disease or of continuing exertion
Fatigue is not alleviated by rest
Fatigue results in a substantial reduction in previous occupational, educational, social, and personal activities
Four or more of the following symptoms, concurrently present for 6 months:
impaired memory or concentration, sore throat, tender cervical or axillary lymph nodes, muscle pain, pain in several joints, new headaches, unrefreshing sleep, or malaise after exertion

Exclusion Criteria

Medical condition explaining fatigue
Major depressive disorder (psychotic features) or bipolar disorder
Schizophrenia, dementia, or delusional disorder
Anorexia nervosa, bulimia nervosa
Alcohol or substance abuse
Severe obesity (BMI >40)

EPIDEMIOLOGY

CFS is seen worldwide, with adult prevalence rates varying between 0.2 and 0.4%. In the United States, the prevalence is higher in women, members of minority groups (African and Native Americans), and individuals with lower levels of education and occupational status. Approximately 75% of all CFS patients are women. The mean age of onset is between 29 and 35 years. It is probable that many patients go undiagnosed and/or do not seek help.

ETIOLOGY

There are numerous hypotheses about the etiology of CFS; there is no definitively identified cause. Distinguishing between predisposing, precipitating, and perpetuating factors in CFS helps to provide a framework for understanding this complex condition ([Table 52-2](#)).

Predisposing factors

Physical inactivity and trauma in childhood tend to increase the risk of CFS in adults. Neuroendocrine dysfunction may be associated with childhood trauma, reflecting a biological correlate of vulnerability. Psychiatric illness and physical hyperactivity in adulthood raise the risk of CFS in later life. Twin studies suggest a familial predisposition to CFS, but no causative genes have been identified.

Precipitating factors

Physical or psychological stress may elicit the onset of CFS. Most patients report an infection (usually a flu-like illness or infectious mononucleosis) as the trigger of their fatigue. Relatively high percentages of CFS follow Q fever and Lyme disease. However, no differences were

TABLE 52-2**PREDISPOSING, PRECIPITATING, AND PERPETUATING FACTORS IN CHRONIC FATIGUE SYNDROME****TIME****Predisposing Factors**

Childhood trauma (sexual, physical, emotional abuse; emotional and physical neglect)
 Physical inactivity during childhood
 Premorbid psychiatric illness or psychopathology
 Premorbid hyperactivity

**Precipitating Factors**

Somatic events: infection (mononucleosis, Q fever, Lyme disease), surgery, pregnancy
 Psychosocial stress, life events

**Perpetuating Factors**

(Non)acknowledgement by physician
 Negative self-efficacy
 Strong physical attributions
 Strong focus on bodily symptoms
 Fear of fatigue
 (Lack of) social support
 Low physical activity pattern

found in Epstein-Barr virus load and immunologic reactivity in individuals who developed CFS and those who did not. While antecedent infections are associated with CFS, a direct microbial causality is unproven and unlikely. A recent study identified a murine leukemia virus-related retrovirus (XMRV); however, several subsequent studies have failed to confirm this result. Patients also often report other precipitating somatic events such as serious injury, surgery, pregnancy, or childbirth. Serious life events such as the loss of a loved one or a job, military combat, and other stressful situations may also precipitate CFS. A third of all patients cannot recall a trigger.

Perpetuating factors

Once CFS has developed, numerous factors may impede recovery. Physicians may contribute to chronicity by ordering unnecessary diagnostic procedures, by persistently suggesting psychological causes, and by not acknowledging CFS as a diagnosis.

A patient's focus on illness and avoidance of activities may perpetuate symptoms. A firm belief in a physical cause, a strong focus on bodily sensations, and a poor sense of control over symptoms may also prolong or exacerbate the fatigue and functional impairment. In most patients, inactivity is caused by negative illness

perceptions rather than by poor physical fitness. Solicitous behavior of others may reinforce a patient's illness-related perceptions and behavior. A lack of social support is another known perpetuating factor.

PATHOPHYSIOLOGY

The pathophysiology of CFS is unclear. Neuroimaging studies have reported that CFS is associated with reduced gray matter volume, associated with a decline in physical activity; these changes were partially reversed following cognitive behavioral therapy (CBT). In addition, functional MRI data have suggested that abnormal patterns of activation correlate with self-reported problems with information processing. Neurophysiologic studies have shown altered CNS activation patterns during muscle contraction.

Evidence for immunologic dysfunction is inconsistent. Modest elevations in titers of antinuclear antibodies, reductions in immunoglobulin subclasses, deficiencies in mitogen-driven lymphocyte proliferation, reductions in natural killer cell activity, disturbances in cytokine production, and shifts in lymphocyte subsets have been described. None of these immune findings appear in most patients, nor do any correlate with the severity of CFS. In theory, symptoms of CFS could result from excessive production of a cytokine, such as interleukin 1, that induces asthenia and other flu-like symptoms; however, compelling data in support of this hypothesis are lacking.

There is some evidence that CFS patients have mild hypocortisolism, the degree of which is associated with a poorer response to CBT.

Discrepancies in perceived and actual cognitive performance are a consistent finding in patients with CFS.

DIAGNOSIS

In addition to a thorough history, a systematic physical examination is warranted to exclude disorders causing fatigue (e.g., endocrine disorders, neoplasms, heart failure, etc.). The heart rate of CFS patients is often slightly above normal. Laboratory tests primarily serve to exclude other diagnoses; there is no test that can diagnose CFS. The following laboratory screen usually suffices: complete blood count; ESR, CRP; serum creatinine, electrolytes, calcium and iron; blood glucose; creatine kinase; liver function tests; TSH; anti-gliadin antibodies; urinalysis. Serology for viral or bacterial infections is usually not helpful. No specific abnormalities have been identified on MRI or CT scans. CFS is a constellation of symptoms without any pathognomonic features, and remains a diagnosis of exclusion.

Bipolar disorders, schizophrenia, and substance abuse exclude a diagnosis of CFS, as do eating disorders,

unless these have been resolved 5 years or longer before symptom onset. Also, CFS is excluded if the chronic fatigue developed immediately after a depressive episode. Depression developing in the course of the fatigue, however, does not preclude CFS. Co-occurring psychiatric disorder, especially anxiety and mood disorders, is seen in 30–60% of all cases.

INITIAL MANAGEMENT

In cases of suspected CFS, the clinician should acknowledge the impact of the patient’s symptoms on daily functioning. Disbelief or denial can provoke an exacerbation of genuine symptoms, which in turn strengthens the clinician’s disbelief, leading to an unfortunate cycle of miscommunication. The possibility of CFS should be considered if a patient fulfils all criteria (Table 52-1) and if other diagnoses have been excluded.

The patient should be asked to describe the symptoms (fatigue and accompanying symptoms) and their duration as well as the consequences (reduction in daily activities). To assess symptom severity and the extent of daily-life impairment, the patient should describe a typical day, from waking to retiring, and to contrast this with an average day prior to symptom onset. Next, potential fatigue-precipitating factors are sought. The severity of fatigue is difficult to assess quantitatively; a brief questionnaire is often helpful (Fig. 52-1).

The patient should be informed of the current understanding of precipitating and perpetuating factors and effective treatments, and be offered general advice about disease management. If CBT for CFS is not available

as an initial treatment option (see below) and depression and anxiety are present, these symptoms should be treated. For patients with headache, diffuse pain, and feverishness, nonsteroidal anti-inflammatory drugs may be helpful. Even modest improvements in symptoms can make an important difference in the patient’s degree of self-sufficiency and ability to appreciate life’s pleasures.

Controlled therapeutic trials have established that acyclovir, fludrocortisone, galantamine, modafinil, and IV immunoglobulin, among other agents, offer no significant benefit in CFS. Countless anecdotes circulate regarding other traditional and nontraditional therapies. It is important to guide patients away from those therapeutic modalities that are toxic, expensive, or unreasonable.

The patient should be encouraged to maintain regular sleep patterns, to remain as active as possible, and to gradually return to previous exercise and activity (work) levels.

TREATMENT

Chronic Fatigue Syndrome

CBT and graded exercise therapy (GET) have been found to be the only beneficial interventions in CFS. Some patient groups argue against these approaches because of the implication that CFS is a purely mental disorder. CBT is a psychotherapeutic approach directed at changing condition-related cognitions and behaviors. CBT for CFS aims at changing a patient’s perpetuating factors by exploiting various techniques and components.

How have you felt during the last two weeks?

Please rate all four statements and per statement check the box that reflects your situation best.

1. I feel tired

Yes, that is true

No, that is not true

2. I tire easily

Yes, that is true

No, that is not true

3. I feel fit

Yes, that is true

No, that is not true

4. Physically I feel exhausted

Yes, that is true

No, that is not true

Scoring:

1, 2 and 4:

Yes, that is true

7

6

5

4

3

2

1

No, that is not true

3: Reversed

Sum scores >18 indicate severe fatigue

FIGURE 52-1
Shortened fatigue questionnaire (SFQ).

It includes educating the patient about the etiologic model, setting goals, restoring fixed bedtime and wake-up time, challenging and changing fatigue- and activity-related cognitions, reducing symptom focusing, spreading activities evenly throughout the day, gradually increasing physical activity, planning a return to work, and resuming other activities. The intervention, which typically consists of 12–14 sessions spread over 6 months, helps CFS patients gain control over their symptoms.

GET is based on the model of deconditioning and exercise intolerance and usually involves a home exercise program that continues for 3–5 months. Walking or cycling is systematically increased, with set target heart rates. Evidence that deconditioning is the basis for symptoms in CFS is lacking, however. The primary component of CBT and GET that results in a reduction in fatigue is the change in the patient's perception of fatigue and focus on symptoms.

CBT is generally the more complex treatment, which might explain why CBT studies tend to yield better improvement rates than GET trials.

Not all patients benefit from CBT or GET. Predictors of poor outcome are somatic comorbidity, current disability claims, and severe pain. CBT offered in an early stage of the illness reduces the burden of CFS for the patient as well as society in terms of decreased medical and disability-related costs.

PROGNOSIS

Full recovery from untreated CFS is rare: the median annual recovery rate is 5% (range 0–31%) and the improvement rate 39% (range 8–63%). Patients with an underlying psychiatric disorder and those who continue to attribute their symptoms to an undiagnosed medical condition have poorer outcomes.

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SECTION V

PSYCHIATRIC DISORDERS

CHAPTER 53

BIOLOGY OF PSYCHIATRIC DISORDERS



Robert O. Messing ■ John H. Rubenstein ■ Eric J. Nestler

Psychiatric disorders are central nervous system diseases characterized by disturbances in emotion, cognition, motivation, and socialization. As a result of their high prevalence, early onset, and persistence, they contribute substantially to the burden of illness worldwide. Most psychiatric disorders are heterogeneous syndromes that currently lack well-defined neuropathology and *bona fide* biological markers. Therefore, diagnoses continue to be made solely from clinical observations using criteria in the *Diagnostic and Statistical Manual of Mental Disorders* of the American Psychiatric Association (2000), 4th edition, text revision (DSM-IVTR). Recent advances in neuroimaging are beginning to provide evidence of brain pathology, which may one day be used for diagnosis and for following treatment. Family, twin, and adoption studies have shown that all common psychiatric syndromes are highly heritable, with genetic risk comprising 20–90% of disease vulnerability. The epidemiology, genetics, and biology of four common psychiatric disorders—autism, schizophrenia, mood disorders, and drug addiction—are presented in this chapter. A detailed discussion of the clinical manifestations and treatment of schizophrenia and mood disorders can be found in Chap. 54. Further discussion of alcoholism can be found in Chap. 56, opiate addiction in Chap. 57, and cocaine and other drugs of abuse in Chap. 58.

AUTISM SPECTRUM DISORDERS

The DSM-IVTR criteria for autism spectrum disorders (ASDs) require delays or abnormal functioning in social interactions, language as used in social communication, and symbolic or imaginative play, with onset prior to age 3. In addition to abnormal social behavior, ASDs are frequently, but not always, associated with reduced IQ and epilepsy. Individuals who exhibit some autism-like

symptoms with relatively preserved cognitive functioning and language skills are described as having Asperger's syndrome.

EPIDEMIOLOGY

There has been a dramatic increase in the diagnosis of ASDs, from ~1/1000 (1950s–1990s) to a current level of ~1/150. Whether this increase reflects increased disease prevalence remains uncertain; ongoing studies are searching for genetic, environmental, and sociologic mechanisms that may have contributed to this change. In the 1950s–1960s, psychological factors were held to underlie autism. This conception was largely debunked by the 1970s, with the demonstration that prenatal rubella and phenylketonuria can cause ASDs, and with evidence for the genetic etiology of ASDs from twin studies. There is ongoing public concern that vaccines in general, or mercury-based preservatives in vaccines, can cause ASDs; however, large epidemiologic analyses have not supported this as an etiology. Whether environmental factors, such as perinatal infection and various toxins, for example, ethanol, illicit drugs, medications, and mutagenic agents, play a role is unclear.

NEUROPATHOLOGY AND NEUROIMAGING

ASDs show no defining neuroanatomic phenotype that would indicate neurodevelopmental abnormalities. However, structural neuroimaging and histologic studies of postmortem brain provide evidence for anatomic defects. There is a modest increase in cerebrum growth (~10%; affecting both the white and grey matter) during early childhood (years 1–3), with the largest effect in the frontal lobes; the growth rate then decreases with age. Cerebellar size is increased by about 7% in children under age 5 years, but is decreased in older patients, and there are reduced (~30%) numbers of cerebellar Purkinje

neurons. Finally, there is reduced cell size and increased cell density in the limbic areas of the brain.

GENETICS

ASDs are highly heritable; concordance rates in monozygotic twins (~60–90%) are roughly tenfold higher than in dizygotic twins and siblings, and first-degree relatives show about fiftyfold increased risk

for autism compared with prevalence in the general population. For unknown reasons, ASDs affect four times as many boys as girls. ASDs are also genetically heterogeneous. More than 20 known mutations, including copy number variations, account for about 10–20% of all cases, though none of these causes individually accounts for more than 1–2% (**Table 53-1**). Many of the genes linked to ASDs can also cause other illnesses. For instance, mutations in MeCP2,

TABLE 53-1

EXAMPLES OF GENES IMPLICATED IN AUTISM

GENE SYMBOL	GENE NAME	FUNCTION
PTEN	Phosphatase and tensin homolog	Signal transduction Synaptic function
TSC1	Tuberous sclerosis 1	Signal transduction Translation and protein stability Synaptic function
TSC2	Tuberous sclerosis 2	Signal transduction Translation and protein stability Synaptic function
FMR1	Fragile X mental retardation 1	Translation and protein stability Synaptic function
UBE3A	Ubiquitin protein ligase E3A	Translation and protein stability Synaptic function
CNTN3	Contactin 3	Synaptic function
CNTN4	Contactin 4	Synaptic function
CNTNAP2	Contactin-associated protein-like 2	Synaptic function
NLGN3	Neurologin 3	Synaptic function
NLGN4	Neurologin 4	Synaptic function
NRXN1	Neurexin 1	Synaptic function
PCDH10	Protocadherin 10	Synaptic function
SHANK3	Shank 3	Synaptic function
SLC6A4	Serotonin transporter	Neurotransmitter signaling
AVPR1	Arginine vasopressin receptor 1	Neurotransmitter signaling
OXTR	Oxytocin receptor	Neurotransmitter signaling
CACNA1C	Voltage-gated calcium channel–alpha 1C subunit	Ion channel
CACNA1H	Voltage-gated calcium channel–alpha 1H subunit	Ion channel
SCN1A	Sodium channel, voltage-gated, type I, alpha subunit	Ion channel
SCN2A	Sodium channel, voltage-gated, type II, alpha subunit	Ion channel
SLC9A9	Sodium/hydrogen exchanger	Ion channel
DHCR7	7-Dehydrocholesterol reductase	Metabolism
PAH	Phenylalanine hydroxylase	Metabolism
ARX	Arx transcription factor	Gene expression
En2	Engrailed 2 transcription factor	Gene expression
MeCP2	Methyl CpG-binding protein 2 (Rett syndrome)	Gene expression
RNF8	Ring finger protein 8	Gene expression

FMR1, and TSC1&2 (see Table 53-1 for abbreviations) can cause mental retardation without ASDs, and alleles of certain genes, for example, neurexin 1, are associated with both ASDs and schizophrenia. It is likely that many cases of ASDs result from more complex genetic mechanisms, including inheritance of multiple genetic variants or epigenetic modifications.

PATHOGENESIS

Despite the genetic heterogeneity of ASDs, there are some common themes that may explain pathogenesis. These include mutations in proteins involved in the formation and function of synapses, control over the size and projections of neurons, production and signaling of neurotransmitters and neuromodulators, the function of ion channels, general cell metabolism, gene expression, and protein synthesis (see Table 53-1). Many of these mutations have a clear relationship to activity-dependent neural responses and can affect the development of neural systems that underlie cognition and social behaviors. They may be detrimental by altering the balance of excitatory vs. inhibitory synaptic signaling in local and extended circuits, and by altering the mechanisms that control brain growth. Another class of mutations affects genes (e.g., PTEN and Tsc) that negatively regulate signaling from several types of extracellular stimuli, including those transduced by receptor tyrosine kinases. Their dysregulation can have pleiotropic effects, including altering brain and neuronal growth as well as synaptic development and function. With further understanding of pathogenesis and the definition of specific ASD subtypes, there is reason to believe that effective therapies will be identified, as in the case of dietary treatments for phenylketonuria. In addition, work in mouse models (e.g., with fragile X or Rett syndrome mutations) has suggested that autism-like behavioral abnormalities can be reversed even in fully developed adult animals by reversing the underlying pathology, which holds out hope for many affected individuals.

SCHIZOPHRENIA

Schizophrenia appears to be a heterogeneous collection of many distinct diseases, which remain poorly defined but linked by common clinical features. Three major symptom clusters are seen in schizophrenia: positive, negative, and cognitive symptoms. Positive symptoms include hallucinations and delusions, experiences that are not characteristic of normal mental life. Negative symptoms represent deficits in normal functions such as blunted affect, impoverished speech, asocial behavior,

and diminished motivation. Cognitive symptoms include deficits in working memory and cognitive control of behavior that often prove extremely disabling. Current antipsychotic drugs are efficacious for positive symptoms only and generally lack efficacy for negative and cognitive symptoms.

EPIDEMIOLOGY

Schizophrenia is common, affecting males and females roughly equally, with a worldwide prevalence of approximately 1%. Environmental risks are thought to include prenatal exposure to viral infection (influenza), prenatal poor nutrition, perinatal hypoxia, psychotropic drug use (in particular, cannabis), and psychological stress. Advanced paternal age, birth order, and season of birth have also been implicated. However, none of these environmental influences has a specific or strong association with most cases of schizophrenia.

NEUROPATHOLOGY AND NEUROIMAGING

The best-established neuropathologic finding in schizophrenia is enlargement of the lateral ventricles of the cerebral hemispheres. This is accompanied by a reduction in cortical thickness. These abnormalities are not specific to schizophrenia and are seen in many other conditions, including many neurodegenerative disorders. However, there is a general consensus that the reduction in cortical thickness in schizophrenia is associated with increased cell packing density and reduced neuropil (defined as axons, dendrites, and glial cell processes) without an overt change in neuronal cell number. Specific classes of interneurons in prefrontal cortex consistently show reduced expression of the gene encoding the enzyme glutamic acid decarboxylase 1 (*GAD1*), which synthesizes γ -aminobutyric acid (GABA), the principal inhibitory neurotransmitter in the brain. Functional imaging studies, by positron emission tomography (PET) or functional magnetic resonance imaging (MRI), show evidence of reduced metabolic or neural activity in the dorsolateral prefrontal cortex at rest and when performing psychological tests of executive function, including working memory. Alleles of two candidate risk genes (catechol-O-methyltransferase [COMT] and metabotropic glutamate receptor 3 [mGluR3]) are reported to affect dorsolateral prefrontal cortex activity, but these findings need to be replicated in larger samples. Similar pathologic and brain imaging abnormalities are seen in several other brain regions, in particular, the hippocampus. There are also numerous reports of abnormalities in myelin and oligodendrocytes in the cerebral cortex of patients with schizophrenia.

GENETICS

Twins studies establish the heritability of schizophrenia, with co-inheritance at ~50% for monozygotic twins and ~10% for dizygotic twins. Genomewide linkage and association studies, and studies of copy number variation, have identified many regions and alleles that confer increased disease risk, particularly near genes on chromosome 22 (disrupted in schizophrenia 1 [DISC1], COMT, neuregulin 1, the neuregulin receptor ERBB4, and the DiGeorge [or velocardiofacial syndrome] region), and on chromosome 16p. The DiGeorge region deletions produce, in heterozygous form, a psychotic disorder with variable clinical features and a moderate to strong degree of penetrance. In contrast, the contribution of each of the individual genes to schizophrenia remains to be established with certainty. Moreover, the responsible genes within the DiGeorge region have not yet been identified. What is clear is that none of these other alleles produce schizophrenia with a high degree of penetrance. The current view in the field is that multiple rare alleles, many or most with limited penetrance, likely contribute to risk of schizophrenia. As for ASDs, the same allele may be a risk factor for multiple disorders. For instance, duplication of chromosome 16p is associated with both schizophrenia and autism, while DiGeorge region deletions and the *DISC1* locus on chromosome 22 are associated with schizophrenia, autism, and bipolar disorder.

PATHOGENESIS

There are several prevailing hypotheses about neurochemical mechanisms underlying schizophrenia. A reduction in the function of cortical and perhaps hippocampal GABAergic interneurons fits with reduced expression of glutamic acid decarboxylase. However, it is unknown whether this is a primary or compensatory feature of the disorder. Nevertheless, defects in parvalbumin-expressing GABAergic interneurons are known to reduce gamma-frequency activity on the EEG, which is a feature of many people with schizophrenia. Reduced excitatory neurotransmitter (glutamate) function is posited based on psychotic and cognitive symptoms generated in humans exposed to ketamine or phencyclidine, which are non-competitive antagonists of the NMDA subtype of glutamate receptors. There are reports of altered levels of glutamate receptors or associated proteins in the brains of individuals with schizophrenia examined postmortem, but no findings have yet been widely replicated. Finally, overactivity of dopamine neurotransmission at D₂-type dopamine receptors is proposed based on the ability of D₂ antagonists (an action common to all current antipsychotic agents; see Chap. 54) to ameliorate the positive symptoms of schizophrenia. Excessive dopamine release

in the striatum elicited by an acute dose of amphetamine has been demonstrated by PET imaging in some patients with schizophrenia. However, it is unclear whether this abnormality reflects the underlying illness or a lasting effect of antipsychotic medications. In contrast, reduced activity of dopamine at D₁ dopamine receptors in the prefrontal cortex has been implicated in working memory deficits based on the cognitive effects of D₁ receptor agonists and antagonists in the illness. Nevertheless, inferring something about disease pathogenesis from the actions of psychotropic drugs, for example, as with the glutamate and dopamine hypotheses, is fraught with artifact.

Efforts to understand how defects in these neurotransmitter systems might generate similar behavioral phenotypes have led to intriguing hypotheses. For instance, in the hippocampus, reduced glutamate transmission (based on a hypothesized deficit in glutamate release or glutamate receptors) onto GABAergic interneurons could lead to reduced glutamic acid decarboxylase expression, reduced gamma oscillations, and reduced inhibition onto excitatory neurons. These events in turn could lead to increased dopamine release from the ventral tegmental area, with dopamine antagonists thereby helping to reset the system to its nonpathologic state. It must be emphasized that these are working models only, and a true pathophysiology (or pathophysiologies) for schizophrenia remains to be established.

Overlaid on these neurotransmitter-based hypotheses is speculation as to how mutations in any of several genes implicated, however tentatively, in schizophrenia lead to the associated pathologic and behavioral abnormalities. DISC1 was originally discovered based on its association with schizophrenia in an Icelandic family. However, as stated earlier, DISC1 has since been variably associated with other neuropsychiatric conditions and its role in schizophrenia remains uncertain. The DISC1 protein has been implicated in several cellular functions, including neuronal growth and maturation, neurite outgrowth, and even the proliferation of new neurons during development. Neuregulin 1 (NRG1), a member of the EGF family of growth factors, and its receptor ERBB4 have also been implicated in schizophrenia in several genetic studies. Interestingly, NRG1 and ERBB4 play important roles in the maturation of GABAergic interneurons in cerebral cortex, and regulate dopamine transmission to several limbic brain regions. Moreover, loss of NRG1-ERBB4 in mice leads to reduced neuropil, thus phenocopying a pathologic finding in schizophrenia. Another gene of potential interest encodes Reelin, a secreted extracellular matrix serine protease. There are unconfirmed reports of association of schizophrenia with the Reelin locus on chromosome 7, and of reduced Reelin expression in the cerebral cortex of schizophrenic subjects, possibly related to increased methylation of the Reelin gene promoter.

Reelin is important during development in the migration of newly born neurons to their appropriate layers of cerebral cortex. In the adult brain, the protein is enriched in cortical GABAergic interneurons and has been implicated in regulating NMDA glutamate receptor function. It is, therefore, easy to imagine how abnormalities in DISC1, NRG1, or Reelin may be related to GABAergic, glutamatergic, and dopaminergic mechanisms in schizophrenia, and to associated pathologic abnormalities, but all such connections are currently speculative.

MOOD DISORDERS

Mood disorders are divided into depressive and bipolar disorders. Depressive disorders include the major depressive disorders, dysthymia, and more minor forms of depression. These disorders are heterogeneous syndromes, each composed of several diseases with presumably distinct pathophysiologies that remain to be elucidated.

EPIDEMIOLOGY

Mood disorders are common, with a prevalence of ~1–2% for bipolar disorder, ~5% for major depression, and ~15–20% for milder forms of depression. Between 40–50% of the risk for depression appears to be genetic. Nongenetic factors as diverse as stress and emotional trauma, viral infections, and even stochastic (random) processes during brain development have been implicated in the etiology. Depressive syndromes can occur in the context of general medical conditions such as endocrine disturbances (hyper- or hypocortisolemia, hyper- or hypothyroidism), autoimmune diseases, Parkinson’s disease, traumatic brain injury, certain cancers, asthma, diabetes, and stroke. Depression and obesity/metabolic syndrome are important risk factors for each other. In predisposed individuals, stressful life events can lead to clear-cut depressive episodes, while severe stress can induce posttraumatic stress disorder (PTSD), instead of depression. Bipolar disorder is characterized by episodes of mania and depression and is one of the most heritable of psychiatric illnesses, with genetic risk of ~80%. Stress and disrupted circadian rhythms can promote the manic episodes, during which patients exhibit extremely elevated mood, abnormal thought patterns, and sometimes psychosis. Several of these clinical signs can resemble certain features of schizophrenia; indeed, recent epidemiologic and genetic research has questioned the DSM-IVTR designations of bipolar disorder, schizophrenia, and schizoaffective disorder as distinct syndromes.

NEUROPATHOLOGY AND NEUROIMAGING

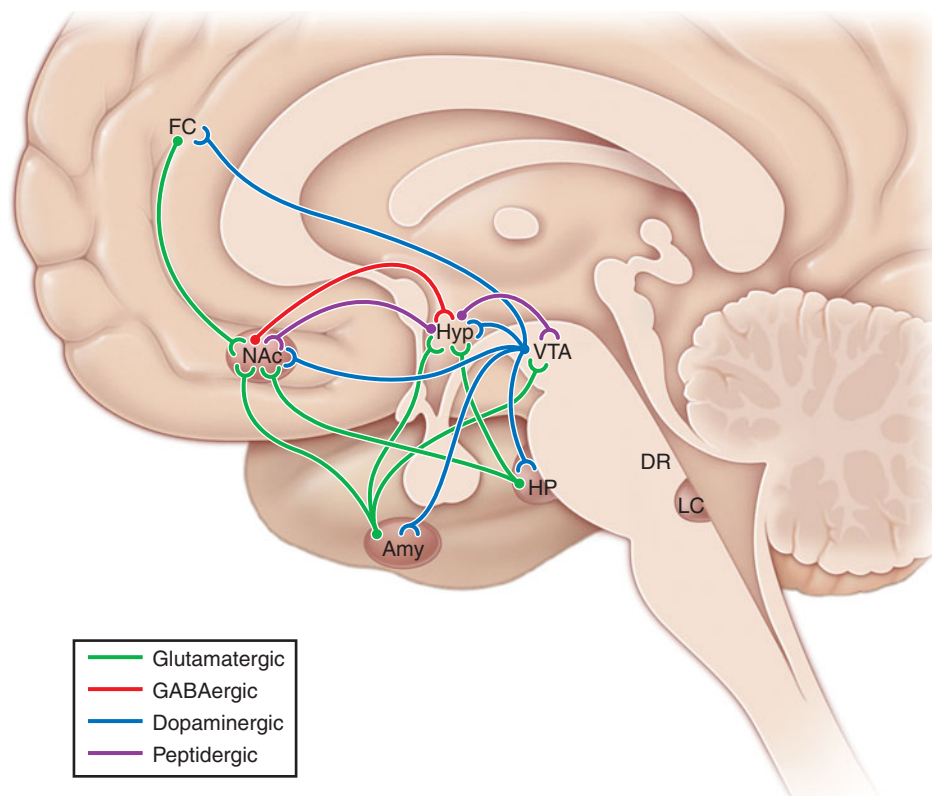
Brain imaging studies in humans are defining the neural circuitry of mood within the brain’s limbic system (Fig. 53-1). Integral to this system are the nucleus accumbens (important for brain reward—see later in the chapter under Substance Use Disorders), amygdala, hippocampus, and regions of prefrontal cortex. Given that many symptoms of depression (so-called neurovegetative symptoms) involve physiologic functions, a key role for the hypothalamus is also presumed. Depressed individuals show a small reduction in hippocampal size. PET and functional MRI have revealed increased activation of the amygdala by negative stimuli and reduced activation of the nucleus accumbens by rewarding stimuli. There is also evidence for altered activity in prefrontal cortex, for example, hyperactivity of subgenual area 25 in anterior cingulate cortex. Deep brain stimulation (DBS) of either the nucleus accumbens or subgenual area 25 elevates mood in normal and depressed individuals. While there are numerous reports of pathologic findings within these various regions postmortem, there is to date no defined neuropathology of depression.

GENETICS

Although depression and bipolar disorder are highly heritable, the specific genes that comprise this risk remain unknown. As noted earlier, some of the genes implicated in autism or schizophrenia seem to cause bipolar disorder in some families. Large genomewide association studies have identified genes for diacylglycerol kinase η (*DGKH*), ankyrin G (*ANK3*), an L-type voltage-gated calcium channel (*CACNA1C*), and a gene-rich region on chromosome 16p12 as being associated with bipolar disorder, but these findings await confirmation by additional studies. Numerous susceptibility genes have also been implicated in linkage and association studies, but none has yet been definitively established as a bona fide depression gene. However, a few genes with variants that may modify depression risk are worthy of mention since they may be linked to mechanisms of pathogenesis (discussed later). These include genes for the type 1 receptor for corticotrophin-releasing factor (*CRHR1*); the glucocorticoid receptor gene (*GR*); *FKBP5*, which encodes a chaperone protein for the glucocorticoid receptor; the serotonin transporter gene (*SLA6A4*); the catechol-O-methyltransferase gene (*COMT*); and brain-derived neurotrophic factor (*BDNF*).

PATHOGENESIS

Human and animal research in depression has focused on the long-term effects of chronic stress on the brain

**FIGURE 53-1**

Neural circuitry of depression and addiction. The figure shows a simplified summary of a series of limbic circuits in brain that regulate mood and motivation and are implicated in depression and addiction. Shown in the figure are the hippocampus (HP) and amygdala (Amy), regions of prefrontal cortex, nucleus accumbens (NAc), and hypothalamus (Hyp). Only a subset of the known interconnections among these brain regions is shown. Also shown is the innervation of several of these brain regions by monoaminergic neurons.

The ventral tegmental area (VTA) provides dopaminergic input to each of the limbic structures. Norepinephrine (from the locus coeruleus or LC) and serotonin (from the dorsal raphe [DR] and other raphe nuclei) innervate all of the regions shown. In addition, there are strong connections between the hypothalamus and the VTA-NAc pathway. Important peptidergic projections from the hypothalamus include those from the arcuate nucleus that release β -endorphin and melanocortin and from the lateral hypothalamus that release orexin.

and their reversal by antidepressant medications; prominent examples are discussed here. A subset of depressed patients show elevated levels of cortisol associated with increased production of corticotrophin-releasing factor from the hypothalamus and perhaps other brain regions (e.g., amygdala). In animals, sustained elevations in glucocorticoids impair hippocampal function, in part *via* direct damage to hippocampal neurons, which is consistent with reduced hippocampal volumes seen in some depressed humans. As the hippocampus exerts the major inhibitory influence over the hypothalamic-pituitary-adrenal axis, impairment of hippocampal function would lead to still further increases in glucocorticoid secretion, establishing a pathologic feed-forward loop.

Stress-induced damage to the hippocampus, and perhaps other limbic regions (e.g., amygdala), in animals is also mediated in part by reduced levels of BDNF and other growth factors and cytokines. Furthermore, stress

leads to a decrease in the birth of new neurons in the adult hippocampus. Interestingly, antidepressant treatments reverse these effects of stress, and the antidepressant effects of these medications seem to depend, in part, on their ability to promote hippocampal neurogenesis in animal models of depression. The clinical ramifications of such observations are unproven, although similar regulation of adult hippocampal neurogenesis may be important for certain forms of learning and memory.

Another important target of stress in animals is the nucleus accumbens, where stress regulation of numerous signaling events (dopaminergic transmission and BDNF signaling are two examples) exert potent effects on depression-like behavioral abnormalities. While a reduction in BDNF in the hippocampus promotes depression-like behaviors, an induction of BDNF in the nucleus accumbens promotes depression; similar changes in BDNF expression have been observed in

postmortem brains of depressed patients. Thus the role of BDNF in regulating mood is highly brain-region specific.

In contrast to depression, animal models of mania as well as bipolar disorder have proved much more elusive. Mice with loss-of-function mutations in the *Clock* or *GluR6* glutamate receptor genes or transgenic mice that overexpress glycogen synthase kinase 3 β (GSK3 β) show manic-like behavioral abnormalities, although the relevance of these observations to human mania remains unknown.

The observation that tricyclic antidepressants (e.g., imipramine) inhibit serotonin and/or norepinephrine reuptake, and that monoamine oxidase inhibitors (e.g., tranylcypromine) are effective antidepressants, initially led to the view that depression is caused by a deficiency of these monoamines. However, this hypothesis has not been well substantiated, although variants in the serotonin transporter, and in the *COMT* gene, have been associated with altered mood states in some individuals. Nevertheless, these medications, particularly the tricyclics, have formed the basis of antidepressant discovery efforts, with virtually all of today's marketed antidepressants being SSRIs (e.g., fluoxetine, sertraline, citalopram), serotonin, and norepinephrine reuptake inhibitors (SNRIs) (e.g., venlafaxine, duloxetine), or norepinephrine reuptake inhibitors (NRIs) (e.g., atomoxetine).

A cardinal feature of all antidepressant medications is that long-term administration is needed for their mood-elevating effects. This means that their short-term actions, namely promotion of serotonin or norepinephrine function, is not per se antidepressant but rather induces a cascade of adaptations in the brain that underlie their clinical effects. The nature of these therapeutic drug-induced adaptations has not been identified with certainty. Presumably, the rich innervation of the brain's limbic circuitry by serotonin and norepinephrine (Fig. 53-1) provide the anatomic basis of their therapeutic actions.

Lithium is a highly effective drug for bipolar disorder, and competes with magnesium to inhibit magnesium-dependent enzymes, including GSK3 β and several enzymes involved in phosphoinositide signaling leading to activation of protein kinase C. These findings have led to discovery programs focused on developing GSK3 and PKC inhibitors as potential novel treatments for mood disorders. Another commonly prescribed drug for bipolar disorder is valproic acid, which has pleiotropic effects, including inhibition of histone deacetylases (HDACs). Histone acetylation promotes transcriptional activation through posttranslational modification of N-terminal lysine residues in histones and thereby causes chromatin decondensation. HDAC inhibitors have shown some antidepressant effects in animal models of depression. Another form of epigenetic control of

gene expression is methylation of cytosine residues in DNA, which inhibits gene transcription. DNA methylation has been shown to be important for inherited maternal effects on emotional behavior. Thus, rats born to mothers that exhibit low levels of nurturing behavior show increased anxiety and reduced expression of hippocampal glucocorticoid receptors due to increased methylation of the receptor gene. They pass these traits on to their offspring, but cross-fostering by mothers that display high levels of nurturing reverses them. As research into epigenetic mechanisms progresses, there is hope that it may become possible to identify specific depression-associated alterations in human chromatin.

SUBSTANCE USE DISORDERS

The DSM-IVTR uses the terms *substance dependence* and *substance abuse* to describe substance use disorders. It is unfortunate that the term *substance dependence* instead of addiction is used, because dependence can develop without addiction, and addiction involves much more than dependence per se. *Physical dependence* develops through resetting of homeostatic cellular mechanisms to permit normal function despite the continued presence of a drug; when drug intake is terminated abruptly, a withdrawal syndrome emerges. Withdrawal from alcohol or other sedative-hypnotics causes nervous system hyperactivity, whereas withdrawal from psychostimulants produces fatigue and sedation. *Tolerance* is a reduction in response to a drug, which like dependence, develops after repeated use. It results from a change in drug metabolism (pharmacokinetic tolerance) or cell signaling (pharmacodynamic tolerance). It is important to recognize that many nonaddictive medications induce tolerance and physical dependence, including β -adrenergic antagonists (e.g., propranolol) and α_2 -adrenergic agonists (e.g., clonidine).

What sets drugs of abuse apart is their unique ability to produce *euphoria*, a positive emotional state characterized by intensely pleasant feelings that are rewarding and *reinforcing* since they motivate users to take the drug repeatedly. Tolerance develops to the rewarding properties of most abused drugs during periods of heavy use, which promotes the use of higher drug doses. In addition, *psychological (or motivational) dependence* develops through the resetting of cellular mechanisms within reward-related regions of the brain and leads to negative emotional symptoms resembling depression during drug withdrawal. Addictive drugs can also cause *sensitization*, an increased drug effect upon repeated use, as exemplified by the paranoid psychosis induced by chronic use of cocaine or other psychostimulants (e.g., amphetamine). Addiction, therefore, results from drug-induced changes

in reward-related regions of the brain that lead to a complex mixture of tolerance, sensitization, and motivational dependence, in addition to powerful conditioning effects of these drugs mediated by the brain's memory circuits.

EPIDEMIOLOGY

Substance use disorders, especially those involving alcohol and tobacco, are very prevalent. The World Health Organization (WHO) estimates that more than 76 million people worldwide have alcohol use disorders and ~1.3 billion people smoke tobacco products (~1 billion men, 250 million women). The most widely used illicit drug in the United States is marijuana, with ~17% of 18–25-year-olds reporting regular use. Estimates of the annual economic burden of substance use disorders in the United States, including health- and crime-related costs and losses in productivity, exceed \$500 billion.

NEUROPATHOLOGY AND NEUROIMAGING

Imaging studies in humans demonstrate that addictive drugs, as well as craving for them, activate the brain's reward circuitry (discussed later). However, there is no established pathology associated with addiction risk. Patients who abuse alcohol or psychostimulants show reduced gray matter in the prefrontal cortex. Functional MRI or PET studies show reduced activity in anterior cingulate and orbitofrontal cortex during tasks of attention and inhibitory control. Damage to these cortical areas may contribute to addiction by impairing decision making and increasing impulsivity.

GENETICS

Substance use disorders are highly heritable, with genetic risk estimated to be 0.4 to 0.7; however, the specific genes that comprise this risk remain largely unknown. The best-established genetic contribution to addiction is the protective effect that mutations in alcohol-metabolizing enzymes have on risk for alcoholism. Mutations that increase alcohol dehydrogenase (ADH) activity and decrease aldehyde dehydrogenase (ALDH) activity are additive and promote accumulation of acetaldehyde following ingestion of alcohol. This produces intoxication at low doses and a flushing reaction that is unpleasant, resembling the reaction to disulfiram, a drug used to prevent relapse. These variants are common among people of East Asian descent, and individuals expressing these variants rarely abuse alcohol.

Genes that promote risk for addiction have begun to emerge from large family and population studies, but all genes identified to date represent only a very small fraction of the overall genetic risk for addiction. The best

established susceptibility loci are regions on chromosomes 4 and 5 containing GABA_A receptor gene clusters linked to alcohol use disorders and the nicotinic acetylcholine receptor gene cluster on chromosome 15 associated with nicotine and alcohol dependence. There are reports of numerous other addiction susceptibility genes (e.g., variants in COMT, the μ -opioid receptor, and the serotonin transporter), but further work is needed to validate these findings. In addition, several genes have been implicated in impulsivity, which is strongly associated with substance abuse. These include variants in genes for the D₄ dopamine receptor, the dopamine transporter, monoamine oxidase A, COMT, and the 5-HT_{1B} serotonin receptor.

PATHOGENESIS

Work in rodents and nonhuman primates has established the brain's reward regions as key neural substrates for the acute actions of drugs of abuse and for addiction induced by repeated drug administration (Fig. 53-1). Midbrain dopamine neurons in the ventral tegmental area (VTA) function normally as rheostats of reward: They are activated by natural rewards (food, sex, social interaction) or even by the expectation of such rewards, and many are suppressed by the absence of an expected reward or by aversive stimuli. These neurons thereby transmit crucial survival signals to the rest of the limbic brain to promote reward-related behavior, including motor responses to seek and obtain the rewards (nucleus accumbens), memories of reward-related cues (amygdala, hippocampus), and executive control of obtaining rewards (prefrontal cortex).

Drugs of abuse alter neurotransmission through initial actions at different classes of ion channels, neurotransmitter receptors, or neurotransmitter transporters (Table 53-2). Although the initial targets differ, the actions of these drugs converge on the brain's reward circuitry by promoting dopamine neurotransmission in the nucleus accumbens and other limbic targets of the VTA. In addition, some drugs promote activation of opioid and cannabinoid receptors, which modulate this reward circuitry. By these mechanisms, drugs of abuse produce powerful rewarding signals, which, after repeated drug administration, corrupt the brain's reward circuitry in ways that promote addiction. Three major pathologic adaptations have been described. First, drugs produce tolerance and dependence in reward circuits, which promote escalating drug intake and a negative emotional state during drug withdrawal that promotes relapse. Second, sensitization to the rewarding effects of the drugs and associated cues is seen during prolonged abstinence and also triggers relapse. Third, executive function is impaired in such a way as to increase impulsivity and compulsivity, both of which promote relapse.

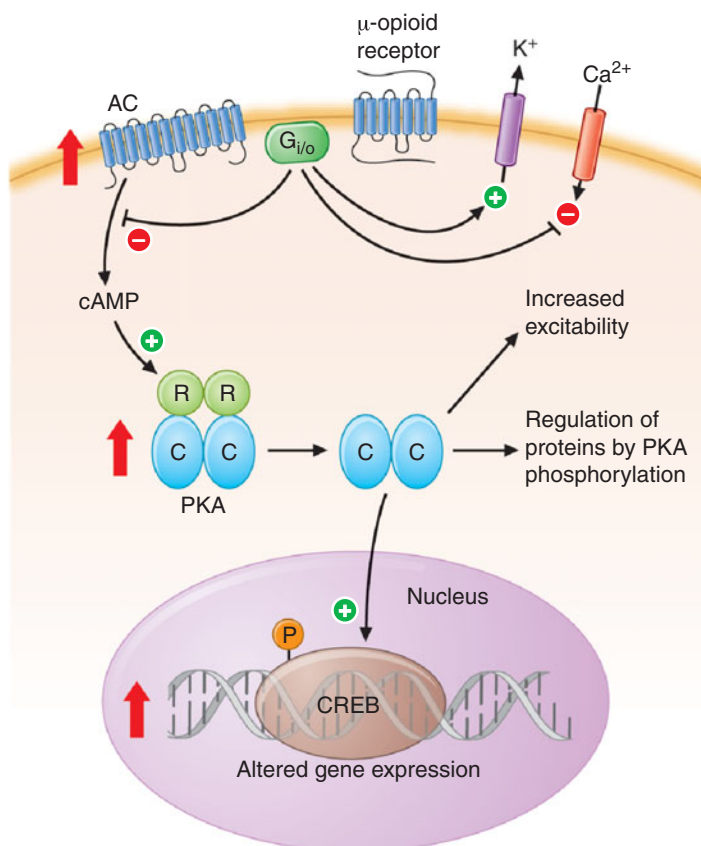
TABLE 53-2

INITIAL ACTIONS OF DRUGS OF ABUSE		
DRUG	NEUROTRANSMITTER AFFECTED	DRUG TARGET (ACTION)
Opiates	Endorphins, enkephalins	μ- and δ-opioid receptors (agonist)
Psychostimulants (cocaine, amphetamine methamphetamine)	Dopamine	Dopamine transporter (antagonist—cocaine; reverse transport—amphetamine, methamphetamine)
Nicotine	Acetylcholine	Nicotinic cholinergic receptors (agonist)
Ethanol	GABA Glutamate Acetylcholine Serotonin —	GABA _A receptors (positive allosteric modulator) NMDA glutamate receptors (antagonist) Nicotinic cholinergic receptors (allosteric modulator) 5HT-3 receptor (positive allosteric modulator) Calcium-activated K ⁺ channel (activator)
Marijuana	Endocannabinoids (anandamide, 2-arachidonoylglycerol)	CB ₁ receptor (agonist)
Phencyclidine	Glutamate	NMDA glutamate receptor (antagonist)

Repeated intake of abused drugs induces specific changes in cellular signal transduction, synaptic strength (long-term potentiation or depression), and neuronal structure (altered dendritic branching or cell soma size) within the brain’s reward circuitry. These modifications are mediated in part by changes in gene expression, achieved by drug regulation of transcription factors (e.g., CREB [cAMP response element binding protein] and ΔFosB) and their target genes. Together, these drug-induced adaptations underlie alterations in numerous neurotransmitter systems (e.g., glutamate, GABA, dopamine), growth factors (e.g., BDNF), neuropeptides (e.g. corticotrophin releasing factor), and intracellular signaling cascades. These adaptations provide opportunities for developing treatments targeted to drug-addicted individuals. The fact that the spectrum of these adaptations partly differ depending on the particular addictive substance used creates opportunities for treatments that are specific for different classes of addictive drugs and that may, therefore, be less likely to disturb basic mechanisms that govern motivation and reward.

Increasingly, causal relationships are being established between individual molecular-cellular adaptations

and specific behavioral abnormalities that characterize the addicted state. For example, acute activation of μ-opioid receptors by morphine or other opiates activates G_{i/o} proteins leading to inhibition of adenylyl cyclase, resulting in reduced cAMP production, protein kinase A (PKA) activation, and activation of the transcription factor CREB. Repeated administration of these drugs (Fig. 53-2) evokes a homeostatic response resulting in upregulation of adenylyl cyclases, increased production of cAMP, and increased activation of PKA and CREB. Such upregulation of cAMP signaling has been identified in the locus coeruleus, periaqueductal gray, VTA, and nucleus accumbens and contributes to opiate craving and signs of opiate withdrawal. The fact that endogenous opioid peptides do not produce tolerance and dependence while morphine and heroin do may relate to the recent observation that, unlike endogenous opioids, morphine and heroin are weak inducers of μ-opioid receptor desensitization and endocytosis. Therefore, these drugs cause prolonged receptor activation and inhibition of adenylyl cyclases, which provides a powerful stimulus for the upregulation of cAMP signaling that characterizes the opiate-dependent state.

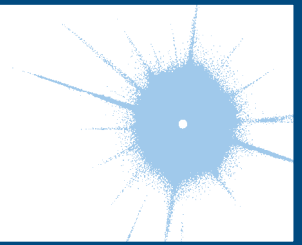
**FIGURE 53-2**

Opiate action in the locus coeruleus (LC). Binding of opiate agonists to μ -opioid receptors catalyzes nucleotide exchange on G_i and G_o proteins, leading to inhibition of adenylyl cyclase, neuronal hyperpolarization via activation of K^+ channels, and inhibition of neurotransmitter release via inhibition of Ca^{2+} channels. Activation of $G_{i/o}$ also inhibits adenylyl cyclase (AC), reducing protein kinase A (PKA) activity and phosphorylation of several PKA substrate proteins, thereby altering their function. For example, opiates reduce phosphorylation of the cAMP response element-binding protein (CREB), which appears to initiate long-term

changes in neuronal function. Chronic administration of opiates increases levels of AC isoforms, PKA catalytic (C) and regulatory (R) subunits, and the phosphorylation of several proteins, including CREB (indicated by red arrows). These changes contribute to the altered phenotype of the drug-addicted state. For example, the excitability of LC neurons is increased by enhanced cAMP signaling, although the ionic basis of this effect remains unknown. Activation of CREB causes upregulation of AC isoforms and tyrosine hydroxylase, the rate-limiting enzyme in catecholamine biosynthesis.

CHAPTER 54

MENTAL DISORDERS



Victor I. Reus

Mental disorders are common in medical practice and may present either as a primary disorder or as a comorbid condition. The prevalence of mental or substance use disorders in the United States is approximately 30%, only one-third of whom are currently receiving treatment. Global burden of disease statistics indicate that 4 of the 10 most important causes of disease worldwide are psychiatric in origin.

The revised fourth edition for use by primary care physicians of the *Diagnostic and Statistical Manual (DSM-IV-PC)* provides a useful synopsis of mental disorders most likely to be seen in primary care practice. The current system of classification is multiaxial and includes the presence or absence of a major mental disorder (axis I), any underlying personality disorder (axis II), general medical condition (axis III), psychosocial and environmental problems (axis IV), and overall rating of general psychosocial functioning (axis V).

Changes in health care delivery underscore the need for primary care physicians to assume responsibility for the initial diagnosis and treatment of the most common mental disorders. Prompt diagnosis is essential to ensure that patients have access to appropriate medical services and to maximize the clinical outcome. Validated patient-based questionnaires have been developed that systematically probe for signs and symptoms associated with the most prevalent psychiatric diagnoses and guide the clinician into targeted assessment. Prime MD (and a self-report form, the PHQ) and the Symptom-Driven Diagnostic System for Primary Care (SDDS-PC) are inventories that require only 10 min to complete and link patient responses to the formal diagnostic criteria of anxiety, mood, somatoform, and eating disorders and to alcohol abuse or dependence.

A physician who refers patients to a psychiatrist should know not only when doing so is appropriate but also how to refer, since societal misconceptions and the stigma of mental illness impede the process. Primary

care physicians should base referrals to a psychiatrist on the presence of signs and symptoms of a mental disorder and not simply on the absence of a physical explanation for a patient's complaint. The physician should discuss with the patient the reasons for requesting the referral or consultation and provide reassurance that he or she will continue to provide medical care and work collaboratively with the mental health professional. Consultation with a psychiatrist or transfer of care is appropriate when physicians encounter evidence of psychotic symptoms, mania, severe depression, or anxiety; symptoms of post-traumatic stress disorder (PTSD); suicidal or homicidal preoccupation; or a failure to respond to first-order treatment. The biology of psychiatric and addictive disorders is discussed in Chap. 53.

ANXIETY DISORDERS

Anxiety disorders, the most prevalent psychiatric illnesses in the general community, are present in 15–20% of medical clinic patients. Anxiety, defined as a subjective sense of unease, dread, or foreboding, can indicate a primary psychiatric condition or can be a component of, or reaction to, a primary medical disease. The primary anxiety disorders are classified according to their duration and course and the existence and nature of precipitants.

When evaluating the anxious patient, the clinician must first determine whether the anxiety antedates or postdates a medical illness or is due to a medication side effect. Approximately one-third of patients presenting with anxiety have a medical etiology for their psychiatric symptoms, but an anxiety disorder can also present with somatic symptoms in the absence of a diagnosable medical condition.

PANIC DISORDER

Clinical manifestations

Panic disorder is defined by the presence of recurrent and unpredictable panic attacks, which are distinct episodes of intense fear and discomfort associated with a variety of physical symptoms, including palpitations, sweating, trembling, shortness of breath, chest pain, dizziness, and a fear of impending doom or death (Table 54-1). Paresthesias, gastrointestinal distress, and feelings of unreality are also common. Diagnostic criteria require at least 1 month of concern or worry about the attacks or a change in behavior related to them. The lifetime prevalence of panic disorder is 1–3%. Panic attacks have a sudden onset, developing within 10 min and usually resolving over the course of an hour, and they occur in an unexpected fashion. The frequency and severity of panic attacks vary, ranging from once a week to clusters of attacks separated by months of well-being. The first attack is usually outside the home, and onset is typically in late adolescence to early adulthood. In some individuals, anticipatory anxiety develops over time and results in a generalized fear and a progressive avoidance of places or situations in which a panic attack might recur. *Agoraphobia*, which occurs commonly in patients with panic disorder, is an acquired irrational fear of being in places where one might feel trapped or unable to escape (Table 54-2). Typically, it leads the patient into a progressive restriction in lifestyle and, in a literal sense, in geography. Frequently, patients are embarrassed that they are housebound and dependent on the company of others to go out into the world and do not volunteer this information; thus physicians

TABLE 54-1

DIAGNOSTIC CRITERIA FOR PANIC ATTACK

A discrete period of intense fear or discomfort, in which four or more of the following symptoms developed abruptly and reached a peak within 10 min:

1. Palpitations, pounding heart, or accelerated heart rate
2. Sweating
3. Trembling or shaking
4. Sensations of shortness of breath or smothering
5. Feeling of choking
6. Chest pain or discomfort
7. Nausea or abdominal distress
8. Feeling dizzy, unsteady, lightheaded, or faint
9. Derealization (feelings of unreality) or depersonalization (being detached from oneself)
10. Fear of losing control or going crazy
11. Fear of dying
12. Paresthesias (numbness or tingling sensations)
13. Chills or hot flushes

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TABLE 54-2

DIAGNOSTIC CRITERIA FOR AGORAPHOBIA

1. Anxiety about being in places or situations from which escape might be difficult (or embarrassing) or in which help may not be available in the event of having an unexpected or situationally predisposed panic attack or panic-like symptoms. Agoraphobic fears typically involve characteristic clusters of situations that include being outside the home alone; being in a crowd or standing in a line; being on a bridge; and traveling in a bus, train, or automobile.
2. The situations are avoided (e.g., travel is restricted) or else are endured with marked distress or with anxiety about having a panic attack or panic-like symptoms, or require the presence of a companion.
3. The anxiety or phobic avoidance is not better accounted for by another mental disorder such as social phobia (e.g., avoidance limited to social situations because of fear of embarrassment), specific phobia (e.g., avoidance limited to a single situation like elevators), obsessive-compulsive disorder (e.g., avoidance of dirt in someone with an obsession about contamination), posttraumatic stress disorder (e.g., avoidance of stimuli associated with a severe stressor), or separation anxiety disorder (e.g., avoidance of leaving home or relatives).

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will fail to recognize the syndrome if direct questioning is not pursued.

Differential diagnosis

A diagnosis of panic disorder is made after a medical etiology for the panic attacks has been ruled out. A variety of cardiovascular, respiratory, endocrine, and neurologic conditions can present with anxiety as the chief complaint. Patients with true panic disorder will often focus on one specific feature to the exclusion of others. For example, 20% of patients who present with syncope as a primary medical complaint have a primary diagnosis of a mood, anxiety, or substance-abuse disorder, the most common being panic disorder. The differential diagnosis of panic disorder is complicated by a high rate of comorbidity with other psychiatric conditions, especially alcohol and benzodiazepine abuse, which patients initially use in an attempt at self-medication. Some 75% of panic disorder patients will also satisfy criteria for major depression at some point in their illness.

When the history is nonspecific, physical examination and focused laboratory testing must be used to rule out anxiety states resulting from medical disorders such as pheochromocytoma, thyrotoxicosis, or hypoglycemia. Electrocardiogram (ECG) and echocardiogram may

detect some cardiovascular conditions associated with panic such as paroxysmal atrial tachycardia and mitral valve prolapse. In two studies, panic disorder was the primary diagnosis in 43% of patients with chest pain who had normal coronary angiograms and was present in 9% of all outpatients referred for cardiac evaluation. Panic disorder has also been diagnosed in many patients referred for pulmonary function testing or with symptoms of irritable bowel syndrome.

Etiology and pathophysiology

The etiology of panic disorder is unknown but appears to involve a genetic predisposition, altered autonomic responsivity, and social learning. Panic disorder shows familial aggregation; the disorder is concordant in 30–45% of monozygotic twins, and genomewide screens have identified suggestive risk loci. Acute panic attacks appear to be associated with increased noradrenergic discharges in the locus coeruleus. Intravenous infusion of sodium lactate evokes an attack in two-thirds of panic disorder patients, as do the α_2 -adrenergic antagonist yohimbine, cholecystokinin tetrapeptide (CCK-4), and carbon dioxide inhalation. It is hypothesized that each of these stimuli activates a pathway involving noradrenergic neurons in the locus coeruleus and serotonergic neurons in the dorsal raphe. Agents that block serotonin reuptake can prevent attacks. Panic-disorder patients have a heightened sensitivity to somatic symptoms, which triggers increasing arousal, setting off the panic attack; accordingly, therapeutic intervention involves altering the patient’s cognitive interpretation of anxiety-producing experiences as well as preventing the attack itself.

TREATMENT **Panic Disorder**

Achievable goals of treatment are to decrease the frequency of panic attacks and to reduce their intensity. The cornerstone of drug therapy is antidepressant medication (Tables 54-3 through 54-5). Selective serotonin reuptake inhibitors (SSRIs) benefit the majority of panic disorder patients and do not have the adverse effects of tricyclic antidepressants (TCAs). Fluoxetine, paroxetine, sertraline, and the selective serotonin-norepinephrine reuptake inhibitor (SNRI) venlafaxine have received approval from the U.S. Food and Drug Administration (FDA) for this indication. These drugs should be started at one-third to one-half of their usual antidepressant dose (e.g., 5–10 mg fluoxetine, 25–50 mg sertraline, 10 mg paroxetine, 37.5 mg venlafaxine). Monoamine oxidase inhibitors (MAOIs) are also effective and may specifically benefit patients who have comorbid features of atypical depression (i.e., hypersomnia and weight gain). Insomnia,

orthostatic hypotension, and the need to maintain a low-tyramine diet (avoidance of cheese and wine) have limited their use, however. Antidepressants typically take 2–6 weeks to become effective, and doses may need to be adjusted based upon the clinical response.

Because of anticipatory anxiety and the need for immediate relief of panic symptoms, benzodiazepines are useful early in the course of treatment and sporadically thereafter (Table 54-6). For example, alprazolam, starting at 0.5 mg qid and increasing to 4 mg/d in divided doses, is effective, but patients must be monitored closely, as some develop dependence and begin to escalate the dose of this medication. Clonazepam, at a final maintenance dose of 2–4 mg/d, is also helpful; its longer half-life permits twice-daily dosing, and patients appear less likely to develop dependence on this agent.

Early psychotherapeutic intervention and education aimed at symptom control enhances the effectiveness of drug treatment. Patients can be taught breathing techniques, be educated about physiologic changes that occur with panic, and learn to expose themselves voluntarily to precipitating events in a treatment program spanning 12–15 sessions. Homework assignments and monitored compliance are important components of successful treatment. Once patients have achieved a satisfactory response, drug treatment should be maintained for 1–2 years to prevent relapse. Controlled trials indicate a success rate of 75–85%, although the likelihood of complete remission is somewhat lower.

GENERALIZED ANXIETY DISORDER

Clinical manifestations

Patients with generalized anxiety disorder (GAD) have persistent, excessive, and/or unrealistic worry associated with muscle tension, impaired concentration, autonomic arousal, feeling “on edge” or restless, and insomnia (Table 54-7). Onset is usually before age 20 years, and a history of childhood fears and social inhibition may be present. The lifetime prevalence of GAD is 5–6%; the risk is higher in first-degree relatives of patients with the diagnosis. Interestingly, family studies indicate that GAD and panic disorder segregate independently. More than 80% of patients with GAD also suffer from major depression, dysthymia, or social phobia. Comorbid substance abuse is common in these patients, particularly alcohol and/or sedative/hypnotic abuse. Patients with GAD worry excessively over minor matters, with life-disrupting effects; unlike in panic disorder, complaints of shortness of breath, palpitations, and tachycardia are relatively rare.

TABLE 54-3

ANTIDEPRESSANTS			
NAME	USUAL DAILY DOSE, mg	SIDE EFFECTS	COMMENTS
SSRIs			
Fluoxetine (Prozac)	10–80	Headache; nausea and other GI effects; jitteriness; insomnia; sexual dysfunction; can affect plasma levels of other medicines (except sertraline); akathisia rare	Once-daily dosing, usually in the morning; fluoxetine has very long half-life; must not be combined with MAOIs
Sertraline (Zoloft)	50–200		
Paroxetine (Paxil)	20–60		
Fluvoxamine (Luvox)	100–300		
Citalopram (Celexa)	20–60		
Escitalopram (Lexapro)	10–30		
TCAs			
Amitriptyline (Elavil)	150–300	Anticholinergic (dry mouth, tachycardia, constipation, urinary retention, blurred vision); sweating; tremor; postural hypotension; cardiac conduction delay; sedation; weight gain	Once-daily dosing, usually qhs; blood levels of most TCAs available; can be lethal in O.D. (lethal dose = 2 g); nortriptyline best tolerated, especially by elderly
Nortriptyline (Pamelor)	50–200		
Imipramine (Tofranil)	150–300		
Desipramine (Norpramin)	150–300		
Doxepin (Sinequan)	150–300		
Clomipramine (Anafranil)	150–300		
Mixed Norepinephrine/Serotonin Reuptake Inhibitors and Receptor Blockers			
Venlafaxine (Effexor)	75–375	Nausea; dizziness; dry mouth; headaches; increased blood pressure; anxiety and insomnia	Bid-tid dosing (extended release available); lower potential for drug interactions than SSRIs; contraindicated with MAOIs
Desvenlafaxine, (Pristiq)	50–400	Nausea, dizziness, insomnia	Primary metabolite of venlafaxine. No increased efficacy with higher dosing
Duloxetine (Cymbalta)	40–60	Nausea, dizziness, headache, insomnia, constipation	May have utility in treatment of neuropathic pain and stress incontinence
Mirtazapine (Remeron)	15–45	Somnolence; weight gain; neutropenia rare	Once daily dosing
Mixed-Action Drugs			
Bupropion (Wellbutrin)	250–450	Jitteriness; flushing; seizures in at-risk patients; anorexia; tachycardia; psychosis	Tid dosing, but sustained release also available; fewer sexual side effects than SSRIs or TCAs; may be useful for adult ADD
Trazodone (Desyrel)	200–600	Sedation; dry mouth; ventricular irritability; postural hypotension; priapism rare	Useful in low doses for sleep because of sedating effects with no anticholinergic side effects
Nefazodone (Serzone)	300–600	Sedation; headache; dry mouth; nausea; constipation	Discontinued sale in United States and several other countries due to risk of liver failure
Amoxapine (Asendin)	200–600	Sexual dysfunction	Lethality in overdose; EPS possible
MAOIs			
Phenelzine (Nardil)	45–90	Insomnia; hypotension; anorgasmia; weight gain; hypertensive crisis; toxic reactions with SSRIs; narcotics	May be more effective in patients with atypical features or treatment-refractory depression
Tranylcypromine (Parnate)	20–50		
Isocarboxazid (Marplan)	20–60		
Transdermal selegiline (Emsam)	6–12	Local skin reaction, hypertension	No dietary restrictions with 6 mg dose

Abbreviations: ADD, attention deficit disorder; EPS, extrapyramidal symptoms; MAOIs, monoamine oxidase inhibitors; SSRIs, selective serotonin reuptake inhibitors; TCAs, tricyclic antidepressants.

TABLE 54-4
MANAGEMENT OF ANTIDEPRESSANT SIDE EFFECTS

SYMPTOMS	COMMENTS AND MANAGEMENT STRATEGIES
Gastrointestinal	
Nausea, loss of appetite	Usually short-lived and dose-related; consider temporary dose reduction or administration with food and antacids
Diarrhea	Famotidine, 20–40 mg/d
Constipation	Wait for tolerance; try diet change, stool softener, exercise; avoid laxatives
Sexual dysfunction	Consider dose reduction; drug holiday
Anorgasmia/impotence; impaired ejaculation	Bethanechol, 10–20 mg, 2 h before activity, or cyproheptadine, 4–8 mg 2 h before activity, or bupropion, 100 mg bid or amantadine, 100 mg bid/tid
Orthostasis	Tolerance unlikely; increase fluid intake, use calf exercises/support hose; fludrocortisone, 0.025 mg/d
Anticholinergic	Wait for tolerance
Dry mouth, eyes	Maintain good oral hygiene; use artificial tears, sugar-free gum
Tremor/jitteriness	Antiparkinsonian drugs not effective; use dose reduction/slow increase; lorazepam, 0.5 mg bid, or propranolol, 10–20 mg bid
Insomnia	Schedule all doses for the morning; trazodone, 50–100 mg qhs
Sedation	Caffeine; schedule all dosing for bedtime; bupropion, 75–100 mg in afternoon
Headache	Evaluate diet, stress, other drugs; try dose reduction; amitriptyline, 50 mg/d
Weight gain	Decrease carbohydrates; exercise; consider fluoxetine
Loss of therapeutic benefit over time	Related to tolerance? Increase dose or drug holiday; add amantadine, 100 mg bid, buspirone, 10 mg tid, or pindolol, 2.5 mg bid

Etiology and pathophysiology

All anxiogenic agents act on the γ -aminobutyric acid (GABA)_A receptor/chloride ion channel complex, implicating this neurotransmitter system in the pathogenesis of anxiety and panic attacks. Benzodiazepines are thought to bind two separate GABA_A receptor sites: type I, which has a broad neuroanatomic distribution, and type II, which is concentrated in the hippocampus, striatum, and neocortex. The antianxiety effects of the various

TABLE 54-5
POSSIBLE DRUG INTERACTIONS WITH SELECTIVE SEROTONIN REUPTAKE INHIBITORS

AGENT	EFFECT
Monoamine oxidase inhibitors	Serotonin syndrome—absolute contraindication
Serotonergic agonists, e.g., tryptophan, fenfluramine	Potential serotonin syndrome
Drugs that are metabolized by P450 isoenzymes: tricyclics, other SSRIs, antipsychotics, beta blockers, codeine, triazolobenzodiazepines, calcium channel blockers	Delayed metabolism resulting in increased blood levels and potential toxicity
Drugs that are bound tightly to plasma proteins, e.g., warfarin	Increased bleeding secondary to displacement
Drugs that inhibit the metabolism of SSRIs by P450 isoenzymes, e.g., quinidine	Increased SSRI side effects

Abbreviation: SSRIs, selective serotonin reuptake inhibitors.

benzodiazepines and side effects such as sedation and memory impairment are influenced by their relative binding to type I and type II receptor sites. Serotonin [5-hydroxytryptamine (5HT)] and 3 α -reduced neuroactive steroids (allosteric modulators of GABA_A) also appear to have a role in anxiety, and buspirone, a partial 5HT_{1A} receptor agonist, and certain 5HT_{2A} and 5HT_{2C} receptor antagonists (e.g., nefazodone) may have beneficial effects.

TREATMENT Generalized Anxiety Disorder

A combination of pharmacologic and psychotherapeutic interventions is most effective in GAD, but complete symptomatic relief is rare. A short course of a benzodiazepine is usually indicated, preferably lorazepam, oxazepam, or temazepam. (The first two of these agents are metabolized via conjugation rather than oxidation and thus do not accumulate if hepatic function is impaired.) Treatment should be initiated at the lowest dose possible and prescribed on an as-needed basis as symptoms warrant. Benzodiazepines differ in their milligram per kilogram potency, half-life, lipid solubility, metabolic pathways, and presence of active metabolites. Agents that are absorbed rapidly and are lipid soluble, such as diazepam, have a rapid onset of action and a higher abuse potential. Benzodiazepines should generally not be prescribed for >4–6 weeks because of

TABLE 54-6

ANXIOLYTICS

NAME	EQUIVALENT PO DOSE, mg	ONSET OF ACTION	HALF-LIFE, h	COMMENTS
Benzodiazepines				
Diazepam (Valium)	5	Fast	20–70	Active metabolites; quite sedating
Flurazepam (Dalmane)	15	Fast	30–100	Flurazepam is a prodrug; metabolites are active; quite sedating
Triazolam (Halcion)	0.25	Intermediate	1.5–5	No active metabolites; can induce confusion and delirium, especially in elderly
Lorazepam (Ativan)	1	Intermediate	10–20	No active metabolites; direct hepatic glucuronide conjugation; quite sedating
Alprazolam (Xanax)	0.5	Intermediate	12–15	Active metabolites; not too sedating; may have specific antidepressant and antipanic activity; tolerance and dependence develop easily
Chlordiazepoxide (Librium)	10	Intermediate	5–30	Active metabolites; moderately sedating
Oxazepam (Serax)	15	Slow	5–15	No active metabolites; direct glucuronide conjugation; not too sedating
Temazepam (Restoril)	15	Slow	9–12	No active metabolites; moderately sedating
Clonazepam (Klonopin)	0.5	Slow	18–50	No active metabolites; moderately sedating
Nonbenzodiazepines				
Buspirone (BuSpar)	7.5	2 weeks	2–3	Active metabolites; tid dosing—usual daily dose 10–20 mg tid; nonsedating; no additive effects with alcohol; useful for controlling agitation in demented or brain-injured patients

TABLE 54-7

DIAGNOSTIC CRITERIA FOR GENERALIZED ANXIETY DISORDER

- A. Excessive anxiety and worry (apprehensive expectation), occurring more days than not for at least 6 months, about a number of events or activities (such as work or school performance).
- B. The person finds it difficult to control the worry.
- C. The anxiety and worry are associated with three (or more) of the following six symptoms (with at least some symptoms present for more days than not for the past 6 months): (1) restlessness or feeling keyed up or on edge; (2) being easily fatigued; (3) difficulty concentrating or mind going blank; (4) irritability; (5) muscle tension; (6) sleep disturbance (difficulty falling or staying asleep, or restless unsatisfying sleep).
- D. The focus of the anxiety and worry is not confined to features of an axis I disorder [e.g., the anxiety or worry is not about having a panic attack (as in panic disorder), being embarrassed in public (as in social phobia), being contaminated (as in obsessive-compulsive disorder), being away from home or close relatives (as in separation anxiety disorder), gaining weight (as in anorexia nervosa), having multiple physical complaints (as in somatization disorder), or having a serious illness (as in hypochondriasis)], and the anxiety and worry do not occur exclusively during posttraumatic stress disorder.
- E. The anxiety, worry, or physical symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- F. The disturbance is not due to the direct physiologic effects of a substance (e.g., a drug of abuse, a medication) or a general medical condition (e.g., hyperthyroidism) and does not occur exclusively during a mood disorder, a psychotic disorder, or a pervasive developmental disorder.

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the development of tolerance and the risk of abuse and dependence. Withdrawal must be closely monitored as relapses can occur. It is important to warn patients that concomitant use of alcohol or other sedating drugs may be neurotoxic and impair their ability to function. An optimistic approach that encourages the patient to clarify environmental precipitants, anticipate his or her reactions, and plan effective response strategies is an essential element of therapy.

Adverse effects of benzodiazepines generally parallel their relative half-lives. Longer-acting agents, such as diazepam, chlorthalidopoxide, flurazepam, and clonazepam, tend to accumulate active metabolites, with resultant sedation, impairment of cognition, and poor psychomotor performance. Shorter-acting compounds, such as alprazolam and oxazepam, can produce daytime anxiety, early morning insomnia, and, with discontinuation, rebound anxiety and insomnia. Although patients develop tolerance to the sedative effects of benzodiazepines, they are less likely to habituate to the adverse psychomotor effects. Withdrawal from the longer half-life benzodiazepines can be accomplished through gradual, step-wise dose reduction (by 10% every 1–2 weeks) over 6–12 weeks. It is usually more difficult to taper patients off shorter-acting benzodiazepines. Physicians may need to switch the patient to a benzodiazepine with a longer half-life or use an adjunctive medication such as a beta blocker or carbamazepine, before attempting to discontinue the benzodiazepine. Withdrawal reactions vary in severity and duration; they can include depression, anxiety, lethargy, diaphoresis, autonomic arousal, and, rarely, seizures.

Buspirone is a nonbenzodiazepine anxiolytic agent. It is nonsedating, does not produce tolerance or dependence, does not interact with benzodiazepine receptors or alcohol, and has no abuse or disinhibition potential. However, it requires several weeks to take effect and requires thrice-daily dosing. Patients who were previously responsive to a benzodiazepine are unlikely to rate buspirone as equally effective, but patients with head injury or dementia who have symptoms of anxiety and/or agitation may do well with this agent. Escitalopram, paroxetine, and venlafaxine are FDA approved for the treatment of GAD, usually at doses that are comparable to their efficacy in major depression. Benzodiazepines are contraindicated during pregnancy and breast-feeding.

Anticonvulsants with GABAergic properties may also be effective against anxiety. Gabapentin, oxcarbazepine, tiagabine, pregabalin, and divalproex have all shown some degree of benefit in a variety of anxiety-related syndromes. Agents that selectively target GABA_A receptor subtypes are currently under development, and it is hoped that these will lack the sedating, memory-impairing, and addicting properties of benzodiazepines.

PHOBIC DISORDERS

Clinical manifestations

The cardinal feature of phobic disorders is a marked and persistent fear of objects or situations, exposure to which results in an immediate anxiety reaction. The patient avoids the phobic stimulus, and this avoidance usually impairs occupational or social functioning. Panic attacks may be triggered by the phobic stimulus or may occur spontaneously. Unlike patients with other anxiety disorders, individuals with phobias usually experience anxiety only in specific situations. Common phobias include fear of closed spaces (claustrophobia), fear of blood, and fear of flying. Social phobia is distinguished by a specific fear of social or performance situations in which the individual is exposed to unfamiliar individuals or to possible examination and evaluation by others. Examples include having to converse at a party, use public restrooms, and meet strangers. In each case, the affected individual is aware that the experienced fear is excessive and unreasonable given the circumstance. The specific content of a phobia may vary across gender, ethnic, and cultural boundaries.

Phobic disorders are common, affecting ~10% of the population. Full criteria for diagnosis are usually satisfied first in early adulthood, but behavioral avoidance of unfamiliar people, situations, or objects dating from early childhood is common.

In one study of female twins, concordance rates for agoraphobia, social phobia, and animal phobia were found to be 23% for monozygotic twins and 15% for dizygotic twins. A twin study of fear conditioning, a model for the acquisition of phobias, demonstrated a heritability of 35–45%, and a genomewide linkage scan identified a risk locus on chromosome 14 in a region previously implicated in a mouse model of fear. Animal studies of fear conditioning have indicated that processing of the fear stimulus occurs through the lateral nucleus of the amygdala, extending through the central nucleus and projecting to the periaqueductal gray region, lateral hypothalamus, and paraventricular hypothalamus.

TREATMENT Phobic Disorders

Beta blockers (e.g., propranolol, 20–40 mg orally 2 h before the event) are particularly effective in the treatment of “performance anxiety” (but not general social phobia) and appear to work by blocking the peripheral manifestations of anxiety such as perspiration, tachycardia, palpitations, and tremor. MAOIs alleviate social phobia independently of their antidepressant activity, and paroxetine, sertraline, and venlafaxine have received FDA approval for treatment of social anxiety. Benzodiazepines

can be helpful in reducing fearful avoidance, but the chronic nature of phobic disorders limits their usefulness.

Behaviorally focused psychotherapy is an important component of treatment, as relapse rates are high when medication is used as the sole treatment. Cognitive-behavioral strategies are based upon the finding that distorted perceptions and interpretations of fear-producing stimuli play a major role in perpetuation of phobias. Individual and group therapy sessions teach the patient to identify specific negative thoughts associated with the anxiety-producing situation and help to reduce the patient's fear of loss of control. In desensitization therapy, hierarchies of feared situations are constructed and the patient is encouraged to pursue and master gradual exposure to the anxiety-producing stimuli.

Patients with social phobia, in particular, have a high rate of comorbid alcohol abuse, as well as of other psychiatric conditions (e.g., eating disorders), necessitating the need for parallel management of each disorder if anxiety reduction is to be achieved.

STRESS DISORDERS

Clinical manifestations

Patients may develop anxiety after exposure to extreme traumatic events such as the threat of personal death or injury or the death of a loved one. The reaction may occur shortly after the trauma (*acute stress disorder*) or be delayed and subject to recurrence (PTSD) (Table 54-8). In both syndromes, individuals experience associated symptoms of detachment and loss of emotional responsiveness. The patient may feel depersonalized and unable to recall specific aspects of the trauma, though typically it is reexperienced through intrusions in thought, dreams, or flashbacks, particularly when cues of the original event are present. Patients often actively avoid stimuli that precipitate recollections of the trauma and demonstrate a resulting increase in vigilance, arousal, and startle response. Patients with stress disorders are at risk for the development of other disorders related to anxiety, mood, and substance abuse (especially alcohol). Between 5 and 10% of Americans will at some time in their life satisfy criteria for PTSD, with women more likely to be affected than men.

Risk factors for the development of PTSD include a past psychiatric history and personality characteristics of high neuroticism and extroversion. Twin studies show a substantial genetic influence on all symptoms associated with PTSD, with less evidence for an environmental effect.

Etiology and pathophysiology

It is hypothesized that in PTSD there is excessive release of norepinephrine from the locus coeruleus in

TABLE 54-8

DIAGNOSTIC CRITERIA FOR POSTTRAUMATIC STRESS DISORDER

- A. The person has been exposed to a traumatic event in which both of the following were present:
 1. The person experienced, witnessed, or was confronted with an event or events that involved actual or threatened death or serious injury, or a threat to the physical integrity of self or others.
 2. The person's response involved intense fear, helplessness, or horror.
- B. The traumatic event is persistently reexperienced in one (or more) of the following ways:
 1. Recurrent and intrusive distressing recollections of the event, including images, thoughts, or perceptions.
 2. Recurrent distressing dreams of the event.
 3. Acting or feeling as if the traumatic event were recurring (includes a sense of reliving the experience, illusions, hallucinations, and dissociative flashback episodes, including those that occur on awakening or when intoxicated).
 4. Intense psychological distress at exposure to internal or external cues that symbolize or resemble an aspect of the traumatic event.
 5. Physiologic reactivity on exposure to internal or external cues that symbolize or resemble an aspect of the traumatic event.
- C. Persistent avoidance of stimuli associated with the trauma and numbing of general responsiveness (not present before the trauma), as indicated by three or more of the following:
 1. Efforts to avoid thoughts, feelings, or conversations associated with the trauma
 2. Efforts to avoid activities, places, or people that arouse recollections of the trauma
 3. Inability to recall an important aspect of the trauma
 4. Markedly diminished interest or participation in significant activities
 5. Feeling of detachment or estrangement from others
 6. Restricted range of affect (e.g., unable to have loving feelings)
 7. Sense of a foreshortened future (e.g., does not expect to have a career, marriage, children, or a normal life span)
- D. Persistent symptoms of increased arousal (not present before the trauma), as indicated by two (or more) of the following:
 1. Difficulty falling or staying asleep
 2. Irritability or outbursts of anger
 3. Difficulty concentrating
 4. Hypervigilance
 5. Exaggerated startle response
- E. Duration of the disturbance (symptoms in criteria B, C, and D) is more than 1 month
- F. The disturbance causes clinically significant distress or impairment in social, occupational, or other important areas of functioning.

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response to stress and increased noradrenergic activity at projection sites in the hippocampus and amygdala. These changes theoretically facilitate the encoding of fear-based memories. Greater sympathetic responses to cues associated with the traumatic event occur in PTSD, although pituitary adrenal responses are blunted.

TREATMENT

Stress Disorders

Acute stress reactions are usually self-limited, and treatment typically involves the short-term use of benzodiazepines and supportive/expressive psychotherapy. The chronic and recurrent nature of PTSD, however, requires a more complex approach employing drug and behavioral treatments. PTSD is highly correlated with peritraumatic dissociative symptoms and the development of an acute stress disorder at the time of the trauma. TCAs such as imipramine and amitriptyline, the MAOI phenelzine, and the SSRIs can all reduce anxiety, symptoms of intrusion, and avoidance behaviors, as can prazosin, an α_1 antagonist. Propranolol and opiates such as morphine, given during the acute stress period may have beneficial effects in preventing the development of PTSD. Trazodone, a sedating antidepressant, is frequently used at night to help with insomnia (50–150 mg qhs). Carbamazepine, valproic acid, or alprazolam have also independently produced improvement in uncontrolled trials. Psychotherapeutic strategies for PTSD help the patient overcome avoidance behaviors and demoralization and master fear of recurrence of the trauma; therapies that encourage the patient to dismantle avoidance behaviors through stepwise focusing on the experience of the traumatic event are the most effective.

OBSESSIVE-COMPULSIVE DISORDER

Clinical manifestations

Obsessive-compulsive disorder (OCD) is characterized by obsessive thoughts and compulsive behaviors that impair everyday functioning. Fears of contamination and germs are common, as are handwashing, counting behaviors, and having to check and recheck such actions as whether a door is locked. The degree to which the disorder is disruptive for the individual varies, but in all cases obsessive-compulsive activities take up >1 h per day and are undertaken to relieve the anxiety triggered by the core fear. Patients often conceal their symptoms, usually because they are embarrassed by the content of their thoughts or the nature of their actions. Physicians must ask specific questions regarding recurrent thoughts and behaviors, particularly if physical clues such as chafed and reddened hands or patchy hair loss (from repetitive hair pulling, or trichotillomania) are present.

Comorbid conditions are common, the most frequent being depression, other anxiety disorders, eating disorders, and tics. OCD has a lifetime prevalence of 2–3% worldwide. Onset is usually gradual, beginning in early adulthood, but childhood onset is not rare. The disorder usually has a waxing and waning course, but some cases may show a steady deterioration in psychosocial functioning.

Etiology and pathophysiology

A genetic contribution to OCD is suggested by twin studies. A genomewide association study (GWAS) reported linkage to chromosome 2p23.2; however, no susceptibility gene for OCD has been identified to date. Family studies show an aggregation of OCD with Tourette’s disorder, and both are more common in males and in first-born children.

The anatomy of obsessive-compulsive behavior is thought to include the orbital frontal cortex, caudate nucleus, and globus pallidus. The caudate nucleus appears to be involved in the acquisition and maintenance of habit and skill learning, and interventions that are successful in reducing obsessive-compulsive behaviors also decrease metabolic activity measured in the caudate.

TREATMENT

Obsessive-Compulsive Disorder

Clomipramine, fluoxetine, fluvoxamine, and sertraline are approved for the treatment of OCD. Clomipramine is a TCA that is often tolerated poorly owing to anticholinergic and sedative side effects at the doses required to treat the illness (25–250 mg/d); its efficacy in OCD is unrelated to its antidepressant activity. Fluoxetine (5–60 mg/d), fluvoxamine (25–300 mg/d), and sertraline (50–150 mg/d) are as effective as clomipramine and have a more benign side effect profile. Only 50–60% of patients with OCD show adequate improvement with pharmacotherapy alone. In treatment-resistant cases, augmentation with other serotonergic agents such as buspirone, or with a neuroleptic or benzodiazepine may be beneficial and in severe cases deep brain stimulation has been found to be effective. When a therapeutic response is achieved, long-duration maintenance therapy is usually indicated.

For many individuals, particularly those with time-consuming compulsions, behavior therapy will result in as much improvement as that afforded by medication. Effective techniques include the gradual increase in exposure to stressful situations, maintenance of a diary to clarify stressors, and homework assignments that substitute new activities for compulsive behaviors.

MOOD DISORDERS

Mood disorders are characterized by a disturbance in the regulation of mood, behavior, and affect. Mood disorders are subdivided into (1) depressive disorders, (2) bipolar disorders, and (3) depression in association with medical illness or alcohol and substance abuse (Chaps. 56 through 58). Major depressive disorder (MDD) is differentiated from bipolar disorder by the absence of a manic or hypomanic episode. The relationship between pure depressive syndromes and bipolar disorders is not well understood; MDD is more frequent in families of bipolar individuals, but the reverse is not true. In the Global Burden of Disease Study conducted by the World Health Organization, unipolar major depression ranked fourth among all diseases in terms of disability-adjusted life-years and was projected to rank second by the year 2020. In the United States, lost productivity directly related to mood disorders has been estimated at \$55.1 billion per year.

DEPRESSION IN ASSOCIATION WITH MEDICAL ILLNESS

Depression occurring in the context of medical illness is difficult to evaluate. Depressive symptomatology may reflect the psychological stress of coping with the disease, may be caused by the disease process itself or by the medications used to treat it, or may simply coexist in time with the medical diagnosis.

Virtually every class of *medication* includes some agent that can induce depression. Antihypertensive drugs, anticholesterolemic agents, and antiarrhythmic agents are common triggers of depressive symptoms. Iatrogenic depression should also be considered in patients receiving glucocorticoids, antimicrobials, systemic analgesics, antiparkinsonian medications, and anticonvulsants. To decide whether a causal relationship exists between pharmacologic therapy and a patient's change in mood, it may sometimes be necessary to undertake an empirical trial of an alternative medication.

Between 20 and 30% of *cardiac* patients manifest a depressive disorder; an even higher percentage experience depressive symptomatology when self-reporting scales are used. Depressive symptoms following unstable angina, myocardial infarction, cardiac bypass surgery, or heart transplant impair rehabilitation and are associated with higher rates of mortality and medical morbidity. Depressed patients often show decreased variability in heart rate (an index of reduced parasympathetic nervous system activity), which may predispose individuals to ventricular arrhythmia and increased morbidity. Depression also appears to increase the risk of developing coronary heart disease, possibly through increased platelet aggregation. TCAs are contraindicated in patients with

bundle branch block, and TCA-induced tachycardia is an additional concern in patients with congestive heart failure. SSRIs appear not to induce ECG changes or adverse cardiac events and thus are reasonable first-line drugs for patients at risk for TCA-related complications. SSRIs may interfere with hepatic metabolism of anticoagulants, however, causing increased anticoagulation.

In patients with *cancer*, the mean prevalence of depression is 25%, but depression occurs in 40–50% of patients with cancers of the pancreas or oropharynx. This association is not due to the effect of cachexia alone, as the higher prevalence of depression in patients with pancreatic cancer persists when compared to those with advanced gastric cancer. Initiation of antidepressant medication in cancer patients has been shown to improve quality of life as well as mood. Psychotherapeutic approaches, particularly group therapy, may have some effect on short-term depression, anxiety, and pain symptoms.

Depression occurs frequently in patients with *neurologic disorders*, particularly cerebrovascular disorders, Parkinson's disease, dementia, multiple sclerosis, and traumatic brain injury. One in five patients with left-hemisphere stroke involving the dorsolateral frontal cortex experiences major depression. Late-onset depression in otherwise cognitively normal individuals increases the risk of a subsequent diagnosis of Alzheimer's disease. Both TCA and SSRI agents are effective against these depressions, as are stimulant compounds and, in some patients, MAOIs.

The reported prevalence of depression in patients with *diabetes mellitus* varies from 8 to 27%, with the severity of the mood state correlating with the level of hyperglycemia and the presence of diabetic complications. Treatment of depression may be complicated by effects of antidepressive agents on glycemic control. MAOIs can induce hypoglycemia and weight gain, while TCAs can produce hyperglycemia and carbohydrate craving. SSRIs, like MAOIs, may reduce fasting plasma glucose, but they are easier to use and may also improve dietary and medication compliance.

Hypothyroidism is frequently associated with features of depression, most commonly depressed mood and memory impairment. Hyperthyroid states may also present in a similar fashion, usually in geriatric populations. Improvement in mood usually follows normalization of thyroid function, but adjunctive antidepressant medication is sometimes required. Patients with subclinical hypothyroidism can also experience symptoms of depression and cognitive difficulty that respond to thyroid replacement.

The lifetime prevalence of depression in *HIV-positive* individuals has been estimated at 22–45%. The relationship between depression and disease progression is multifactorial and likely to involve psychological and social factors, alterations in immune function, and central nervous system (CNS) disease. Chronic hepatitis C

infection is also associated with depression, which may worsen with interferon- α treatment.

Some chronic disorders of uncertain etiology, such as chronic fatigue syndrome (Chap. 52) and fibromyalgia, are strongly associated with depression and anxiety; patients may benefit from antidepressant treatment or anticonvulsant agents such as pregabalin.

DEPRESSIVE DISORDERS

Clinical manifestations

Major depression is defined as depressed mood on a daily basis for a minimum duration of 2 weeks (Table 54-9). An episode may be characterized by sadness, indifference, apathy, or irritability and is usually associated with changes in sleep patterns, appetite, and weight; motor agitation or retardation; fatigue; impaired concentration and decision making; feelings of shame or guilt; and thoughts of death or dying. Patients with depression have a profound loss of pleasure in all enjoyable activities, exhibit early morning awakening, feel that the dysphoric mood state is qualitatively different from sadness, and often notice a diurnal variation in mood (worse in morning hours).

Approximately 15% of the population experiences a major depressive episode at some point in life, and 6–8% of all outpatients in primary care settings satisfy diagnostic criteria for the disorder. Depression is often undiagnosed, and, even more frequently, it is treated inadequately. If a physician suspects the presence of a major depressive episode, the initial task is to determine whether it represents unipolar or bipolar depression or is one of the 10–15% of cases that are secondary to general medical illness or substance abuse. Physicians should also assess the risk of suicide by direct questioning, as patients are often reluctant to verbalize such thoughts without prompting. If specific plans are uncovered or if significant risk factors exist (e.g., a past history of suicide attempts, profound hopelessness, concurrent medical illness, substance abuse, or social isolation), the patient must be referred to a mental health specialist for immediate care. The physician should specifically probe each of these areas in an empathic and hopeful manner, being sensitive to denial and possible minimization of distress. The presence of anxiety, panic, or agitation significantly increases near-term suicidal risk. Approximately 4–5% of all depressed patients will commit suicide; most will have sought help from physicians within 1 month of their deaths.

In some depressed patients, the mood disorder does not appear to be episodic and is not clearly associated with either psychosocial dysfunction or change from the individual’s usual experience in life. Dysthymic disorder consists of a pattern of chronic (at least 2 years), ongoing, mild depressive symptoms that are less severe

TABLE 54-9
CRITERIA FOR A MAJOR DEPRESSIVE EPISODE

- A. Five (or more) of the following symptoms have been present during the same 2-week period and represent a change from previous functioning; at least one of the symptoms is either (1) depressed mood or (2) loss of interest or pleasure. **Note:** Do not include symptoms that are clearly due to a general medical condition, or mood-incongruent delusions or hallucinations.
 - 1. Depressed mood most of the day, nearly every day, as indicated by either subjective report (e.g., feels sad or empty) or observation made by others (e.g., appears tearful)
 - 2. Markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day (as indicated by either subjective account or observation made by others)
 - 3. Significant weight loss when not dieting or weight gain (e.g., a change of >5% of body weight in a month), or decrease or increase in appetite nearly every day
 - 4. Insomnia or hypersomnia nearly every day
 - 5. Psychomotor agitation or retardation nearly every day (observable by others, not merely subjective feelings of restlessness or being slowed down)
 - 6. Fatigue or loss of energy nearly every day
 - 7. Feelings of worthlessness or excessive or inappropriate guilt (which may be delusional) nearly every day (not merely self-reproach or guilt about being sick)
 - 8. Diminished ability to think or concentrate, or indecisiveness, nearly every day (either by subjective account or as observed by others)
 - 9. Recurrent thoughts of death (not just fear of dying), recurrent suicidal ideation without a specific plan, or a suicide attempt or a specific plan for committing suicide
- B. The symptoms do not meet criteria for a mixed episode
- C. The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning
- D. The symptoms are not due to the direct physiologic effects of a substance (e.g., a drug of abuse, a medication) or a general medical condition (e.g., hypothyroidism)
- E. The symptoms are not better accounted for by bereavement (i.e., after the loss of a loved one), the symptoms persist for >2 months or are characterized by marked functional impairment, morbid preoccupation with worthlessness, suicidal ideation, psychotic symptoms, or psychomotor retardation

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and less disabling than those found in major depression; the two conditions are sometimes difficult to separate, however, and can occur together (“double depression”). Many patients who exhibit a profile of pessimism, disinterest, and low self-esteem respond to antidepressant treatment. Dysthymic disorder exists in ~5% of primary care patients. The term *minor depression* is used for individuals who experience at least two depressive

symptoms for 2 weeks but who do not meet the full criteria for major depression. Despite its name, minor depression is associated with significant morbidity and disability and also responds to pharmacologic treatment.

Depression is approximately twice as common in women as in men, and the incidence increases with age in both sexes. Twin studies indicate that the liability to major depression of early onset (before age 25) is largely genetic in origin. Negative life events can precipitate and contribute to depression, but genetic factors influence the sensitivity of individuals to these stressful events. In most cases, both biologic and psychosocial factors are involved in the precipitation and unfolding of depressive episodes. The most potent stressors appear to involve death of a relative, assault, or severe marital or relationship problems.

Unipolar depressive disorders usually begin in early adulthood and recur episodically over the course of a lifetime. The best predictor of future risk is the number of past episodes; 50–60% of patients who have a first episode have at least one or two recurrences. Some patients experience multiple episodes that become more severe and frequent over time. The duration of an untreated episode varies greatly, ranging from a few months to ≤ 1 year. The pattern of recurrence and clinical progression in a developing episode are also variable. Within an individual, the nature of episodes (e.g., specific presenting symptoms, frequency and duration) may be similar over time. In a minority of patients, a severe depressive episode may progress to a psychotic state; in elderly patients, depressive symptoms may be associated with cognitive deficits mimicking dementia (“pseudodementia”). A seasonal pattern of depression, called *seasonal affective disorder*, may manifest with onset and remission of episodes at predictable times of the year. This disorder is more common in women, whose symptoms are anergy, fatigue, weight gain, hypersomnia, and episodic carbohydrate craving. The prevalence increases with distance from the equator, and improvement may occur by altering light exposure.

Etiology and pathophysiology

Although evidence for genetic transmission of unipolar depression is not as strong as in bipolar disorder, monozygotic twins have a higher concordance rate (46%) than dizygotic siblings (20%), with little support for any effect of a shared family environment.

Neuroendocrine abnormalities that reflect the neurovegetative signs and symptoms of depression include: (1) increased cortisol and corticotropin-releasing hormone (CRH) secretion, (2) an increase in adrenal size, (3) a decreased inhibitory response of glucocorticoids to dexamethasone, and (4) a blunted response of thyroid-stimulating hormone (TSH) level to infusion of thyroid-releasing hormone (TRH). Antidepressant treatment leads

to normalization of these abnormalities. Major depression is also associated with an upregulation of proinflammatory cytokines, which also normalizes with antidepressant treatment.

Diurnal variations in symptom severity and alterations in circadian rhythmicity of a number of neurochemical and neurohumoral factors suggest that biologic differences may be secondary to a primary defect in regulation of biologic rhythms. Patients with major depression show consistent findings of a decrease in rapid eye movement (REM) sleep onset (REM latency), an increase in REM density, and, in some subjects, a decrease in stage IV delta slow-wave sleep.

Although antidepressant drugs inhibit neurotransmitter uptake within hours, their therapeutic effects typically emerge over several weeks, implicating adaptive changes in second messenger systems and transcription factors as possible mechanisms of action.

The pathogenesis of depression is discussed in detail in Chap. 53.

TREATMENT Depressive Disorders

Treatment planning requires coordination of short-term strategies to induce remission combined with longer term maintenance designed to prevent recurrence. The most effective intervention for achieving remission and preventing relapse is medication, but combined treatment, incorporating psychotherapy to help the patient cope with decreased self-esteem and demoralization, improves outcome (**Fig. 54-1**). Approximately 40% of primary care patients with depression drop out of treatment and discontinue medication if symptomatic improvement is not noted within a month, unless additional support is provided. Outcome improves with (1) increased intensity and frequency of visits during the first 4–6 weeks of treatment, (2) supplemental educational materials, and (3) psychiatric consultation as indicated. Despite the widespread use of SSRIs and other second-generation antidepressant drugs, there is no convincing evidence that this class of antidepressant is more efficacious than TCAs. Between 60 and 70% of all depressed patients respond to any drug chosen, if it is given in a sufficient dose for 6–8 weeks. There is no ideal antidepressant; no current compound combines rapid onset of action, moderate half-life, a meaningful relationship between dose and blood level, a low side effect profile, minimal interaction with other drugs, and safety in overdose.

A rational approach to selecting which antidepressant to use involves matching the patient's preference and medical history with the metabolic and side effect profile of the drug (Tables 54-4 and 54-5). A previous response, or a family history of a positive response, to

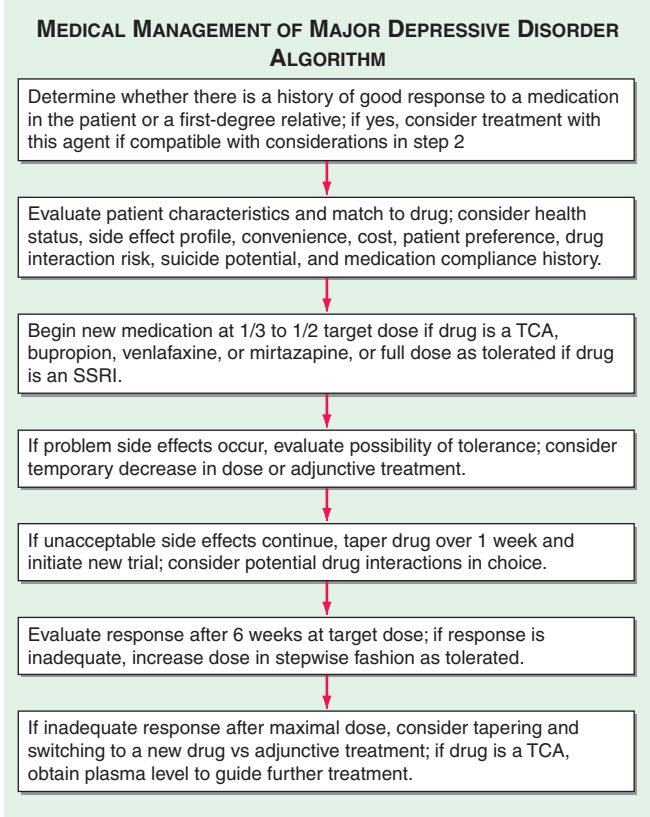


FIGURE 54-1
A guideline for the medical management of major depressive disorder. SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

a specific antidepressant often suggests that that drug be tried first. Before initiating antidepressant therapy, the physician should evaluate the possible contribution of comorbid illnesses and consider their specific treatment. In individuals with suicidal ideation, particular attention should be paid to choosing a drug with low toxicity if taken in overdose. The SSRIs, and other newer antidepressant drugs are distinctly safer in this regard; nevertheless, the advantages of TCAs have not been completely superseded. The existence of generic equivalents make TCAs relatively cheap, and for secondary tricyclics, particularly nortriptyline and desipramine, well-defined relationships among dose, plasma level, and therapeutic response exist. The steady-state plasma level achieved for a given drug dose can vary more than tenfold between individuals and plasma levels may help in interpreting apparent resistance to treatment and/or unexpected drug toxicity. The principal side effects of TCAs are antihistamine (sedation) and anticholinergic (constipation, dry mouth, urinary hesitancy, blurred vision). TCAs are contraindicated in patients with serious cardiovascular risk factors and overdoses of tricyclic agents can be lethal, with desipramine carrying the greatest risk. It is judicious to prescribe only a 10-day

supply when suicide is a risk. Most patients require a daily dose of 150–200 mg of imipramine or amitriptyline or its equivalent to achieve a therapeutic blood level of 150–300 ng/mL and a satisfactory remission; some patients show a partial effect at lower doses. Geriatric patients may require a low starting dose and slow escalation. Ethnic differences in drug metabolism are significant, with Hispanic, Asian, and black patients generally requiring lower doses than whites to achieve a comparable blood level. P450 profiling using genetic chip technology may be clinically useful in predicting individual sensitivity.

Second-generation antidepressants include amoxapine, maprotiline, trazodone, and bupropion. Amoxapine is a dibenzoxazepine derivative that blocks norepinephrine and serotonin reuptake and has a metabolite that shows a degree of dopamine blockade. Long-term use of this drug carries a risk of tardive dyskinesia. Maprotiline is a potent noradrenergic reuptake blocker that has little anticholinergic effect but may produce seizures. Bupropion is a novel antidepressant whose mechanism of action is thought to involve enhancement of noradrenergic function. It has no anticholinergic, sedating, or orthostatic side effects and has a low incidence of sexual side effects. It may, however, be associated with stimulant-like side effects, may lower seizure threshold, and has an exceptionally short half-life, requiring frequent dosing. An extended-release preparation is available.

SSRIs such as fluoxetine, sertraline, paroxetine, citalopram, and escitalopram cause a lower frequency of anticholinergic, sedating, and cardiovascular side effects but a possibly greater incidence of gastrointestinal complaints, sleep impairment, and sexual dysfunction than do TCAs. Akathisia, involving an inner sense of restlessness and anxiety in addition to increased motor activity, may also be more common, particularly during the first week of treatment. One concern is the risk of “serotonin syndrome,” thought to result from hyperstimulation of brainstem 5HT_{1A} receptors and characterized by myoclonus, agitation, abdominal cramping, hyperpyrexia, hypertension, and potentially death. Serotonergic agonists taken in combination should be monitored closely for this reason. Considerations such as half-life, compliance, toxicity, and drug-drug interactions may guide the choice of a particular SSRI. Fluoxetine and its principal active metabolite, norfluoxetine, for example, have a combined half-life of almost 7 days, resulting in a delay of 5 weeks before steady-state levels are achieved and a similar delay for complete drug excretion once its use is discontinued. All the SSRIs may impair sexual function, resulting in diminished libido, impotence, or difficulty in achieving orgasm. Sexual dysfunction frequently results in noncompliance and should be asked about specifically. Sexual dysfunction can sometimes be ameliorated by lowering the dose, by instituting weekend

drug holidays (two or three times a month), or by treatment with amantadine (100 mg tid), bethanechol (25 mg tid), buspirone (10 mg tid), or bupropion (100–150 mg/d). Paroxetine appears to be more anticholinergic than either fluoxetine or sertraline, and sertraline carries a lower risk of producing an adverse drug interaction than the other two. Rare side effects of SSRIs include angina due to vasospasm and prolongation of the prothrombin time. Escitalopram is the most specific of currently available SSRIs and appears to have no specific inhibitory effects on the P450 system.

Venlafaxine, desvenlafaxine, and duloxetine block the reuptake of both norepinephrine and serotonin but produce relatively little in the way of traditional tricyclic side effects. Unlike the SSRIs, venlafaxine has a relatively linear dose-response curve. Patients should be monitored for a possible increase in diastolic blood pressure, and multiple daily dosing is required because of the drug's short half-life. An extended-release form is available and has a somewhat lower incidence of gastrointestinal side effects. Mirtazapine is a TCA that has a unique spectrum of activity. It increases noradrenergic and serotonergic neurotransmission through a blockade of central α_2 -adrenergic receptors and postsynaptic 5HT₂ and 5HT₃ receptors. It is also strongly antihistaminic and, as such, may produce sedation.

With the exception of citalopram and escitalopram, each of the SSRIs may inhibit one or more cytochrome P450 enzymes. Depending on the specific isoenzyme involved, the metabolism of a number of concomitantly administered medications can be dramatically affected. Fluoxetine and paroxetine, for example, by inhibiting 2D6, can cause dramatic increases in the blood level of type 1C antiarrhythmics, while sertraline, by acting on 3A4, may alter blood levels of carbamazepine, or digoxin.

The MAOIs are highly effective, particularly in atypical depression, but the risk of hypertensive crisis following intake of tyramine-containing food or sympathomimetic drugs makes them inappropriate as first-line agents. Transdermal selegiline may avert this risk at low dose. Common side effects include orthostatic hypotension, weight gain, insomnia, and sexual dysfunction. MAOIs should not be used concomitantly with SSRIs, because of the risk of serotonin syndrome, or with TCAs, because of possible hyperadrenergic effects.

Electroconvulsive therapy is at least as effective as medication, but its use is reserved for treatment-resistant cases and delusional depressions. Transcranial magnetic stimulation (TMS) is approved for treatment-resistant depression and has been shown to have efficacy in several controlled trials. Vagus nerve stimulation (VNS) has also recently been approved for treatment-resistant depression, but its degree of efficacy is controversial. Deep brain stimulation is another treatment that is being used experimentally in treatment-resistant cases.

Regardless of the treatment undertaken, the response should be evaluated after ~2 months. Three-quarters of patients show improvement by this time, but if remission is inadequate the patient should be questioned about compliance and an increase in medication dose should be considered if side effects are not troublesome. If this approach is unsuccessful, referral to a mental health specialist is advised. Strategies for treatment then include selection of an alternative drug, combinations of antidepressants, and/or adjunctive treatment with other classes of drugs, including lithium, thyroid hormone, atypical antipsychotic agents, and dopamine agonists. A large randomized trial (STAR-D) was unable to show preferential efficacy, but the addition of atypical antipsychotic drugs has received FDA approval. Patients whose response to an SSRI wanes over time may benefit from the addition of buspirone (10 mg tid) or pindolol (2–5 mg tid) or small amounts of a TCA such as desipramine (25 mg bid or tid). Most patients will show some degree of response but aggressive treatment should be pursued until remission is achieved, and drug treatment should be continued for at least 6–9 more months to prevent relapse. In patients who have had two or more episodes of depression, indefinite maintenance treatment should be considered.

It is essential to educate patients both about depression and the benefits and side effects of medications they are receiving. Advice about stress reduction and cautions that alcohol may exacerbate depressive symptoms and impair drug response are helpful. Patients should be given time to describe their experience, their outlook, and the impact of the depression on them and their families. Occasional empathic silence may be as helpful for the treatment alliance as verbal reassurance. Controlled trials have shown that cognitive-behavioral and interpersonal therapies are effective in improving psychological and social adjustment and that a combined treatment approach is more successful than medication alone for many patients.

BIPOLAR DISORDER

Clinical manifestations

Bipolar disorder is characterized by unpredictable swings in mood from mania (or hypomania) to depression. Some patients suffer only from recurrent attacks of *mania*, which in its pure form is associated with increased psychomotor activity; excessive social extroversion; decreased need for sleep; impulsivity and impairment in judgment; and expansive, grandiose, and sometimes irritable mood (Table 54-10). In severe mania, patients may experience delusions and paranoid thinking indistinguishable from schizophrenia. One-half of patients with bipolar disorder present with a mixture of psychomotor agitation and activation with dysphoria,

TABLE 54-10

CRITERIA FOR A MANIC EPISODE

- A. A distinct period of abnormally and persistently elevated, expansive, or irritable mood, lasting at least 1 week (or any duration if hospitalization is necessary)
- B. During the period of mood disturbance, three (or more) of the following symptoms have persisted (four if the mood is only irritable) and have been present to a significant degree:
 - 1. Inflated self-esteem or grandiosity
 - 2. Decreased need for sleep (e.g., feels rested after only 3 h of sleep)
 - 3. More talkative than usual or pressure to keep talking
 - 4. Flight of ideas or subjective experience that thoughts are racing
 - 5. Distractibility (i.e., attention too easily drawn to unimportant or irrelevant external stimuli)
 - 6. Increase in goal-directed activity (either socially, at work or school, or sexually) or psychomotor agitation
 - 7. Excessive involvement in pleasurable activities that have a high potential for painful consequences (e.g., engaging in unrestrained buying sprees, sexual indiscretions, or foolish business investments)
- C. The symptoms do not meet criteria for a mixed episode.
- D. The mood disturbance is sufficiently severe to cause marked impairment in occupational functioning or in usual social activities or relationships with others, or to necessitate hospitalization to prevent harm to self or others, or there are psychotic features.
- E. The symptoms are not due to the direct physiologic effects of a substance (e.g., a drug of abuse, a medication, or other treatment) or a general medical condition (e.g., hyperthyroidism).

Note: Manic-like episodes that are clearly caused by somatic antidepressant treatment (e.g., medication, electroconvulsive therapy, light therapy) should not count toward a diagnosis of bipolar I disorder.
Source: Reprinted with permission from the *Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition, Text Revision. Copyright © 2000 American Psychiatric Association.

anxiety, and irritability. It may be difficult to distinguish *mixed mania* from *agitated depression*. In some bipolar patients (*bipolar II disorder*), the full criteria for mania are lacking, and the requisite recurrent depressions are separated by periods of mild activation and increased energy (hypomania). In *cyclothymic disorder*, there are numerous hypomanic periods, usually of relatively short duration, alternating with clusters of depressive symptoms that fail, either in severity or duration, to meet the criteria of major depression. The mood fluctuations are chronic and should be present for at least 2 years before the diagnosis is made.

Manic episodes typically emerge over a period of days to weeks, but onset within hours is possible, usually in the early morning hours. An untreated episode of either depression or mania can be as short as several weeks or last as long as 8–12 months, and rare patients have an unrelenting chronic course. The term *rapid cycling* is used

for patients who have four or more episodes of either depression or mania in a given year. This pattern occurs in 15% of all patients, almost all of whom are women. In some cases, rapid cycling is linked to an underlying thyroid dysfunction and, in others, it is iatrogenically triggered by prolonged antidepressant treatment. Approximately one-half of patients have sustained difficulties in work performance and psychosocial functioning, with depressive phases being more responsible for impairment than mania.

Bipolar disorder is common, affecting ~1.5% of the population in the United States. Onset is typically between 20 and 30 years of age, but many individuals report premorbid symptoms in late childhood or early adolescence. The prevalence is similar for men and women; women are likely to have more depressive and men more manic episodes over a lifetime.

Differential diagnosis

The differential diagnosis of mania includes secondary mania induced by stimulant or sympathomimetic drugs, hyperthyroidism, AIDS, neurologic disorders, such as Huntington’s or Wilson’s disease, and cerebrovascular accidents. Comorbidity with alcohol and substance abuse is common, either because of poor judgment and increased impulsivity or because of an attempt to self-treat the underlying mood symptoms and sleep disturbances.

Etiology and pathophysiology

Genetic predisposition to bipolar disorder is evident from family studies; the concordance rate for monozygotic twins approaches 80%. Patients with bipolar disorder also appear to have altered circadian rhythmicity, and lithium may exert its therapeutic benefit through a resynchronization of intrinsic rhythms keyed to the light/dark cycle. A detailed discussion of the pathogenesis of bipolar disorder is presented in Chap. 53.

TREATMENT **Bipolar Disorder**

(Table 54-11) Lithium carbonate is the mainstay of treatment in bipolar disorder, although sodium valproate and olanzapine are equally effective in acute mania, as is lamotrigine in the depressed phase. The response rate to lithium carbonate is 70–80% in acute mania, with beneficial effects appearing in 1–2 weeks. Lithium also has a prophylactic effect in prevention of recurrent mania and, to a lesser extent, in the prevention of recurrent depression. A simple cation, lithium is rapidly absorbed from the gastrointestinal tract and remains unbound to plasma or tissue proteins. Some 95% of a given dose is excreted unchanged through the kidneys within 24 h.

TABLE 54-11

CLINICAL PHARMACOLOGY OF MOOD STABILIZERS

AGENT AND DOSING	SIDE EFFECTS AND OTHER EFFECTS
Lithium	Common Side Effects
Starting dose: 300 mg bid or tid Therapeutic blood level: 0.8–1.2 meq/L	Nausea/anorexia/diarrhea, fine tremor, thirst, polyuria, fatigue, weight gain, acne, folliculitis, neutrophilia, hypothyroidism Blood level is increased by thiazides, tetracyclines, and NSAIDs Blood level is decreased by bronchodilators, verapamil, and carbonic anhydrase inhibitors <i>Rare side effects:</i> Neurotoxicity, renal toxicity, hypercalcemia, ECG changes
Valproic Acid	Common Side Effects
Starting dose: 250 mg tid Therapeutic blood level: 50–125 µg/mL	Nausea/anorexia, weight gain, sedation, tremor, rash, alopecia Inhibits hepatic metabolism of other medications <i>Rare side effects:</i> Pancreatitis, hepatotoxicity, Stevens-Johnson syndrome
Carbamazepine/ Oxcarbazepine	Common Side Effects
Starting dose: 200 mg bid for carbamazepine, 150 mg bid for oxcarbazepine Therapeutic blood level: 4–12 µg/mL for carbamazepine	Nausea/anorexia, sedation, rash, dizziness/ataxia Carbamazepine, but not oxcarbazepine, induces hepatic metabolism of other medications <i>Rare side effects:</i> Hyponatremia, agranulocytosis, Stevens-Johnson syndrome
Lamotrigine	Common Side Effects
Starting dose: 25 mg/d	Rash, dizziness, headache, tremor, sedation, nausea <i>Rare side effect:</i> Stevens-Johnson syndrome

Abbreviations: NSAIDs, nonsteroidal anti-inflammatory drugs; ECG, electrocardiogram.

Serious side effects from lithium are rare, but minor complaints such as gastrointestinal discomfort, nausea, diarrhea, polyuria, weight gain, skin eruptions, alopecia, and edema are common. Over time, urine-concentrating ability may be decreased, but significant nephrotoxicity does not usually occur. Lithium exerts an antithyroid

effect by interfering with the synthesis and release of thyroid hormones. More serious side effects include tremor, poor concentration and memory, ataxia, dysarthria, and incoordination. There is suggestive, but not conclusive, evidence that lithium is teratogenic, inducing cardiac malformations in the first trimester.

In the treatment of acute mania, lithium is initiated at 300 mg bid or tid, and the dose is then increased by 300 mg every 2–3 days to achieve blood levels of 0.8–1.2 meq/L. Because the therapeutic effect of lithium may not appear until after 7–10 days of treatment, adjunctive usage of lorazepam (1–2 mg every 4 h) or clonazepam (0.5–1 mg every 4 h) may be beneficial to control agitation. Antipsychotics are indicated in patients with severe agitation who respond only partially to benzodiazepines. Patients using lithium should be monitored closely, since the blood levels required to achieve a therapeutic benefit are close to those associated with toxicity.

Valproic acid may be better than lithium for patients who experience rapid cycling (i.e., more than four episodes a year) or who present with a mixed or dysphoric mania. Tremor and weight gain are the most common side effects; hepatotoxicity and pancreatitis are rare toxicities.

Carbamazepine and oxcarbazepine, although not formally approved by the FDA for bipolar disorder, have clinical efficacy in the treatment of acute mania. Second-generation antipsychotic drugs (olanzapine, quetiapine, risperidone, ziprasidone, aripiprazole, and asenapine) have also been shown to be effective, either alone or in combination with a mood stabilizer. An increased risk of weight gain and other metabolic abnormalities is a concern with these agents.

The recurrent nature of bipolar mood disorder necessitates maintenance treatment. A sustained blood lithium level of at least 0.8 meq/L is important for optimal prophylaxis and has been shown to reduce risk of suicide, a finding not yet apparent for other mood stabilizers. Compliance is frequently an issue and often requires enlistment and education of concerned family members. Efforts to identify and modify psychosocial factors that may trigger episodes are important, as is an emphasis on lifestyle regularity. Antidepressant medications are sometimes required for the treatment of severe breakthrough depressions, but their use should generally be avoided during maintenance treatment because of the risk of precipitating mania or accelerating the cycle frequency. Loss of efficacy over time may be observed with any of the mood-stabilizing agents. In such situations, an alternative agent or combination therapy is usually helpful.

Consensus guidelines for the treatment of acute mania and bipolar depression are described in [Table 54-12](#).

TABLE 54-12
CONSENSUS GUIDELINES ON THE DRUG TREATMENT OF ACUTE MANIA AND BIPOLAR DEPRESSION

CONDITION	PREFERRED AGENTS
Euphoric mania	Lithium
Mixed/dysphoric mania	Valproic acid
Mania with psychosis	Valproic acid with olanzapine, aripiprazole, conventional antipsychotic, or risperidone
Hypomania	Lithium, lamotrigine, or valproic acid alone
Severe depression with psychosis	Venlafaxine, bupropion, or paroxetine <i>plus</i> lithium <i>plus</i> olanzapine, or risperidone; consider ECT
Severe depression without psychosis	Bupropion, paroxetine, sertraline, venlafaxine, or citalopram <i>plus</i> lithium
Mild to moderate depression	Lithium or lamotrigine alone; add bupropion if needed

Abbreviation: ECT, electroconvulsive therapy.
Source: From GS Sachs et al: Postgrad Med April, 2000.

SOMATOFORM DISORDERS

CLINICAL MANIFESTATIONS

Patients with multiple somatic complaints that cannot be explained by a known medical condition or by the effects of alcohol or of recreational or prescription drugs are commonly seen in primary care practice; one survey indicated a prevalence of such complaints of 5%. In *somatization disorder*, the patient presents with multiple physical complaints referable to different organ systems (**Table 54-13**). Onset is usually before age 30 years, and the disorder is persistent. Formal diagnostic criteria require the recording of at least four pain, two gastrointestinal, one sexual, and one pseudoneurologic symptom. Patients with somatization disorder often present with dramatic complaints, but the complaints are inconsistent. Symptoms of comorbid anxiety and mood disorder are common and may be the result of drug interactions due to regimens initiated independently by different physicians. Patients with somatization disorder may be impulsive and demanding and frequently qualify for a formal comorbid psychiatric diagnosis. In *conversion disorder*, the symptoms focus on deficits that involve motor or sensory function and on psychological factors that initiate or exacerbate the medical presentation. Like somatization disorder, the deficit is not intentionally produced or simulated, as is the case in factitious disorder (malingering). In *hypochondriasis*, the essential feature is a belief

TABLE 54-13
DIAGNOSTIC CRITERIA FOR SOMATIZATION DISORDER

- A. A history of many physical complaints beginning before age 30 years that occur over a period of several years and result in treatment being sought or significant impairment in social, occupational, or other important areas of functioning.
- B. Each of the following criteria must have been met, with individual symptoms occurring at any time during the course of the disturbance:
 - 1. *Four pain symptoms*: a history of pain related to at least four different sites or functions (e.g., head, abdomen, back, joints, extremities, chest, rectum, during menstruation, during sexual intercourse, or during urination)
 - 2. *Two gastrointestinal symptoms*: a history of at least two gastrointestinal symptoms other than pain (e.g., nausea, bloating, vomiting other than during pregnancy, diarrhea, or intolerance of several different foods)
 - 3. *One sexual symptom*: a history of at least one sexual or reproductive symptom other than pain (e.g., sexual indifference, erectile or ejaculatory dysfunction, irregular menses, excessive menstrual bleeding, vomiting throughout pregnancy)
 - 4. *One pseudoneurologic symptom*: a history of at least one symptom or deficit suggesting a neurologic condition not limited to pain (conversion symptoms such as impaired coordination or balance, paralysis or localized weakness, difficulty swallowing or lump in throat, aphonia, urinary retention, hallucinations, loss of touch or pain sensation, double vision, blindness, deafness, seizures; dissociative symptoms such as amnesia; or loss of consciousness other than fainting)
- C. Either of the following:
 - 1. After appropriate investigation, each of the symptoms in criterion B cannot be fully explained by a known general medical condition or the direct effects of a substance (e.g., a drug of abuse, a medication)
 - 2. When there is a related general medical condition, the physical complaints or resulting social or occupational impairment are in excess of what would be expected from the history, physical examination, or laboratory findings
- D. The symptoms are not intentionally produced or feigned (as in factitious disorder or malingering).

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of serious medical illness that persists despite reassurance and appropriate medical evaluation. As with somatization disorder, patients with hypochondriasis have a history of poor relationships with physicians stemming from their sense that they have been evaluated and treated inappropriately or inadequately. Hypochondriasis can be disabling in intensity and is persistent, with waxing and waning symptomatology.

In *factitious illnesses*, the patient consciously and voluntarily produces physical symptoms of illness. The term *Munchausen's syndrome* is reserved for individuals with particularly dramatic, chronic, or severe factitious illness. In true factitious illness, the sick role itself is gratifying. A variety of signs, symptoms, and diseases have been either simulated or caused by factitious behavior, the most common including chronic diarrhea, fever of unknown origin, intestinal bleeding or hematuria, seizures, and hypoglycemia. Factitious disorder is usually not diagnosed until 5–10 years after its onset, and it can produce significant social and medical costs. In *malinger*, the fabrication derives from a desire for some external reward such as a narcotic medication or disability reimbursement.

TREATMENT Somatoform Disorders

Patients with somatization disorders are frequently subjected to many diagnostic tests and exploratory surgeries in an attempt to find their “real” illness. Such an approach is doomed to failure and does not address the core issue. Successful treatment is best achieved through behavior modification, in which access to the physician is tightly regulated and adjusted to provide a sustained and predictable level of support that is less clearly contingent on the patient's level of presenting distress. Visits can be brief and should not be associated with a need for a diagnostic or treatment action. Although the literature is limited, some patients with somatization disorder may benefit from antidepressant treatment.

Any attempt to confront the patient usually creates a sense of humiliation and causes the patient to abandon treatment from that caregiver. A better strategy is to introduce psychological causation as one of a number of possible explanations and to include factitious illness as an option in the differential diagnoses that are discussed. Without directly linking psychotherapeutic intervention to the diagnosis, the patient can be offered a face-saving means by which the pathologic relationship with the health care system can be examined and alternative approaches to life stressors developed.

PERSONALITY DISORDERS

Clinical manifestations

Personality disorders are characteristic patterns of thinking, feeling, and interpersonal behavior that are relatively inflexible and cause significant functional impairment or subjective distress for the individual. The observed behaviors are not secondary to another mental disorder, nor are they precipitated by substance abuse or a general medical condition. This distinction is often difficult to make in clinical practice, as

personality change may be the first sign of serious neurologic, endocrine, or other medical illness. Patients with frontal lobe tumors, for example, can present with changes in motivation and personality while the results of the neurologic examination remain within normal limits. Individuals with personality disorders are often regarded as “difficult patients” in clinical medical practice because they are seen as excessively demanding and/or unwilling to follow recommended treatment plans. Although DSM-IV portrays personality disorders as qualitatively distinct categories, there is an alternative perspective that personality characteristics vary as a continuum between normal functioning and formal mental disorder.

Personality disorders have been grouped into three overlapping clusters. *Cluster A* includes paranoid, schizoid, and schizotypal personality disorders. It includes individuals who are odd and eccentric and who maintain an emotional distance from others. Individuals have a restricted emotional range and remain socially isolated. Patients with schizotypal personality disorder frequently have unusual perceptual experiences and express magical beliefs about the external world. The essential feature of paranoid personality disorder is a pervasive mistrust and suspiciousness of others to an extent that is unjustified by available evidence. *Cluster B* disorders include antisocial, borderline, histrionic, and narcissistic types and describe individuals whose behavior is impulsive, excessively emotional, and erratic. *Cluster C* incorporates avoidant, dependent, and obsessive-compulsive personality types; enduring traits are anxiety and fear. The boundaries between cluster types are to some extent artificial, and many patients who meet criteria for one personality disorder also meet criteria for aspects of another. The risk of a comorbid major mental disorder is increased in patients who qualify for a diagnosis of personality disorder.

TREATMENT Personality Disorders

Dialectical behavior therapy (DBT) is a cognitive-behavioral approach that focuses on behavioral change while providing acceptance, compassion, and validation of the patient. Several randomized trials have demonstrated the efficacy of DBT in the treatment of personality disorders. Antidepressant medications and low-dose antipsychotic drugs have some efficacy in cluster A personality disorders, while anticonvulsant mood-stabilizing agents and MAOIs may be considered for patients with cluster B diagnoses who show marked mood reactivity, behavioral dyscontrol, and/or rejection hypersensitivity. Anxious or fearful cluster C patients often respond to medications used for axis I anxiety disorders (discussed earlier). It is important that the physician and the patient have rea-

sonable expectations vis-à-vis the possible benefit of any medication used and its side effects. Improvement may be subtle and observable only over time.

SCHIZOPHRENIA

Clinical manifestations

Schizophrenia is a heterogeneous syndrome characterized by perturbations of language, perception, thinking, social activity, affect, and volition. There are no pathognomonic features. The syndrome commonly begins in late adolescence, has an insidious (and less commonly acute) onset, and, often, a poor outcome, progressing from social withdrawal and perceptual distortions to recurrent delusions and hallucinations. Patients may present with positive symptoms (such as conceptual disorganization, delusions, or hallucinations) or negative symptoms (loss of function, anhedonia, decreased emotional expression, impaired concentration, and diminished social engagement) and must have at least two of these for a 1-month period and continuous signs for at least 6 months to meet formal diagnostic criteria. As individuals age, positive psychotic symptoms tend to attenuate and some measure of social and occupational function may be regained. “Negative” symptoms predominate in one-third of the schizophrenic population and are associated with a poor long-term outcome and a poor response to drug treatment. However, marked variability in the course and individual character of symptoms is typical.

The four main subtypes of schizophrenia are catatonic, paranoid, disorganized, and residual. Many individuals have symptoms of more than one type. *Catatonic-type* describes patients whose clinical presentation is dominated by profound changes in motor activity, negativism, and echolalia or echopraxia. *Paranoid-type* describes patients who have a prominent preoccupation with a specific delusional system and who otherwise do not qualify as having *disorganized-type* disease, in which disorganized speech and behavior are accompanied by a superficial or silly affect. In *residual-type* disease, negative symptomatology exists in the absence of delusions, hallucinations, or motor disturbance. The term *schizophreniform disorder* describes patients who meet the symptom requirements but not the duration requirements for schizophrenia, and *schizoaffective disorder* is used for those who manifest symptoms of schizophrenia and independent periods of mood disturbance. Prognosis depends not on symptom severity but on the response to antipsychotic medication. A permanent remission without recurrence does occasionally occur. About 10% of schizophrenic patients commit suicide.

Schizophrenia is present in 0.85% of individuals worldwide, with a lifetime prevalence of ~1–1.5%. An

estimated 300,000 episodes of acute schizophrenia occur annually in the United States, resulting in direct and indirect costs of \$62.7 billion.

Differential diagnosis

The diagnosis is principally one of exclusion, requiring the absence of significant associated mood symptoms, any relevant medical condition, and substance abuse. Drug reactions that cause hallucinations, paranoia, confusion, or bizarre behavior may be dose-related or idiosyncratic; parkinsonian medications, clonidine, quinacrine, and procaine derivatives are the most common prescription medications associated with these symptoms. Drug causes should be ruled out in any case of newly emergent psychosis. The general neurologic examination in patients with schizophrenia is usually normal, but motor rigidity, tremor, and dyskinesias are noted in one-quarter of untreated patients.

Epidemiology and pathophysiology

Epidemiologic surveys identify several risk factors for schizophrenia, including genetic susceptibility, early developmental insults, winter birth, and increasing parental age. Genetic factors are involved in at least a subset of individuals who develop schizophrenia. Schizophrenia is observed in ~6.6% of all first-degree relatives of an affected proband. If both parents are affected, the risk for offspring is 40%. The concordance rate for monozygotic twins is 50%, compared to 10% for dizygotic twins. Schizophrenia-prone families are also at risk for other psychiatric disorders, including schizoaffective disorder and *schizotypal* and *schizoid personality disorders*, the latter terms designating individuals who show a lifetime pattern of social and interpersonal deficits characterized by an inability to form close interpersonal relationships, eccentric behavior, and mild perceptual distortions. The pathogenesis of schizophrenia is discussed in detail in Chap. 53.

TREATMENT Schizophrenia

Antipsychotic agents (Table 54-14) are the cornerstone of acute and maintenance treatment of schizophrenia and are effective in the treatment of hallucinations, delusions, and thought disorders, regardless of etiology. The mechanism of action involves, at least in part, binding to dopamine D₂/D₃ receptors in the ventral striatum; the clinical potencies of traditional antipsychotic drugs parallel their affinities for the D₂ receptor, and even the newer “atypical” agents exert some degree of D₂ receptor blockade. All neuroleptics induce expression of the immediate-early gene *c-fos* in the nucleus accumbens, a dopaminergic site connecting prefrontal and limbic

TABLE 54-14

ANTIPSYCHOTIC AGENTS

NAME	USUAL PO DAILY DOSE, mg	SIDE EFFECTS	SEDATION	COMMENTS
First-Generation Antipsychotics				
Low-potency				
Chlorpromazine (Thorazine)	100–1000	Anticholinergic effects; orthostasis; photosensitivity; cholestasis; QT prolongation	+++	EPSEs usually not prominent; can cause anticholinergic delirium in elderly patients
Thioridazine (Mellaril)	100–600			
Midpotency				
Trifluoperazine (Stelazine)	2–50	Fewer anticholinergic side effects	++	Well tolerated by most patients
Perphenazine (Trilafon)	4–64	Fewer EPSEs than with higher potency agents.	++	
Loxapine (Loxitane)	30–100	Frequent EPSEs	++	
Molindone (Moban)	30–100	Frequent EPSEs	0	Little weight gain
High potency				
Haloperidol (Haldol)	5–20	No anticholinergic side effects; EPSEs often prominent	0/+	Often prescribed in doses that are too high; long-acting injectable forms of haloperidol and fluphenazine available
Fluphenazine (Prolixin)	1–20	Frequent EPSEs	0/+	
Thiothixene (Navane)	2–50	Frequent EPSEs	0/+	
Second-Generation Antipsychotics				
Clozapine (Clozaril)	150–600	Agranulocytosis (1%); weight gain; seizures; drooling; hyperthermia	++	Requires weekly WBC count for first 6 months, then biweekly if stable
Risperidone (Risperdal)	2–8	Orthostasis	+	Requires slow titration; EPSEs observed with doses >6 mg qd
Olanzapine (Zyprexa)	10–30	Weight gain	++	Mild prolactin elevation
Quetiapine (Seroquel)	350–800	Sedation; weight gain; anxiety	+++	Bid dosing
Ziprasidone (Geodon)	120–200	Orthostatic hypotension	+ / ++	Minimal weight gain; increases QT interval
Aripiprazole (Abilify)	10–30	Nausea, anxiety, insomnia	0/+	Mixed agonist/antagonist
Paliperidone (Invega)	3–12	Restlessness, EPSEs	+	Active metabolite of risperidone
Iloperidone (Fanapt)	12–24	Dizziness, hypotension	0/+	Requires dose titration
Asenapine (Saphris)	10–20	Dizziness, EPSEs, weight gain	++	Sublingual tablets; bid dosing
Lurasidone (Latuda)	40–80	Nausea, EPSEs	++	Uses CYP3A4

Abbreviations: EPSEs, extrapyramidal side effects; WBC, white blood cell.

cortices. The clinical efficacy of newer atypical neuroleptics, however, may involve *N*-methyl-D-aspartate (NMDA) receptor blockade, α_1 - and α_2 -noradrenergic activity, altering the relationship between 5HT₂ and D₂ receptor activity, as well as faster dissociation of D₂ binding and effects on neuroplasticity.

Conventional neuroleptics differ in their potency and side-effect profile. Older agents such as chlorpromazine and thioridazine, are more sedating and anticholinergic and more likely to cause orthostatic hypotension, while higher potency antipsychotics such as haloperidol,

perphenazine, and thiothixene, are more likely to induce extrapyramidal side effects. The model “atypical” antipsychotic agent is *clozapine*, a dibenzodiazepine that has a greater potency in blocking the 5HT₂ than the D₂ receptor and a much higher affinity for the D₄ than the D₂ receptor. Its principal disadvantage is a risk of blood dyscrasias. Paliperidone is a recently approved agent that is a metabolite of risperidone and shares many of its properties. Unlike other antipsychotics, clozapine does not cause a rise in prolactin level. Approximately 30% of patients who do not benefit from conventional

antipsychotic agents will have a better response to this drug, which also has a demonstrated superiority to other antipsychotic agents in preventing suicide; however, its side-effect profile makes it most appropriate for treatment-resistant cases. *Risperidone*, a benzisoxazole derivative, is more potent at 5HT₂ than D₂ receptor sites, like clozapine, but it also exerts significant α_2 antagonism, a property that may contribute to its perceived ability to improve mood and increase motor activity. Risperidone is not as effective as clozapine in treatment-resistant cases but does not carry a risk of blood dyscrasias. *Olanzapine* is similar neurochemically to clozapine but has a significant risk of inducing weight gain. *Quetiapine* is distinct in having a weak D₂ effect but potent α_1 and histamine blockade. *Ziprasidone* causes minimal weight gain and is unlikely to increase prolactin but may increase QT prolongation. *Aripiprazole* also has little risk of weight gain or prolactin increase but may increase anxiety, nausea, and insomnia as a result of its partial agonist properties.

Antipsychotic agents are effective in 70% of patients presenting with a first episode. Improvement may be observed within hours or days, but full remission usually requires 6–8 weeks. The choice of agent depends principally on the side effect profile and cost of treatment or on a past personal or family history of a favorable response to the drug in question. Atypical agents appear to be more effective in treating negative symptoms and improving cognitive function. An equivalent treatment response can usually be achieved with relatively low doses of any drug selected (i.e., 4–6 mg/d of haloperidol, 10–15 mg of olanzapine, or 4–6 mg/d of risperidone). Doses in this range result in >80% D₂ receptor blockade, and there is little evidence that higher doses increase either the rapidity or degree of response. Maintenance treatment requires careful attention to the possibility of relapse and monitoring for the development of a movement disorder. Intermittent drug treatment is less effective than regular dosing, but gradual dose reduction is likely to improve social functioning in many schizophrenic patients who have been maintained at high doses. If medications are completely discontinued, however, the relapse rate is 60% within 6 months. Long-acting injectable preparations (risperidone) are considered when noncompliance with oral therapy leads to relapses. In treatment-resistant patients, a transition to clozapine usually results in rapid improvement, but a prolonged delay in response in some cases necessitates a 6- to 9-month trial for maximal benefit to occur.

Antipsychotic medications can cause a broad range of side effects, including lethargy, weight gain, postural hypotension, constipation, and dry mouth. Extrapyramidal symptoms such as dystonia, akathisia, and akinesia are also frequent with first-generation agents and

may contribute to poor adherence if not specifically addressed. Anticholinergic and parkinsonian symptoms respond well to trihexyphenidyl, 2 mg bid, or benztropine mesylate, 1–2 mg bid. Akathisia may respond to beta blockers. In rare cases, more serious and occasionally life-threatening side effects may emerge, including hyperprolactinemia, ventricular arrhythmias, gastrointestinal obstruction, retinal pigmentation, obstructive jaundice, and neuroleptic malignant syndrome (characterized by hyperthermia, autonomic dysfunction, muscular rigidity, and elevated creatine phosphokinase levels). The most serious adverse effects of clozapine are agranulocytosis, which has an incidence of 1%, and induction of seizures, which has an incidence of 10%. Weekly white blood cell counts are required, particularly during the first 3 months of treatment.

The risk of type 2 diabetes mellitus appears to be increased in schizophrenia, and second-generation agents as a group produce greater adverse effects on glucose regulation, independent of effects on obesity, than traditional agents. Clozapine, olanzapine, and quetiapine seem more likely to cause hyperglycemia, weight gain, and hypertriglyceridemia than other atypical antipsychotic drugs. Close monitoring of plasma glucose and lipid levels are indicated with the use of these agents.

A serious side effect of long-term use of first generation antipsychotic agents is *tardive dyskinesia*, characterized by repetitive, involuntary, and potentially irreversible movements of the tongue and lips (buccolinguo-masticatory triad), and, in approximately half of cases, choreoathetosis. Tardive dyskinesia has an incidence of 2–4% per year of exposure, and a prevalence of 20% in chronically treated patients. The prevalence increases with age, total dose, and duration of drug administration. The risk associated with second-generation agents appears to be much lower. The cause may involve formation of free radicals and perhaps mitochondrial energy failure. Vitamin E may reduce abnormal involuntary movements if given early in the syndrome.

The CATIE study, a large-scale investigation of the effectiveness of antipsychotic agents in “real world” patients, revealed a high rate of discontinuation of treatment over 18 months. Olanzapine showed greater effectiveness than quetiapine, risperidone, perphenazine, or ziprasidone but also a higher discontinuation rate due to weight gain and metabolic effects. Surprisingly, perphenazine, a first-generation agent, showed little evidence of inferiority to newer drugs.

Drug treatment of schizophrenia is by itself insufficient. Educational efforts directed toward families and relevant community resources have proved to be necessary to maintain stability and optimize outcome. A treatment model involving a multidisciplinary case-management

team that seeks out and closely follows the patient in the community has proved particularly effective.

ASSESSMENT AND EVALUATION OF VIOLENCE

Primary care physicians may encounter situations in which family, domestic, or societal violence is discovered or suspected. Such an awareness can carry legal and moral obligations; many state laws mandate reporting of child, spousal, and elder abuse. Physicians are frequently the first point of contact for both victim and abuser. Approximately 2 million older Americans and 1.5 million U.S. children are thought to experience some form of physical maltreatment each year. Spousal abuse is thought to be even more prevalent. An interview study of 24,000 women in 10 countries found a lifetime prevalence of physical or sexual violence that ranged from 15 to 71%; these individuals are more likely to suffer from depression, anxiety, somatization disorder, and substance abuse and to have attempted suicide. In addition, abused individuals frequently express low self-esteem, vague somatic symptomatology, social isolation, and a passive feeling of loss of control. Although it is essential to treat these elements in the victim, the first obligation is to ensure that the perpetrator has taken responsibility for preventing any further violence. Substance abuse and/or dependence and serious mental illness in the abuser may contribute to the risk of harm and require direct intervention. Depending on the situation, law enforcement agencies, community resources such as support groups and shelters, and individual and family counseling can be appropriate components of a treatment plan. A safety plan should be formulated with

the victim, in addition to providing information about abuse, its likelihood of recurrence, and its tendency to increase in severity and frequency. Antianxiety and antidepressant medications may sometimes be useful in treating the acute symptoms, but only if independent evidence for an appropriate psychiatric diagnosis exists.

MENTAL HEALTH PROBLEMS IN THE HOMELESS

There is a high prevalence of mental disorders and substance abuse among homeless and impoverished individuals. Depending on the definition used, estimates of the total number of homeless individuals in the United States range from 800,000 to 2 million, one-third of whom qualify as having a serious mental disorder. Poor hygiene and nutrition, substance abuse, psychiatric illness, physical trauma, and exposure to the elements combine to make the provision of medical care challenging. Only a minority of these individuals receive formal mental health care; the main points of contact are outpatient medical clinics and emergency departments. Primary care settings represent a critical site in which housing needs, treatment of substance dependence, and evaluation and treatment of psychiatric illness can most efficiently take place. Successful intervention is dependent on breaking down traditional administrative barriers to health care and recognizing the physical constraints and emotional costs imposed by homelessness. Simplifying health care instructions and follow-up, allowing frequent visits, and dispensing medications in limited amounts that require ongoing contact are possible techniques for establishing a successful therapeutic relationship.

CHAPTER 55

NEUROPSYCHIATRIC ILLNESSES IN WAR VETERANS



Charles W. Hoge

Neuropsychiatric sequelae are common in combat veterans. Advances in personal protective body armor, armored vehicles, battlefield resuscitation, and the speed of evacuation to tertiary care have considerably improved the survivability of battlefield injuries, resulting in a greater awareness of the “silent wounds” associated with service in a combat zone. Although psychiatric and neurologic problems have been well documented in veterans of prior wars, the conflicts in Iraq and Afghanistan have been unique in terms of the level of commitment by the U.S. Department of Defense (DoD) and Department of Veterans Affairs (VA), Veterans Health Administration (VHA) to support research as the wars have unfolded, and to utilize that knowledge to guide population-level screening, evaluation, and treatment initiatives.

These conflicts, like previous ones, have produced hundreds of thousands of combat veterans, many of whom have received or will need care in government and civilian medical facilities. Studies have shown that service in the Iraq and Afghanistan theaters is associated with significantly elevated rates of mental disorders. Two conditions in particular have been labeled the signature injuries related to these wars: posttraumatic stress disorder (PTSD) and mild traumatic brain injury (mTBI)—also known as concussion. Although particular emphasis will be given in this chapter to PTSD and concussion/mTBI, it is important to understand that service in all wars is associated with a number of health concerns that coexist and overlap, and a multidisciplinary patient-centered approach to care is necessary.

EPIDEMIOLOGY OF WAR-RELATED PSYCHOLOGICAL AND NEUROLOGIC CONDITIONS

Service members from the current decade of war have faced multiple deployments to two very different high-intensity combat theaters, and the cumulative strain has

negatively impacted marriages, parenting, educational goals, and civilian occupations. The stresses of service in these conflicts have led to a significant increase in the rate of suicide in personnel from the two branches of service involved in the greatest level of ground combat (army, marines).

Service in a war zone can involve extreme physical stress in austere environments, prolonged sleep deprivation, physical injury, exposure to highly life-threatening events and hazards such as explosive devices, sniper fire, ambushes, indirect fire from rockets and mortars, and chemical pollutants. Certain events such as loss of a close friend in combat, leave indelible scars. All of these experiences have additive effects on health, likely mediated through physiologic mechanisms involving dysregulation of neuroendocrine and autonomic nervous system (ANS) functions.

Veterans of virtually all wars have reported elevated rates of generalized and multisystem physical, cognitive, and psychological health concerns that often become the focus of treatment months or years after returning home. These multisystem health concerns include sleep disturbance, memory and concentration problems, headaches, musculoskeletal pain, gastrointestinal symptoms (including gastroesophageal reflux), residual effects of war-time injuries, fatigue, anger, hyperarousal symptoms, high blood pressure, rapid heart rate (sometimes associated with panic symptoms), sexual problems, and symptoms associated with PTSD and depression. In order to provide optimal care to veterans with these symptoms, it is important to understand how the symptoms interrelate, and to consider the possibility that there may be underlying combat-related physiologic effects.

POSTWAR SYMPTOMS

The overlapping and multisystem health symptoms reported by warriors from every generation have been

given different labels, and have led to debates among medical professionals as to whether these are mediated primarily by physical or psychological causes. For example, World War I produced extensive debate about whether “shell shock,” diagnosed in more than 80,000 British soldiers, was neurologic (“commotional” from the brain being shaken in the skull by concussive blasts) or psychological (“emotional” or “neurasthenia”) in origin. World War II veterans were said to suffer from “battle fatigue,” Korean War veterans developed “combat stress reactions,” and Vietnam veterans developed the “post-Vietnam syndrome.” The role of environmental exposure (e.g., agent orange) and psychological causes (alcohol addiction, drug addiction, and PTSD) continue to be debated.

Gulf War I led to extensive debates as to whether Gulf War syndrome, also known as multisystem illness, was best explained by environmental exposures (e.g., oil fires, depleted uranium, nerve gas, multiple vaccinations) or the psychological stress of deployment to a war zone where there was anticipation of high casualty rates from chemical and biologic weapons and stressful training exercises involving the use of impermeable full-body protective uniforms (made from rubber, vinyl, charcoal impregnated polyurethane, and other materials) in desert conditions under extreme temperatures. Although no clinical syndrome was ever definitively confirmed among the nearly 1 million service members who deployed in 1990–1991, studies consistently found that military personnel who served in the Gulf experienced elevations in generalized symptoms across all health domains (e.g. physical, cognitive, neurologic, psychological) compared with service members who deployed elsewhere or did not deploy. In addition, there is good evidence that deployment to the Persian Gulf region during this period was associated with subsequent development of PTSD; other psychiatric disorders including generalized anxiety disorder, depression, and substance abuse, particularly alcohol abuse (Chap. 56); functional gastrointestinal symptoms such as irritable bowel syndrome; and chronic fatigue syndrome (Chap. 52).

The conflicts in Iraq and Afghanistan have led to similar debates as to whether postwar symptoms such as headaches, irritability, sleep disturbance, dizziness, and concentration problems are best attributed to concussion/mTBI or to PTSD. Several studies have shown that either PTSD or depression explains the majority of the postdeployment “postconcussive” symptoms attributed to concussion/mTBI, a finding not well received by some experts in traumatic brain injury (TBI) but consistent with civilian studies on risk factors for developing persistent symptoms after concussion. As in past wars, it has taken years to understand how PTSD and concussion/mTBI interrelate with other deployment-related health concerns, and the implications for designing effective evaluation and treatment strategies.

Veterans understandably may become angry at the suggestion that their postwar health concerns are stress-related or psychological, and thus it is necessary for primary care professionals to be sensitive to this concern.

PTSD

PTSD is the most common mental disorder documented following war-zone service. Studies from the conflicts in Iraq and Afghanistan have found PTSD prevalence rates of 2–6% before deployment (comparable to civilian general population samples) and rates of 6–20% postdeployment, depending primarily on the level of combat frequency and intensity. Many other veterans experience subclinical PTSD symptoms after war-zone service, sometimes termed posttraumatic stress (PTS) or combat stress. These subclinical symptoms can contribute to distress and affect health, even if overall functioning is not as impaired as in the full disorder.

PTSD is defined by the American Psychiatric Association as persistent (>1 month) symptoms occurring after a life-threatening traumatic event in which there was an immediate response of fear, helplessness, or horror. The symptoms must be associated with significant distress or impairment in social or occupational functioning. Symptoms are grouped into three categories: (1) re-experiencing symptoms in which the person has nightmares, flashbacks, or intrusive thoughts and memories connected with the traumatic event; (2) hyperarousal symptoms in which the person is physiologically revved up, hyperalert, startles easily, and experiences sleep disturbance, anger, and/or concentration problems; and (3) avoidance symptoms where the person loses interest in things that previously brought enjoyment, and avoids places, situations, or other stimuli that serve as reminders of the traumatic event (e.g., a crowded mall that triggers heightened alertness to threat). Additional symptoms, currently categorized with the avoidance cluster, but which will likely become a fourth category in future definitions of PTSD, include emotional numbing, feeling distant or cutoff from others, and a foreshortened sense of future (Chap. 54). While PTSD is a clinical symptom-based case definition, it is best to think of PTSD not as an emotional or psychological/psychiatric condition, but rather as a physiologic-based response to life-threatening trauma that is associated with physical, cognitive, emotional, and psychological symptoms.

PTSD has strong biologic correlates, based in fear-conditioning responses to threat and responses to extreme stress involving neuroendocrine dysregulation and ANS reactivity. Numerous studies have shown that PTSD is highly correlated with generalized physical and cognitive symptoms—including hypertension, chronic pain, and cardiovascular disease—as well as cell-mediated immune

dysfunction and shortened life expectancy. PTSD is frequently comorbid with other mental disorders such as major depressive disorder, generalized anxiety, substance use disorders (SUDs), as well as risky behaviors (e.g., aggression, accidents); it has been estimated that up to 80% of patients with PTSD exhibit one or more comorbid conditions. Misuse of alcohol or substances is most prevalent, often reflecting self-medication. PTSD is also associated with tolerance and withdrawal symptoms related to prescription pain and sleep medications, as well as nicotine dependence.

Clinicians should understand the limitations of the definition of PTSD when applied to responses to trauma occurring in the occupational context of military service (similar to police, firefighters, and other first responders). Service members are trained to respond to traumatic events, and relatively rarely report fear, helplessness, or horror, which are characteristic responses of civilian victims of trauma. In addition, the reactions that are labeled as symptoms of PTSD are based on the adaptive survival responses of warriors in a combat environment. For example, physiologic hyperarousal, use of anger, and being able to shut down other emotions are very useful skills in combat, and can be present prior to traumatic events when there is tough realistic training. These responses only become symptoms when they impair functioning after warriors return home.

CONCUSSION/mTBI

TBI (Chap. 36) gained increased recognition during the conflicts in Iraq and Afghanistan because of the widespread exposure of troops to improvised explosive devices. Contributing to heightened concern were high prevalence estimates of deployment-related TBI that did not distinguish concussion/mTBI from moderate or severe TBI; data from animal models of blast suggesting that explosions may cause a different kind of concussion associated with inflammatory changes; and speculation that repetitive blast exposure may lead to future dementia, based on case series of professional athletes (e.g., boxers, football players) exposed to highly repetitive injuries linked to chronic traumatic encephalopathy (previously termed *dementia pugilistica*). Many veterans of Iraq and Afghanistan reported experiencing multiple concussions during deployments, and many also reported ignoring concussions and not seeking treatment at the time of injury.

TBI includes closed and penetrating head injuries; closed head injuries are categorized as mild (mTBI or concussion), moderate, or severe based on the duration of loss of consciousness, duration of posttraumatic amnesia, and the Glasgow coma score (GCS) (Table 36-2). Several studies have estimated that 10–20% of all military personnel deployed to Iraq or Afghanistan sustained one or

more concussion/mTBI events during deployment, most commonly from exposure to blasts.

Although there is a neurophysiologic continuum of injury, there are stark clinical and epidemiologic distinctions between concussion/mTBI and moderate or severe TBI (Table 55-1). Concussion/mTBI is defined as a blow or jolt to the head that results in brief loss of consciousness (LOC) for <30 min (most commonly only a few seconds to minutes), posttraumatic amnesia (PTA) of <24 h (most commonly <1 h), or transient alteration in consciousness (AOC) without loss of consciousness. The majority of concussions in Iraq or Afghanistan have involved AOC without LOC or PTA (which soldiers may refer to as getting their “bell rung”). GCSs are usually normal (15 out of 15). Concussion is treated with rest to allow the brain time to heal, and almost never results in air-evacuation from the battlefield unless there are other injuries.

In contrast, moderate, severe, or penetrating TBI, which account for <1% of all battlefield head injuries in Iraq and Afghanistan, are characterized by LOC ≥30 min (up to permanent coma), PTA ≥24 h (also may be permanent), and GCSs as low as 3 (the minimum value). These virtually always result in air-evacuation from the battlefield and can result in severe long-term neurologic impairment and requirement for rehabilitative care.

Symptoms following concussion/mTBI can include headache; fatigue; concentration, memory, or attention problems; sleep disturbance; irritability; balance difficulties; and tinnitus, among other symptoms. Recovery is usually rapid, with symptoms usually resolving in a few hours to days, but in a small percentage of patients, symptoms may persist for a longer period or become chronic (referred to as persistent “postconcussive symptoms” or “PCS”).

Establishing a clear causal connection between a deployment concussion injury and persistent PCS months or years after return from deployment is difficult and often confounded by other postwar conditions that cause similar symptoms, including injuries not involving the head, other medical disorders, sleep disorders, PTSD, depression, substance use disorders, chronic pain, and the general physiologic effects of wartime service. Contributing to the difficulty in establishing causation is the fact that the concussion/mTBI case definition refers only to the acute injury event and lacks symptoms, time-course or impairment; case definitions for persistent postconcussion syndrome are not well validated. Several studies found that PTSD was a much stronger predictor of postdeployment PCS after combat deployment than concussions/mTBIs, and one study even found that objective neuropsychological impairment after deployment was entirely explained by PTSD. These data do not minimize the importance of

TABLE 55-1

COMPARISON BETWEEN CONCUSSION/mTBI AND MODERATE/SEVERE TBI

	MILD TBI (CONCUSSION)	MOD/SEVERE TBI
Clinical case definition		
Loss of consciousness	<30 min (usually few seconds to minutes)	≥30 min to indefinite
Altered consciousness	<24 h (usually <30 min)	≥24 h to indefinite
Posttraumatic amnesia	<24 h (usually <30 min)	≥24 h to indefinite
Glasgow coma score	13–15 (usually 15)	As low as 3
Focal neurologic signs	None or transient	Frequently present
Traditional neuroimaging (CT/MRI)	Usually negative	Diagnostic
Clinical usefulness of neurocognitive testing after acute injury period	Usually inconclusive	Essential and valuable
Neuronal cell damage	Metabolic/ionic processes associated with axonal swelling, which can lead to disconnection	Direct injury effects plus metabolic/ionic effects
Sequelae, natural history, and recovery	Full recovery expected in majority of individuals; no consensus on natural history; the percentage who develop persistent symptoms is debated	Based directly on injury characteristics; may be severely disabling
Predictors of persistent postconcussive symptoms or disability	Intensely debated; risk factors found to be most predictive include psychiatric conditions (e.g., depression, PTSD) and negative expectations	Not debated, predictors are directly related to injury severity and clinical progress with rehabilitation treatment

concussion/mTBI, but highlight the complex interrelationships of war-related health problems.

Studies of veterans who sustained concussions in Iraq or Afghanistan have suggested that blast mechanisms produce similar clinical outcomes as nonblast mechanisms, in contrast to expectations based on animal models. An explosion can produce serious injury from rapid atmospheric pressure changes (primary blast wave mechanism), as well as from flying debris (secondary blast mechanism) or by being thrown into a hard object (tertiary blast mechanism). Secondary and tertiary mechanisms are similar to other mechanical mechanisms of concussions sustained during accidents. The possibility of a unique head injury from the primary blast wave in otherwise uninjured soldiers appears to be low, but cannot be discounted.

Multisystem health problems that lack clear case definitions do not lend themselves well to uniform public health strategies such as screening. Nevertheless, mass population screening for concussion/mTBI was mandated for all U.S. service members returning from Iraq or Afghanistan and all veterans presenting for care at VA health care facilities. These screening processes, which attempt to apply the acute concussion case definition (lacking symptoms, time-course, or impairment) months or years after injury, led to sharp criticism that they were encouraging clinicians to misattribute common postwar symptoms to concussion/mTBI.

Management is largely symptom-focused, and ideally carried out within primary-care based structures of care. Optimal care avoids unnecessary specialty referrals, use of nonevidence-based interventions, or poor communication that results in negative expectations. Concussion research has shown that negative expectations are one of the most important risk factors for persistent symptoms.

While many questions remain regarding the long-term health effects of concussions (particularly multiple concussions) sustained during deployment, these are important battlefield injuries that require careful attention. However, they need to be addressed within the context of all other war-related health concerns.

STIGMA AND BARRIERS TO CARE

Adding to the complexity of treating veterans is stigma and other barriers to care. Despite extensive education efforts among military leaders and service members, perceptions of stigma have shown little change over the many years of war; warriors are concerned that they will be perceived as weak by peers or leaders if they seek care. Studies showed that less than one-half of service members with serious mental health problems receive needed care. Many factors contribute to this, including the pervasive nature of stigma in society in general (particularly among men), the critical importance of group

APPROACH TO THE PATIENT

Evaluation of Veterans with Neuropsychiatric Health Concerns

Evaluation should begin with a careful occupational history as part of the routine medical evaluation; this includes the number of years served, military occupation, deployment locations and dates, illnesses or injuries resulting from service, and significant combat traumatic experiences that may be continuing to affect the individual (Table 55-2). The clinician should evaluate the degree to which the patient’s current difficulties reflect the normal course of readjusting after the intense occupational experience of combat. It is helpful to reinforce the many strengths associated with being a professional in the military: courage, honor, service to country, resiliency in combat, leadership, ability to work as cohesive workgroup with peers, and demonstrated skills in handling extreme stress.

One of the challenges with current medical practice is that there may be multiple providers with different clinical perspectives. Care should be coordinated through the primary care clinician, with the assistance of a care manager if needed. It is particularly important to continually evaluate all medications prescribed by other practitioners and assess for possible long-term side effects, dependency, or drug-drug interactions. Particular attention should be given to the level of chronic pain and sleep disturbance,

self-medication with alcohol or substances, chronic use of nonsteroidal anti-inflammatory agents (which can contribute to rebound headaches or pain), chronic use of sedative-hypnotic agents, chronic use of narcotic pain medications, and the impact of war-related health concerns on social and occupational functioning.

Screening for PTSD, depression, and alcohol misuse should be performed routinely in all combat veterans. Three screening tools, which are in the public domain, have been validated for use in primary care, and have been used frequently in veterans: the four-question Primary Care PTSD Screen (PC-PTSD), the two-question Patient Health Questionnaire (PHQ-2), and the three-question Alcohol Use Disorders Identification Test-Consumption module (AUDIT-C) (Table 55-3).

Since the clinical definition of an acute concussion/mTBI does not include symptoms, time-course, or impairment, there is currently no clinically validated screening process for use months or years after injury. However, it is important to gather information about all injuries sustained during deployment, including any that resulted in loss or alteration of consciousness or loss of memory around the time of the event. If concussion injuries have occurred, the clinician should assess the number of such injuries, the duration of time unconscious, and injury mechanisms. This should be followed by an assessment of any postconcussive symptoms immediately following the injury event (e.g., headaches, dizziness, tinnitus, nausea, irritability, insomnia, and concentration or memory problems), and the severity and duration of such symptoms.

TABLE 55-2
SPECIFIC CONSIDERATIONS IN THE MEDICAL EVALUATION OF VETERANS

Occupational context of health concerns	Deployment locations and dates, combat experiences or other deployment stressors, frequent moves, separations from family, impact of deployment on civilian occupation (for reservists)
Medical problems during deployment	History of deployment-related injuries (including concussions), environmental exposures, sleep pattern during deployment, use of caffeine/energy drinks, use of other substances
Current medical history	Current symptoms, level of chronic pain, sleep problems, evidence of persistent physiologic hyperarousal (hypertension, tachycardia, panic symptoms, concentration/memory problems, irritability/anger, sleep disturbance), chronic use of caffeine or energy drinks, chronic use of nonsteroidal anti-inflammatory medications, chronic use of narcotic pain medications, chronic use of nonbenzodiazepine sedative-hypnotic medications, chronic use of benzodiazepines for sleep or anxiety
Mental health assessment	Screen for PTSD, major depressive disorder. Ask about suicidal or homicidal ideation, intent, or plans, as well as access to firearms
Alcohol/substance use	Screen for alcohol and substance use disorders, quantity and frequency of use, and evidence of tolerance. Inquire about “self-medication” (e.g., use of alcohol for sleep or to “calm down” or “forget” war-zone experiences)
Functional impairment	Impact of current symptoms on social and occupational functioning. High-risk behaviors (e.g., drinking and driving, reckless driving, aggression)
Social support, impact of military service on marriage and family	Level of social support. Readjustment stress on spouse, children, or other family members

TABLE 55-3

PRIMARY CARE MENTAL HEALTH SCREENING TOOLS

PC-PTSD Screen

1. Have you ever had any experience that was so frightening, horrible, or upsetting that, in the past month, you:

Have had nightmares about it or thought about it when you did not want to?	Yes	No
Tried hard not to think about it or went out of your way to avoid situations that remind you of it?	Yes	No
Were constantly on guard, watchful, or easily startled?	Yes	No
Felt numb or detached from others, activities, or your surroundings?	Yes	No

Note: Two or more “yes” responses (three or more a more specific cutoff) is considered a positive screen.

Source: A Prins et al: The Primary Care PTSD Screen (PC-PTSD): Development and operating characteristics. Prim Care Psychiatr 9:9, 2004.

PHQ-2 Depression Screen

2. Over the last 2 weeks, how often have you been bothered by any of the following problems?

	Not at all (0)	Few or several days (1)	More than half the days (2)	Nearly every day (3)
Little interest or pleasure in doing things.	0	1	2	3
Feeling down, depressed, or hopeless.	0	1	2	3

Note: If either (or both) questions are marked 2 or 3 (“more than half the days or higher), this is considered a positive screen for depression.

Source: K Kroenke et al: The Patient Health Questionnaire-2: Validity of a two-item depression screener. Med Care 41:1284, 2003.

AUDIT-C Alcohol Screen

3a. How often do you have a drink containing alcohol?

Never (0)	Monthly or less (1)	Two or four times a month (2)	Two to three times per week (3)	Four or more times a week (4)
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3b. How many drinks containing alcohol do you have on a typical day when you are drinking?

1 or 2 (0)	3 or 4 (1)	5 or 6 (2)	7 or 9 (3)	10 or more (4)
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3c. How often do you have six or more drinks on one occasion?

Never (0)	Less than Monthly (1)	Monthly (2)	Two to three times per week (3)	Four or more times a week (4)
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Note: A positive AUDIT-C screen is defined as a total score for men ≥ 4 ; for women ≥ 3 . A report of drinking 6 or more drinks on one occasion should prompt an in-depth assessment of drinking.

Source: K Bush et al: The AUDIT Alcohol Consumption Questions (AUDIT-C): An effective brief screening test for problem drinking. Arch Intern Med 158:1789, 1998.

TREATMENT

Neuropsychiatric Illnesses in War Veterans

Given the interrelationship of postwar health concerns, care needs to be carefully coordinated. Specific techniques that have been found to be helpful include scheduling regular primary care visits instead of as needed visits, establishing care management, utilizing good risk-communication principles, establishing a consultative step care approach that draws on the expertise of specialists in a collaborative manner (instead of immediately referring the patient to a specialist and relying on the specialist to provide care), and having behavioral health support directly within primary care clinics (both for referrals and to provide education and support to primary care professionals prescribing treatment for depression or PTSD).

It is important not to implicitly or explicitly convey the message that physical or cognitive symptoms are

psychological or due to stress. Even if depression or anxiety plays a large role in the etiology of physical health symptoms, the treatment approach should be designed within a patient-centered primary care structure, and referrals managed from within this framework. For example, it might help to explain that the primary goal of referral to a mental health professional is to improve sleep and reduce physiologic hyperarousal, which in turn will help with treatment of war-related chronic headaches, concentration problems, or chronic fatigue. If however, the primary care professional conveys the message that the cause of headaches or concentration problems is anxiety or depression, and this conflicts with the patient's own viewpoint, then this could damage therapeutic rapport and in turn exacerbate the symptoms.

Specific questions related to military service (Table 55-2) combined with screening for depression, PTSD, and alcohol use disorders (Table 55-3) should be a routine part of

care for all veterans. A positive screen for depression or PTSD should prompt follow-up questions related to these disorders (or use of a longer screening tool such as the [PHQ-9] or National Center for PTSD Checklist), as well as risk assessment for suicide or homicide. It is important to assess the impact of depression or PTSD symptoms on occupational functioning and interpersonal relationships.

A positive screen for alcohol misuse should prompt a brief motivational intervention that includes bringing attention to the elevated level of drinking, informing the veteran about the effects of alcohol on health, recommending limiting use or abstaining, exploring and setting goals related to drinking behavior, and follow-up and referral to specialty care if needed. This type of brief primary care intervention has been found to be effective, and should be incorporated into routine practice. One way to facilitate dialogue about this topic with veterans is to point out how hyperarousal associated with combat service can lead to increased craving for alcohol as the body searches for ways to modulate this. Veterans may consciously or unconsciously drink more to help with sleep, reduce arousal, or avoid thinking about events that happened “downrange.” A key educational strategy is to help the veteran to learn that drinking to get to sleep actually damages sleep architecture and makes sleep worse (e.g., reduces rapid eye movement [REM] sleep initially followed by rebound REM activity and early morning waking).

SPECIFIC TREATMENT STRATEGIES FOR PTSD AND COMORBID DEPRESSION PTSD and depression are highly comorbid in combat veterans, and the evidence-based treatments are similar, involving either antidepressant medications, cognitive behavioral therapy (CBT), or both. Psychoeducation that assists veterans to understand that their symptoms of PTSD have a basis in adaptive survival mechanisms and skills they exhibited in combat can facilitate therapeutic rapport. Remaining hypervigilant to threat, being able to shut down emotions, being able to function on less sleep, and using anger to help focus and control fear are all adaptive beneficial survival skills in a combat environment. Therefore, PTSD for warriors is both a medical disorder and a set of reactions that have their roots in the physiologic adaptation and skills they successfully applied in combat.

It is important to know that combat is not the only important trauma in a war-zone environment. Rape, assault, and accidents also occur. Rape or assault by a fellow service member, which affects a greater number of women veterans, can be particularly devastating because it destroys the vital feeling of safety that individuals derive from their own unit peers in a war environment.

The treatments for PTSD considered by most consensus guideline committees to have an A level of evidence

include CBTs and medications, specifically selective serotonin reuptake inhibitors (SSRIs) (of which paroxetine and sertraline received FDA approval for PTSD). Although lacking a specific FDA indication, there is increasing evidence that serotonin norepinephrine reuptake inhibitors (SNRIs) (e.g., venlafaxine and duloxetine) and mirtazapine are also effective. (See Table 54-3 for recommended dosages.)

CBT interventions include narrative therapy (often called “imaginal exposure”), in vivo exposure focused on retraining the body not to react to stimuli related to traumatic reminders (e.g., a crowded mall), and techniques to modulate physiologic hyperarousal (e.g., diaphragmatic breathing, progressive muscle relaxation). A number of complementary alternative medicine approaches including acupuncture, mindfulness meditation, yoga, and massage are also being tested in PTSD. Although not evidence-based treatments per se, if they facilitate a relaxation response and alleviation of hyperarousal symptoms, they can be considered useful adjunctive modalities.

There have been no head-to-head comparisons of medication compared with psychotherapy for treatment of PTSD. It is reasonable for primary care clinicians to consider initiating treatment for mild to moderate PTSD symptoms with an SSRI, and refer patients to a mental health professional if there are more severe symptoms, significant comorbidity, safety concerns, or limited response to initial treatment. All PTSD treatments are associated with a sizable proportion of individuals who fail to respond adequately, and it is often necessary to add modalities or switch treatment. SNRIs may be useful alternatives to SSRIs if there has been nonresponse, side effects, or if there is comorbid pain (duloxetine, in particular, has indications for pain). Both SSRIs and SNRIs can increase anxiety initially; patients should be warned about this possibility and treatment should be initiated with the lowest recommended dose (or even one-half of the lowest dose for a few days) and gradually increased thereafter. Mirtazapine use can cause drowsiness and weight gain. Antidepressants also are likely to be useful in comorbid depression, which is common in veterans with PTSD. All antidepressants have potential drug-drug interactions that must be considered.

Many other medications have been used in PTSD, including tricyclic antidepressants, benzodiazepines, atypical antipsychotics, and anticonvulsants. In general, these should be prescribed in conjunction with psychiatric consultation, because of their greater side effects and risks. Benzodiazepines, in particular, should be avoided in combat veterans. Studies have shown that they do not reduce core PTSD symptoms, are likely to exacerbate substance use disorders that are common in veterans with PTSD, and may produce significant rebound anxiety. Veterans with PTSD often report symptomatic relief

upon initiation of a benzodiazepine, but this is generally short lived and associated with a high risk of tolerance and dependence that can worsen recovery. Atypical antipsychotics, which have gained widespread popularity as adjunctive treatment for depression, anxiety, or sleep problems, have significant long-term side effects, including metabolic effects (e.g., glucose dysregulation), weight gain, and cardiovascular risks.

Sleep disturbance should be addressed initially with sleep hygiene education, followed by consideration of an antihistamine, trazodone, or non-benzodiazepine sedative-hypnotic such as zolpidem, eszopiclone, or zaleplon. However, the non-benzodiazepine sedative-hypnotics should be used with caution in veterans as they can lead to tolerance and rebound sleep problems similar to those seen with benzodiazepine use.

TREATMENT STRATEGIES FOR CONCUSSION/mTBI AND POSTDEPLOYMENT POST-CONCUSSIVE SYMPTOMS Concussion/mTBI is best treated at the time of injury with education and rest to allow time for the brain to heal and protect against a second impact syndrome (a rare but life-threatening event involving brain swelling that can occur when a second concussion occurs before the brain has adequately healed from an initial event). Randomized trials have shown that education regarding concussion that informs the patient of what to expect and promotes the expectation of recovery is the most effective treatment in preventing persistent symptoms.

Once service members return from deployment and seek care for postwar health problems, treatment is largely symptom focused, following the principles of patient-centered and collaborative care models. Cognitive rehabilitation, which is very useful in moderate and severe TBI to improve memory, attention, and concentration, has

generally not been shown to be effective for mTBI in randomized clinical studies, though consensus groups have supported its use.

General recommendations for the clinical management of persistent, chronic postconcussive symptoms include treating physical and cognitive health problems based on symptom presentation, coexisting health problems, and individual preferences; and addressing coexisting depression, PTSD, substance use disorders, or other factors that may be contributing to symptom persistence. Headache is the most common symptom associated with concussion/mTBI, and the evaluation and treatment of headache parallels those for other causes of headache (Chap. 8). Stimulant medications for alleviating neurocognitive effects attributed to concussion/mTBI are not recommended. Clinicians should be aware of the potential for cognitive or sedative side effects of certain medications that may be prescribed for depression, anxiety, sleep, or chronic pain.

Treatment of neuropsychiatric problems must be coordinated with care for other war-related health concerns, with the goal of treatment to reduce the severity of symptoms, improve social and occupational functioning, and prevent long-term disability. Understanding the occupational context of war-related health concerns is important in communicating with veterans and developing a comprehensive treatment strategy.

DISCLOSURE

This material has been reviewed by the Walter Reed Army Institute of Research. There is no objection to its presentation and/or publication. The opinions or assertions contained herein are the private views of the author and are not to be construed as official, or as reflecting true views of the Department of the Army or the Department of Defense.

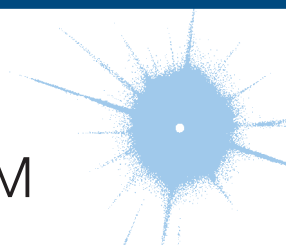
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SECTION VI

ALCOHOLISM AND DRUG DEPENDENCY

CHAPTER 56

ALCOHOL AND ALCOHOLISM



Marc A. Schuckit

INTRODUCTION

Alcohol (beverage ethanol) distributes throughout the body, affecting almost all systems and altering nearly every neurochemical process in the brain. This drug is likely to exacerbate most medical conditions, affect almost any medication metabolized in the liver, and temporarily mimic many medical (e.g., diabetes) and psychiatric (e.g., depression) conditions. Because ~80% of people in Western countries have consumed alcohol, and two-thirds have been drunk in the prior year, the lifetime risk for serious, repetitive alcohol problems is almost 20% for men and 10% for women, regardless of a person's education or income. While low doses of alcohol have some healthful benefits, the intake of more than three standard drinks per day on a regular basis enhances the risk for cancer and vascular disease, and alcohol use disorders decrease the life span by about 10 years. Unfortunately, most clinicians have had only limited education regarding these conditions. This chapter presents a brief overview of clinically useful information about alcohol use, abuse, and dependence.

PHARMACOLOGY AND NUTRITIONAL IMPACT OF ETHANOL

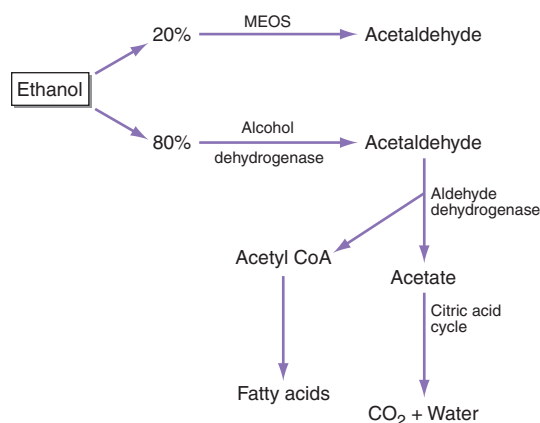
Blood levels of ethanol are expressed as milligrams or grams of ethanol per deciliter (e.g., 100 mg/dL = 0.10 g/dL), with values of ~0.02 g/dL resulting from the ingestion of one typical drink. In round figures, 340 mL (12 oz) of beer, 115 mL (4 oz) of nonfortified wine, and 43 mL (1.5 oz) (a shot) of 80-proof beverage such as whisky, gin, or vodka each contain ~10–15 g of ethanol and represent a standard drink; 0.5 L (1 pint) of 80-proof beverage contains ~160 g (about 16 standard drinks), and 750 mL of wine contains ~60 g of ethanol. These beverages also have additional components

known as *congeners* that affect the drink's taste and might contribute to adverse effects on the body. Congeners include methanol, butanol, acetaldehyde, histamine, tannins, iron, and lead. Alcohol acutely decreases neuronal activity and has similar behavioral effects and cross-tolerance with other depressants, including benzodiazepines and barbiturates.

Alcohol is absorbed from mucous membranes of the mouth and esophagus (in small amounts), from the stomach and large bowel (in modest amounts), and from the proximal portion of the small intestine (the major site). The rate of absorption is increased by rapid gastric emptying (as can be induced by carbonated beverages); by the absence of proteins, fats, or carbohydrates (which interfere with absorption); and by dilution to a modest percentage of ethanol (maximum at ~20% by volume).

Between 2% (at low blood alcohol concentrations) and 10% (at high blood alcohol concentrations) of ethanol is excreted directly through the lungs, urine, or sweat, but the greater part is metabolized to acetaldehyde, primarily in the liver. The most important pathway occurs in the cell cytosol where alcohol dehydrogenase (ADH) produces acetaldehyde, which is then rapidly destroyed by aldehyde dehydrogenase (ALDH) in the cytosol and mitochondria (**Fig. 56-1**). A second pathway in the microsomes of the smooth endoplasmic reticulum (the microsomal ethanol-oxidizing system, or MEOS) is responsible for ≥10% of ethanol oxidation at high blood alcohol concentrations.

While alcohol supplies calories (a drink contains ~300 kJ, or 70–100 kcal), these are devoid of nutrients such as minerals, proteins, and vitamins. In addition, alcohol can also interfere with absorption of vitamins in the small intestine and decreases their storage in the liver with modest effects on folate (folacin or folic acid), pyridoxine (B₆), thiamine (B₁), nicotinic acid (niacin, B₃), and vitamin A.

**FIGURE 56-1**

The metabolism of alcohol. MEOS, microsomal ethanol-oxidizing system.

A heavy ethanol load in a fasting, healthy individual is likely to produce transient hypoglycemia within 6–36 h, secondary to the acute actions of ethanol on gluconeogenesis. This can result in temporary abnormal glucose tolerance tests (with a resulting erroneous diagnosis of diabetes mellitus) until the alcoholic has abstained for 2–4 weeks. Alcohol ketoacidosis, probably reflecting a decrease in fatty acid oxidation coupled with poor diet or recurrent vomiting, can be misdiagnosed as diabetic ketoacidosis. With the former, patients show an increase in serum ketones along with a mild increase in glucose but a large anion gap, a mild to moderate increase in serum lactate, and a β -hydroxybutyrate/lactate ratio of between 2:1 and 9:1 (with normal being 1:1).

In the brain, alcohol affects almost all neurotransmitter systems, with acute actions that are often the opposite of those seen following desistance after a period of heavy drinking. The most prominent actions relate to boosting gamma aminobutyric acid (GABA) activity, especially in GABA_A receptors. Enhancement of this complex chloride channel system contributes to anticonvulsant, sleep-inducing, antianxiety, and muscle relaxation effects of all GABA-boosting drugs. Acutely administered alcohol produces a release of GABA, and continued use of this drug increases density of GABA_A receptors, while alcohol withdrawal states are characterized by decreases in GABA-related activity. Equally important is the ability of acute alcohol to inhibit postsynaptic *N*-methyl-D-aspartate (NMDA) excitatory glutamate receptors, while chronic drinking and desistance are associated with an upregulation of these excitatory receptor subunits. The relationships between greater GABA and diminished NMDA receptor activity during acute intoxication and diminished GABA with enhanced NMDA actions during alcohol withdrawal explain much of intoxication and withdrawal phenomena.

As with all pleasurable activities, drinking alcohol acutely increases dopamine levels in the brain, especially

in the ventral tegmentum and related brain regions, and this effect plays an important role in continued alcohol use, craving, and relapse. The changes in dopamine pathways are also linked to increases in “stress hormones,” including cortisol and adrenocorticotrophic hormone (ACTH) during intoxication and decreases in these hormones during withdrawal. Such alterations are likely to contribute to both feelings of reward during intoxication and depression during falling blood alcohol concentrations. Also closely linked to alterations in dopamine (especially in the nucleus accumbens) are alcohol-induced changes in opioid receptors, with acute alcohol also causing release of beta endorphins.

Additional important neurochemical changes include increases in synaptic levels of serotonin during acute intoxication, and subsequent upregulation of serotonin receptors. Acute increases in nicotinic acetylcholine systems also contribute to the impact of alcohol in the ventral tegmental region, which occur in concert with enhanced dopamine activity. In the same regions, alcohol impacts on cannabinol receptors, with resulting release of dopamine, GABA, and glutamate as well as subsequent effects on brain reward circuits.

BEHAVIORAL EFFECTS, TOLERANCE, AND DEPENDENCE

The acute effects of a drug depend on the dose, the rate of increase in plasma, the concomitant presence of other drugs, and the past experience with the agent. “Legal intoxication” with alcohol in most states requires a blood alcohol concentration of 0.08 g/dL, while levels of 0.04 or even lower are cited in other countries. However, behavioral, psychomotor, and cognitive changes are seen at levels as low as 0.02–0.03 g/dL (i.e., after one to two drinks) (Table 56-1). Deep but disturbed sleep can be seen at twice the legal intoxication level, and death can occur with levels between 0.30 and

TABLE 56-1

EFFECTS OF BLOOD ALCOHOL LEVELS IN THE ABSENCE OF TOLERANCE

BLOOD LEVEL, g/dL	USUAL EFFECT
0.02	Decreased inhibitions, a slight feeling of intoxication
0.08	Decrease in complex cognitive functions and motor performance
0.20	Obvious slurred speech, motor incoordination, irritability, and poor judgment
0.30	Light coma and depressed vital signs
0.40	Death

0.40 g/dL. Beverage alcohol is probably responsible for more overdose deaths than any other drug.

Repeated use of alcohol contributes to acquired tolerance, a complex phenomenon involving at least three types of compensatory mechanisms. (1) After 1–2 weeks of daily drinking, *metabolic or pharmacokinetic tolerance* can be seen, with up to 30% increase in the rate of hepatic ethanol metabolism. This alteration disappears almost as rapidly as it develops. (2) *Cellular or pharmacodynamic tolerance* develops through neurochemical changes that maintain relatively normal physiologic functioning despite the presence of alcohol. Subsequent decreases in blood levels contribute to symptoms of withdrawal. (3) Individuals learn to adapt their behavior so that they can function better than expected under influence of the drug (*learned or behavioral tolerance*).

The cellular changes caused by chronic ethanol exposure may not resolve for several weeks or longer following cessation of drinking. Rapid decreases in blood alcohol levels before that time can result in a withdrawal syndrome, which is most intense during the first 5 days, but some symptoms (e.g., disturbed sleep and anxiety) can take up to 4–6 months to resolve.

THE EFFECTS OF ETHANOL ON ORGAN SYSTEMS

Relatively low doses of alcohol (one or two drinks per day) have potential beneficial effects of increasing high-density lipoprotein cholesterol and decreasing aggregation of platelets, with a resulting decrease in risk for occlusive coronary disease and embolic strokes. Red wine has additional potential health-promoting qualities at relatively low doses due to flavinols and related substances, which may work by inhibiting platelet activation. Modest drinking might also decrease the risk for vascular dementia and, possibly, Alzheimer's disease. However, any potential healthful effects disappear with the regular consumption of three or more drinks per day, and knowledge about the deleterious effects of alcohol can both help the physician to identify patients with alcohol abuse and dependence, and to supply them with information that might help motivate a change in behavior.

NERVOUS SYSTEM

Approximately 35% of drinkers (and a much higher proportion of alcoholics) experience a *blackout*, an episode of temporary anterograde amnesia, in which the person forgets all or part of what occurred during a drinking evening. Another common problem, one seen after as few as one or two drinks shortly before bedtime, is disturbed sleep. Although alcohol might initially help a person to fall asleep, it disrupts sleep throughout the

rest of the night. The stages of sleep are also altered, and time spent in rapid eye movement (REM) and deep sleep is reduced. Alcohol relaxes muscles in the pharynx, which can cause snoring and exacerbate sleep apnea; symptoms of the latter occur in 75% of alcoholic men older than age 60 years. Patients may also experience prominent and sometimes disturbing dreams. All of these sleep problems are more pronounced in alcoholics, and their persistence may contribute to relapse.

Another common consequence of alcohol use is impaired judgment and coordination, increasing the risk of accidents and injury. In the United States, ~40% of drinkers have at some time driven while intoxicated. Heavy drinking can also be associated with headache, thirst, nausea, vomiting, and fatigue the following day, a hangover syndrome that is responsible for much missed time and temporary cognitive deficits at work and school.

The effect of alcohol on the *nervous system* is even more pronounced among alcohol-dependent individuals. Chronic high doses cause *peripheral neuropathy* in ~10% of alcoholics: similar to diabetes, patients experience bilateral limb numbness, tingling, and paresthesias, all of which are more pronounced distally. Approximately 1% of alcoholics develop *cerebellar degeneration or atrophy*. This is a syndrome of progressive unsteady stance and gait often accompanied by mild nystagmus; neuroimaging studies reveal atrophy of the cerebellar vermis. Fortunately, very few alcoholics (perhaps as few as 1 in 500 for the full syndrome) develop *Wernicke's* (ophthalmoparesis, ataxia, and encephalopathy) and *Korsakoff's* (retrograde and anterograde amnesia) *syndromes*, although a higher proportion have one or more neuropathologic findings related to these syndromes. These occur as the result of low levels of thiamine, especially in predisposed individuals, e.g., those with transketolase deficiency. Alcoholics can manifest *cognitive problems* and temporary memory impairment lasting for weeks to months after drinking very heavily for days or weeks. Brain atrophy, evident as ventricular enlargement and widened cortical sulci on MRI and CT scans, occurs in ~50% of chronic alcoholics; these changes are usually reversible if abstinence is maintained. There is no single alcoholic dementia syndrome; rather, this label is used to describe patients who have apparently irreversible cognitive changes (possibly from diverse causes) in the context of chronic alcoholism.

Psychiatric comorbidity

As many as two-thirds of alcohol-dependent individuals meet the criteria for a psychiatric syndrome in the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-IV) of the American Psychiatric Association (Chap. 54). Half of these relate to a preexisting antisocial personality manifesting as impulsivity and disinhibition that contribute to both alcohol

and drug dependence. The lifetime risk is 3% in males, and $\geq 80\%$ of such individuals demonstrate alcohol and/or drug dependence. Another common comorbidity occurs with dependence on illicit substances. The remainder of alcoholics with psychiatric syndromes have preexisting conditions such as schizophrenia or manic-depressive disease and anxiety disorders such as panic disorder. The comorbidities of alcoholism with independent psychiatric disorders might represent an overlap in genetic vulnerabilities, impaired judgment in the use of alcohol from the independent psychiatric condition, or an attempt to use alcohol to alleviate some of the symptoms of the disorder or side effects of medications.

Many psychiatric syndromes can be seen *temporarily* during heavy drinking and subsequent withdrawal. These alcohol-induced conditions include an intense *sadness* lasting for days to weeks in the midst of heavy drinking seen in 40% of alcoholics, which tends to disappear over several weeks of abstinence (alcohol-induced mood disorder); temporary severe *anxiety* in 10–30% of alcoholics, often beginning during alcohol withdrawal, which can persist for a month or more after cessation of drinking (alcohol-induced anxiety disorder); and auditory *hallucinations* and/or paranoid delusions in a person who is alert and oriented, seen in 3–5% of alcoholics (*alcohol-induced psychotic disorder*).

Treatment of all forms of alcohol-induced psychopathology includes helping patients achieve abstinence and offering supportive care, as well as reassurance and “talk therapy” such as cognitive-behavioral approaches. However, with the exception of short-term antipsychotics or similar drugs for substance-induced psychoses, substance-induced psychiatric conditions only rarely require medications. Recovery is likely within several days to 4 weeks of abstinence. Conversely, because alcohol-induced conditions are temporary and do not indicate a need for long-term pharmacotherapy, a history of alcohol intake is an important part of the workup for any patient with one of these psychiatric symptoms.

THE GASTROINTESTINAL SYSTEM

Esophagus and stomach

Alcohol intake can result in inflammation of the esophagus and stomach causing epigastric distress and gastrointestinal bleeding, making alcohol one of the most common causes of hemorrhagic gastritis. Violent vomiting can produce severe bleeding through a Mallory-Weiss lesion, a longitudinal tear in the mucosa at the gastroesophageal junction.

Pancreas and liver

The incidence of acute pancreatitis (~ 25 per 1000 per year) is almost threefold higher in alcoholics than in the

general population, accounting for an estimated 10% or more of the total cases. Alcohol impairs gluconeogenesis in the liver, resulting in a fall in the amount of glucose produced from glycogen, increased lactate production, and decreased oxidation of fatty acids. This contributes to an increase in fat accumulation in liver cells. In healthy individuals these changes are reversible, but with repeated exposure to ethanol, especially daily heavy drinking, more severe changes in the liver occur, including alcohol-induced hepatitis, perivenular sclerosis, and cirrhosis, with the latter observed in an estimated 15% of alcoholics. Perhaps through an enhanced vulnerability to infections, alcoholics have an elevated rate of hepatitis C, and drinking in the context of that disease is associated with more severe liver deterioration.

CANCER

As few as 1.5 drinks per day increases a woman's risk of breast cancer 1.4-fold. For both genders, four drinks per day increases the risk for oral and esophageal cancers approximately threefold and rectal cancers by a factor of 1.5; seven to eight or more drinks per day enhances approximately fivefold the risks for many cancers. These consequences may result directly from cancer-promoting effects of alcohol and acetaldehyde or indirectly by interfering with immune homeostasis.

HEMATOPOIETIC SYSTEM

Ethanol causes an increase in red blood cell size (mean corpuscular volume [MCV]), which reflects its effects on stem cells. If heavy drinking is accompanied by folic acid deficiency, there can also be hypersegmented neutrophils, reticulocytopenia, and a hyperplastic bone marrow; if malnutrition is present, sideroblastic changes can be observed. Chronic heavy drinking can decrease production of white blood cells, decrease granulocyte mobility and adherence, and impair delayed-hypersensitivity responses to novel antigens (with a possible false-negative tuberculin skin test). Associated immune deficiencies can contribute to vulnerability toward infections, including hepatitis and HIV, and interfere with their treatment. Finally, many alcoholics have mild thrombocytopenia, which usually resolves within a week of abstinence unless there is hepatic cirrhosis or congestive splenomegaly.

CARDIOVASCULAR SYSTEM

Acutely, ethanol decreases myocardial contractility and causes peripheral vasodilation, with a resulting mild decrease in blood pressure and a compensatory increase in cardiac output. Exercise-induced increases in cardiac oxygen consumption are higher after alcohol intake.

These acute effects have little clinical significance for the average healthy drinker but can be problematic when persisting cardiac disease is present.

The consumption of three or more drinks per day results in a dose-dependent increase in blood pressure, which returns to normal within weeks of abstinence. Thus, heavy drinking is an important factor in mild to moderate hypertension. Chronic heavy drinkers also have a sixfold increased risk for coronary artery disease, related, in part, to increased low-density lipoprotein cholesterol, and carry an increased risk for cardiomyopathy through direct effects of alcohol on heart muscle. Symptoms of the latter include unexplained arrhythmias in the presence of left ventricular impairment, heart failure, hypocontractility of heart muscle, and dilation of all four heart chambers with associated mural thrombi and mitral valve regurgitation. Atrial or ventricular arrhythmias, especially paroxysmal tachycardia, can also occur temporarily after heavy drinking in individuals showing no other evidence of heart disease—a syndrome known as the “holiday heart.”

GENITOURINARY SYSTEM CHANGES, SEXUAL FUNCTIONING, AND FETAL DEVELOPMENT

Drinking in adolescence can affect normal sexual development and reproductive onset. At any age, modest ethanol doses (e.g., blood alcohol concentrations of 0.06 g/dL) can increase sexual drive but also decrease erectile capacity in men. Even in the absence of liver impairment, a significant minority of chronic alcoholic men show irreversible testicular atrophy with shrinkage of the seminiferous tubules, decreases in ejaculate volume, and a lower sperm count.

The repeated ingestion of high doses of ethanol by women can result in amenorrhea, a decrease in ovarian size, absence of corpora lutea with associated infertility, and an increased risk of spontaneous abortion. Heavy drinking during pregnancy results in the rapid placental transfer of both ethanol and acetaldehyde, which may have serious consequences for fetal development. One severe result is the *fetal alcohol syndrome* (FAS), seen in ~5% of children born to heavy-drinking mothers, which can include any of the following: facial changes with epicanthal eye folds; poorly formed ear concha; small teeth with faulty enamel; cardiac atrial or ventricular septal defects; an aberrant palmar crease and limitation in joint movement; and microcephaly with mental retardation. A less severe condition is the *fetal alcohol spectrum disorder* (FASD), which can include low birth weight, a lower IQ, hyperactive behavior, and some modest cognitive deficits. The amount of ethanol required and the time of vulnerability during pregnancy have not been defined, making it advisable for pregnant women to abstain completely.

OTHER EFFECTS

Between one-half and two-thirds of alcoholics have skeletal muscle weakness caused by acute *alcoholic myopathy*, a condition that improves but which might not fully remit with abstinence. Effects of repeated heavy drinking on the *skeletal system* include changes in calcium metabolism, lower bone density, and decreased growth in the epiphyses, leading to an increased risk for fractures and osteonecrosis of the femoral head. *Hormonal changes* include an increase in cortisol levels, which can remain elevated during heavy drinking; inhibition of vasopressin secretion at rising blood alcohol concentrations and enhanced secretion at falling blood alcohol concentrations (with the final result that most alcoholics are likely to be slightly overhydrated); a modest and reversible decrease in serum thyroxine (T₄); and a more marked decrease in serum triiodothyronine (T₃). Hormone irregularities should be reevaluated as they may disappear after a month of abstinence.

ALCOHOLISM (ALCOHOL ABUSE OR DEPENDENCE)

Because many drinkers occasionally imbibe to excess, temporary alcohol-related pathology is common in nonalcoholics, especially in the late teens to the late twenties. When repeated problems in multiple life areas develop, the individual is likely to meet criteria for alcohol abuse or dependence.

DEFINITIONS AND EPIDEMIOLOGY

Alcohol dependence is defined in DSM-IV as repeated alcohol-related difficulties in at least three of seven life areas that cluster together at about the same time (e.g., over the same 12-month period). Two of these seven items, tolerance and withdrawal, may have special importance as they are associated with a more severe clinical course. Dependence predicts a course of recurrent problems with the use of alcohol and the consequent shortening of the life span by a decade.

Alcohol abuse is defined as repetitive problems with alcohol in any one of four life areas—social, interpersonal, legal, and occupational—or repeated use in hazardous situations such as driving while intoxicated in an individual who is not alcohol dependent. About 50% of those with alcohol abuse continue to have alcohol problems 2–5 years later, but only ~10% of these patients—including adolescents—go on to develop alcohol dependence.

The lifetime risk for alcohol dependence in most Western countries is about 10–15% for men and 5–8% for women. Rates are generally similar in the United

States, Canada, Germany, Australia, and the United Kingdom; tend to be lower in most Mediterranean countries, such as Italy, Greece, and Israel; and may be higher in Ireland, France, and Scandinavia. An even higher lifetime prevalence has been reported for most native cultures, including American Indians, Eskimos, Maori groups, and aboriginal tribes of Australia. These differences reflect both cultural and genetic influences, as described later. In Western countries, the typical alcoholic is more often a blue- or white-collar worker or homemaker. The lifetime risk for alcoholism among physicians is similar to that of the general population.

GENETICS

Approximately 60% of the risk for alcohol use disorders is attributed to genes, as indicated by the fourfold higher risk for alcohol abuse and dependence in children of alcoholics (even if these children were adopted early in life and raised by nonalcoholics) and a higher risk in identical twins as compared to fraternal twins of alcoholics. The genetic variations appear to operate primarily through intermediate characteristics that subsequently relate to the environment in altering the risk for heavy drinking and alcohol problems. These include genes relating to a high risk for all substance use disorders that operate through impulsivity, schizophrenia, and bipolar disorder. Another characteristic, an intense flushing response when drinking, decreases the risk for only alcohol use disorders through gene variations for several alcohol-metabolizing enzymes, especially aldehyde dehydrogenase (a mutation only seen in Asians), and to a lesser extent, variations in alcohol dehydrogenase.

An additional genetically influenced characteristic, a low sensitivity to alcohol, affects the risk for heavy drinking and may operate, in part, through variations in genes relating to potassium channels, GABA, nicotinic, and serotonin systems. A low response per drink is observed early in the drinking career and before alcohol use disorders have developed. All follow-up studies have demonstrated that this need for higher doses of alcohol to achieve desired effects predicts future heavy drinking, alcohol problems, and alcohol use disorders. The impact of a low response to alcohol on adverse drinking outcomes is mediated, at least in part, by a range of environmental influences, including the selection of heavier-drinking friends, the development of more positive expectations of the effects of high doses of alcohol, and suboptimal ways of coping with stress.

NATURAL HISTORY

Although the age of the first drink (~15 years) is similar in most alcoholics and nonalcoholics, a slightly earlier onset of regular drinking and drunkenness, especially

in the context of conduct problems, is associated with a higher risk for later alcohol use disorders. By the early to midtwenties, most nonalcoholic men and women moderate their drinking (perhaps learning from more minor problems), whereas alcoholics are likely to escalate their patterns of drinking despite difficulties. The first major life problem from alcohol often appears in the late teens to early twenties, and a pattern of multiple alcohol difficulties by the midtwenties. Once established, the course of alcoholism is likely to be one of exacerbations and remissions, with little difficulty in temporarily stopping or controlling alcohol use when problems develop, but without help, desistance usually gives way to escalations in alcohol intake and subsequent problems. Following treatment, between half and two-thirds of alcoholics maintain abstinence for years, and often permanently. Even without formal treatment or self-help groups there is also at least a 20% chance of spontaneous remission with long-term abstinence. However, should the alcoholic continue to drink, the life span is shortened by ~10 years on average, with the leading causes of death, heart disease, cancer, accidents, and suicide.

TREATMENT

The approach to treating alcohol-related conditions is relatively straightforward: (1) recognize that at least 20% of all patients have alcohol abuse or dependence; (2) learn how to identify and treat acute alcohol-related conditions; (3) know how to help patients begin to address their alcohol problems; and (4) know enough about treating alcoholism to appropriately refer patients for additional help.

IDENTIFICATION OF THE ALCOHOLIC

Even in affluent locales, ~20% of patients have an alcohol use disorder. These men and women can be identified by asking questions about *alcohol problems* and noting laboratory test results that are likely to be abnormal in the context of regular consumption of six to eight or more drinks per day. The two blood tests with ≥60% sensitivity and specificity for heavy alcohol consumption are γ -glutamyl transferase (GGT) (>35 U) and carbohydrate-deficient transferrin (CDT) (>20 U/L or >2.6%); the combination of the two is likely to be more accurate than either alone. The values for these serologic markers are likely to return toward normal within several weeks of abstinence. Other useful blood tests include high-normal MCVs ($\geq 91 \mu\text{m}^3$) and serum uric acid (>416 mol/L, or 7 mg/dL).

The diagnosis of alcohol abuse or dependence ultimately rests on the documentation of a pattern of repeated difficulties associated with alcohol use. Thus,

in screening it is important to probe for marital or job problems, legal difficulties, histories of accidents, medical problems, evidence of tolerance, etc., and then attempt to tie in use of alcohol or another substance. Some standardized questionnaires can be helpful, including the 10-item Alcohol Use Disorders Identification Test (AUDIT) (Table 56-2), but these are only screening tools, and a face-to-face interview is still required for a meaningful diagnosis.

TABLE 56-2
THE ALCOHOL USE DISORDERS IDENTIFICATION TEST (AUDIT)^a

ITEM	5-POINT SCALE (LEAST TO MOST)
1. How often do you have a drink containing alcohol?	Never (0) to 4+ per week (4)
2. How many drinks containing alcohol do you have on a typical day?	1 or 2 (0) to 10+ (4)
3. How often do you have six or more drinks on one occasion?	Never (0) to daily or almost daily (4)
4. How often during the last year have you found that you were not able to stop drinking once you had started?	Never (0) to daily or almost daily (4)
5. How often during the last year have you failed to do what was normally expected from you because of drinking?	Never (0) to daily or almost daily (4)
6. How often during the last year have you needed a first drink in the morning to get yourself going after a heavy drinking session?	Never (0) to daily or almost daily (4)
7. How often during the last year have you had a feeling of guilt or remorse after drinking?	Never (0) to daily or almost daily (4)
8. How often during the last year have you been unable to remember what happened the night before because you had been drinking?	Never (0) to daily or almost daily (4)
9. Have you or someone else been injured as a result of your drinking?	No (0) to yes, during the last year (4)
10. Has a relative, friend, doctor or other health worker been concerned about your drinking or suggested that you should cut down?	No (0) to yes, during the last year (4)

^aThe AUDIT is scored by summing the 10 values. A score >8 may indicate harmful alcohol use.
Source: Adapted from DF Reinert, GP Allen: *Alcoholism: Clinical & Experimental Research* 26:272, 2002, and from MA Schuckit: *Drug and Alcohol Abuse: A Clinical Guide to Diagnosis and Treatment*, 6th ed. New York, Springer 2006.

TREATMENT Alcohol-Related Conditions

Acute Intoxication The first priority in treating severe intoxication is to assess vital signs and manage respiratory depression, cardiac arrhythmia, or blood pressure instability, if present. The possibility of intoxication with other drugs should be considered by obtaining toxicology screens for opioids or other CNS depressants such as benzodiazepines. Aggressive behavior should be handled by offering reassurance but also by considering the possibility of a show of force with an intervention team. If the aggressive behavior continues, relatively low doses of a short-acting benzodiazepine such as lorazepam (e.g., 1–2 mg PO or IV) may be used and can be repeated as needed, but care must be taken not to destabilize vital signs or worsen confusion. An alternative approach is to use an antipsychotic medication (e.g., 0.5–5 mg of haloperidol PO or IM every 4–8 h as needed, or olanzapine 2.5–10 mg IM repeated at 2 and 6 h, if needed).

Intervention There are two main elements to intervention in a person with alcoholism: motivational interviewing and brief interventions. During motivational interviewing, the clinician helps the patient to think through the assets (e.g., comfort in social situations) and liabilities (e.g., health and interpersonal related problems) of the current pattern of drinking. The patient’s responses are key, and the clinician should listen empathetically, helping to weigh options and encouraging the patient to take responsibility for changes that need to be made. Patients should be reminded that only they can decide to avoid the consequences that will occur without changes in drinking. The process of motivational interviewing has been summarized by the acronym FRAMES: Feedback to the patient; Responsibility to be taken by the patient; Advice, rather than orders, on what needs to be done; Menus of options that might be considered; Empathy for understanding of the patient’s thoughts and feelings; and Self-efficacy, i.e., offering support for the capacity of the patient to succeed in making changes.

Once the patient begins to consider change, the emphasis shifts to brief interventions designed to help the patient understand more about potential action. Discussions focus on consequences of high alcohol consumption, suggested approaches to stopping drinking, and help in recognizing and avoiding situations likely to lead to heavy drinking. Both motivational interviewing and brief interventions can be carried out in 15-min sessions, but because patients do not always change behavior right away, multiple meetings are often required to explain the problem, discuss optimal treatments, and explain the benefits of abstinence.

Alcohol Withdrawal If the patient agrees to stop drinking, sudden decreases in alcohol intake can produce withdrawal symptoms, many of which are the opposite of those produced by intoxication. Features include tremor of the hands (shakes); agitation and anxiety; autonomic nervous system overactivity including an increase in pulse, respiratory rate, and body temperature; and insomnia. These symptoms usually begin within 5–10 h of decreasing ethanol intake, peak on day 2 or 3, and improve by day 4 or 5, although mild levels of these problems may persist for 4–6 months as a protracted abstinence syndrome.

About 2–5% of alcoholics experience a withdrawal seizure, with the risk increasing in the context of concomitant medical problems, misuse of additional drugs, and higher alcohol quantities. The same risk factors also contribute to a similar rate of *delirium tremens* (DTs), where the withdrawal includes delirium (mental confusion, agitation, and fluctuating levels of consciousness) associated with a tremor and autonomic overactivity (e.g., marked increases in pulse, blood pressure, and respirations). The risks for seizures and DTs can be diminished by identifying and treating any underlying medical conditions early in the course of withdrawal.

The first step in treating withdrawal is to perform a thorough physical examination in all alcoholics who are considering stopping drinking, including a search for evidence of liver failure, gastrointestinal bleeding, cardiac arrhythmia, infection, and glucose or electrolyte imbalance. It is also important to offer adequate nutrition and oral multiple B vitamins, including 50–100 mg of thiamine daily for a week or more. Because most alcoholics who enter withdrawal are either normally hydrated or mildly overhydrated, IV fluids should be avoided unless there is a relevant medical problem or significant recent bleeding, vomiting, or diarrhea.

The next step is to recognize that because withdrawal symptoms reflect the rapid removal of a CNS depressant, alcohol, the symptoms can be controlled by administering any depressant in doses that decrease the agitation and then gradually tapering the dose over 3–5 days. While most CNS depressants are effective, benzodiazepines (Chap. 54) have the highest margin of safety and lowest cost and are, therefore, the preferred class of drugs. Short-half-life benzodiazepines can be considered for patients with serious liver impairment or evidence of brain damage, but they must be given every 4 h to avoid abrupt blood-level fluctuations that may increase the risk for seizures. Therefore, most clinicians use drugs with longer half-lives (e.g., chlordiazepoxide), adjusting the dose if signs of withdrawal escalate, and withholding the drug if the patient is sleeping or has evidence of orthostatic hypotension. The average patient requires doses of 25–50 mg of chlordiazepoxide

or 10 mg of diazepam given PO every 4–6 h on the first day, with doses then decreased to zero over the next 5 days. While alcohol withdrawal can be treated in a hospital, patients in good physical condition who demonstrate mild signs of withdrawal despite low blood alcohol concentrations and who have no prior history of DTs or withdrawal seizures can be considered for outpatient detoxification. These patients should return daily for evaluation of vital signs and can be hospitalized if signs and symptoms of withdrawal escalate.

Treatment of the patient with DTs can be challenging, and the condition is likely to run a course of 3–5 days regardless of the therapy employed. The focus of care is to identify and correct medical problems and to control behavior and prevent injuries. Many clinicians recommend the use of high doses of a benzodiazepine (as much as 800 mg/d of chlordiazepoxide has been reported), a treatment that will decrease agitation and raise the seizure threshold but probably does little to improve the confusion. Other clinicians recommend the use of antipsychotic medications, such as haloperidol or olanzapine as discussed earlier, although these drugs have not been directly evaluated for DTs. Antipsychotics are less likely to exacerbate confusion but may increase the risk of seizures; they have no place in the treatment of mild withdrawal symptoms.

Generalized withdrawal seizures rarely require more than giving an adequate dose of benzodiazepines. There is little evidence that anticonvulsants such as phenytoin or gabapentin are more effective in drug-withdrawal seizures, and the risk of seizures has usually passed by the time effective drug levels are reached. The rare patient with status epilepticus must be treated aggressively (Chap. 26).

REHABILITATION OF ALCOHOLICS

An Overview After completing alcoholic rehabilitation, ≥60% of alcoholics, especially middle-class patients, maintain abstinence for at least a year, and many achieve lifetime sobriety. The core of treatment uses cognitive-behavioral approaches to help patients recognize the need to change, while working with them to alter their behaviors to enhance compliance. A key step is to optimize motivation toward abstinence through education about alcoholism and instructions to family members to stop protecting the patient from problems caused by alcohol. After years of heavy drinking, patients also need counseling, vocational rehabilitation, and self-help groups such as Alcoholics Anonymous (AA) to help them learn how to deal with life's stresses while sober. A third component, *relapse prevention*, helps the patient to identify situations in which a return to drinking is likely, formulate ways of managing